Online Policy Optimization in Unknown Nonlinear Systems

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Abstract

1. Introduction

Online policy optimization concerns controllers that learn while operating a system. Decisions are made sequentially along a single, non-episodic trajectory: at each step the agent observes the current state and outcome of past actions, selects an action using a parameterized policy, and immediately incurs a stage cost before moving on. There is no reset or repeated episode in which to re-try alternatives; improvements must be squeezed out of the one trajectory the system actually follows. This viewpoint is attractive in applications such as robotic manipulation, aerial vehicles, and process control, where data arrive during operation and the dynamics and objectives can change over time.

Two themes make this problem challenging in practice. First, the mechanism that maps actions to future outcomes may be partially known, time-varying, or outright unknown, which complicates choosing a reliable update direction. Second, algorithms must be both general and lightweight: they should handle nonlinear, time-varying systems and flexible policy classes, yet run in real-time without large replay buffers or repeated resimulation of counterfactual trajectories. Much of the early literature addressed linear or time-invariant settings; later work relaxed either the modeling assumptions or the generality of policies, but rarely both at once.

To navigate these constraints, the GAPS family adopts a surrogate-objective lens. Instead of evaluating a policy parameter by hypothetically rewinding and re-executing the entire past, the surrogate judges how a single parameter would have performed if it had been used from the start, and a gradient-based update is computed from information gathered along the realized trajectory. The original GAPS algorithm implements this idea with a short, logarithmically growing memory that captures the most influential recent sensitivities. Under a contractive perturbation view of the closed loop, this yields sublinear local regret guarantees. It also avoids heavy resimulation. M-GAPS refines the mechanism with an internal sensitivity recursion, collapsing the history into a constant-size memory and keeping per-step computation constant, which makes the approach suitable for embedded and realtime deployments.

In prior work on unknown dynamics, M-GAPS has been embedded in a meta-framework that couples an online policy optimizer (ALG) with an online estimator (EST). There, theoretical guarantees track how prediction errors in values and state derivatives translate into bounded perturbations of the policy update. M-GAPS standalone removes the estimator layer and studies the unperturbed regime in which the update has direct access to the model components it needs. This isolates the algorithmic phenomena underlying stability and regret, independent of identification quality, and provides a clean baseline against which estimation strategies can later be reintroduced.

Design principles. Two structural ingredients recur in our analysis and design. The first is a contraction- stability requirement for the joint evolution of the system and the algorithm's internal memory under slowly varying parameters: small parameter changes should induce state deviations that shrink over time, ensuring boundedness and rapid forgetting of past perturbations. The second is a slow-variation condition on the parameter path, achieved by standard stepsize choices on a smooth surrogate objective; this keeps the update stable while allowing adaptation to nonstationarity.

Contributions. This project contributes a clarified and tutorialized standalone presentation of M-GAPS, with emphasis on exposition rather than introducing new algorithmic or theoretical results:

- **Standalone regime formalization and correspondence.** We isolate the "standalone" case as a clean baseline for studying the policy-update mechanism without estimation, residual terms, exploration perturbations, or exogenous disturbances, via restating the online, non-episodic single-trajectory setting in terms of known components (g_t, π_t, h_t), and we make the correspondence to the original M-GAPS formulation explicit (what is removed/kept, and why).
- **Motivating worked example (early in the paper).** We develop a concrete nonlinear online control example (e.g., obstacle-avoiding tracking with saturating actuators) that explicitly instantiates the key ingredients (g_t, π_t, h_t), illustrates why online adaptation of policy parameters is useful in practice, and makes the operational role of the surrogate objective and the sensitivity recursion transparent.
- **Tutorial rewrite with proof scaffolding.** We reorganize the standalone analysis into a tutorial format with additional explanations and intermediate derivations (e.g., explicit chain-rule decompositions and term-by-term bounds), expanded/annotated proof details for key lemmas and theorems, consolidated notation and constant tables, and a dependency flowchart that visually connects all the formal statements of the paper, making the logical dependencies and proof pathways between statements immediately clear.

Relation to prior work. Our standpoint connects three threads. From online optimization, we inherit surrogate- based progress measures and projected gradient updates tailored to nonconvex, time- varying objectives. From learning- based control and adaptive control, we adopt the insight that stabilization and performance can be achieved by fitting along the realized trajectory rather than identifying a globally accurate model. From recent online control algorithms, we retain the focus on scalability and implementability, replacing history truncation with an internal recursion that preserves constant memory. In the companion unknown- dynamics setting, the same update can be combined with on- trajectory regression; here we deliberately abstract away estimation to expose the core algorithmic mechanisms.

Paper roadmap. We begin by detailing the standalone setting and assumptions, then present the standalone M-GAPS update and its computational structure. We follow with a comparison between the standalone M-GAPS and the original M-GAPS, and conclude with stability and regret analyses under contraction and slow variation.

2. Problem Setting

We first present the M-GAPS standalone setting. The main focus is on the policy update of θ from M-GAPS but with no perturbation, nonlinear residual, and disturbance.

Thus, in a discrete-time dynamical system, our simplified dynamics become

$$x_{t+1} = g_t(x_t, u_t), \tag{1}$$

where $x_t \in \mathbb{R}^n$ denotes the system state, $u_t \in \mathbb{R}^m$ denotes the control input, and g_t is the dynamical function.

To control this system, the online agent adopts a time-varying control policy π_t that is parameterized by a policy parameter $\theta_t \in \Theta \subseteq \mathbb{R}^d$. Specifically, the online agent picks the control input from the policy class

$$u_t = \pi_t(x_t, \theta_t). \tag{2}$$

The objective of the online agent is to minimize the total cost

$$\sum_{t=0}^{T-1} c_t \tag{3}$$

incurred over a finite horizon, where the stage cost at time step t is given by

$$c_t = h_t(x_t, u_t, \theta_t)^{1} \tag{4}$$

As a minor technical detail, we require that $h_{T-1} \geq 0$ in the proof of Theorem 13 at (12c) in Section 5.1 when upper bounding expressions.

We provide a simple nonlinear control example that can be captured by our online policy optimization framework to help the readers understand the concepts we discussed.

Example 1 (Obstacle-avoiding tracking with saturating actuators) *Consider the problem setting with known dynamics g_t and a time–varying policy $\pi_t(\cdot, \theta_t)$. Let $x_t \in \mathbb{R}^2$ be the state and $u_t \in \mathbb{R}^2$ the control. The dynamics include smooth actuator saturation and a known exogenous drift (disturbance) $d_t \in \mathbb{R}^2$:*

$$x_{t+1} = g_t(x_t, u_t) = x_t + \Delta\, \sigma(u_t) \;+\; d_t, \qquad \sigma(z) = \tanh(z) \text{ (applied elementwise)}, \;\; \Delta > 0.$$

At time t we are given a reference $r_t \in \mathbb{R}^2$ and the center of a moving obstacle $o_t \in \mathbb{R}^2$. Define the repulsive potential/proximity penalty and its gradient

$$\phi_t(x) = \frac{1}{\|x - o_t\|^2 + \varepsilon}, \qquad p_t(x) = \nabla \phi_t(x) = -\frac{2(x - o_t)}{(\|x - o_t\|^2 + \varepsilon)^2}, \quad \varepsilon > 0.$$

The online agent chooses inputs from the policy class

$$u_t = \pi_t(x_t, \theta_t) = \alpha_t\,(r_t - x_t) + \beta_t\, p_t(x_t), \qquad \theta_t = (\alpha_t, \beta_t) \in \Theta := [-\kappa, \kappa]^2,$$

where $\kappa > 0$ is a fixed design bound so Θ is convex and compact.

The stage cost used in the problem setting ($c_t = h_t(x_t, u_t, \theta_t)$) balances tracking, effort, and safety:

$$c_t = h_t(x_t, u_t, \theta_t) = \|x_t - r_t\|_2^2 \;+\; \lambda_u \|u_t\|_2^2 \;+\; \lambda_o\, \phi_t(x_t).$$

This models a ground robot (or planar UAV) that must track a moving waypoint r_t while keeping distance from a pedestrian/vehicle at o_t; the tanh *nonlinearity captures actuator limits, the term d_t captures a known drift (disturbance)(e.g., wind or conveyor motion), and the gains $\theta_t = (\alpha_t, \beta_t)$ adapt online to trade off tracking and obstacle clearance.*

1. We include θ_t in the stage cost for generality, allowing e.g. regularization. $c_t = h_t(x_t, u_t)$ is a special case.

3. Main Result

This section introduces a concrete, simplified online policy optimization algorithm, Memoryless Gradient-based Adaptive Policy Selection (M-GAPS, Algorithm 1), together with the fundamental definitions that formalize the underlying dynamical system, admissible parameter evolution, stability structure, and optimization objectives.

3.1. The M-GAPS Algorithm

We model the controller together with the plant as the following joint recursion:

$$\begin{pmatrix} x_{t+1} \\ y_{t+1} \\ \theta_{t+1} \end{pmatrix} = q_t(x_t, y_t, \theta_t) = \begin{pmatrix} q_t^x(x_t, y_t, \theta_t) \\ q_t^y(x_t, y_t, \theta_t) \\ q_t^\theta(x_t, y_t, \theta_t) \end{pmatrix}, \quad x_t \in \mathbb{R}^n,\ y_t \in \mathbb{R}^p,\ \theta_t \in \Theta \subset \mathbb{R}^d. \tag{5}$$

Here, $y_t \in \mathbb{R}^p$ is an auxiliary state that M-GAPS can use to store something besides the system state x_t and the policy parameter θ_t to help it perform the update. For example, y_t can be a finite memory buffer that stores information from the past. It can also be the integral of past tracking error in an integral controller. To understand (5) intuitively, it is helpful to draw connections with the process of using a gradient-based optimizer to update θ_t iteratively where the exact gradients are available. In more practical scenarios though, the optimizer can only use biased gradient estimations, which still perform well in general.

To make explicit how θ_t affects both the next state and the stage cost, we collect the closed-loop maps:

$$u_t \coloneqq \pi_t(x_t, \theta_t), \tag{6a}$$

$$g_{t+1|t}(x_t, \theta_t) \coloneqq g_t(x_t,\ u_t) = g_t(x_t, \pi_t(x_t, \theta_t)), \tag{6b}$$

$$h_{t|t}(x_t, \theta_t) \coloneqq h_t(x_t,\ u_t,\ \theta_t) = h_t(x_t, \pi_t(x_t, \theta_t), \theta_t). \tag{6c}$$

These operational steps (applying u_t, plant evolution to x_{t+1}, and incurring c_t) are shown inline in Algorithm 1. We then analyze the properties of the induced joint recursion (x_t, y_t, θ_t).

Algorithm 1: Memoryless Gradient-based Adaptive Policy Selection

Parameters: Learning rate η, initial parameter θ_0.
Initialize: Policy parameter θ_0; Internal state $y_0 = \mathbf{0}_{n\times d}$.
for $t = 0, 1, \ldots, T-1$ **do**
  Take inputs x_t, θ_t, g_t, h_t, and π_t.
  Set $u_t \leftarrow \pi_t(x_t, \theta_t)$; $x_{t+1} \leftarrow g_t(x_t, u_t)$; $c_t \leftarrow h_t(x_t, u_t, \theta_t)$. */* Controller action and plant evolution (context, not part of gradient calc) */*
  Use these to obtain $g_{t+1|t}$ and $h_{t|t}$. */* $g_{t+1|t}$ and $h_{t|t}$ are defined in (6). */*
  Update $y_{t+1} \leftarrow \left.\frac{\partial g_{t+1|t}}{\partial x_t}\right|_{x_t,\theta_t} \cdot y_t + \left.\frac{\partial g_{t+1|t}}{\partial \theta_t}\right|_{x_t,\theta_t}$. */* Update partial derivatives accumulator. */*
  Let $G_t \leftarrow \left.\frac{\partial h_{t|t}}{\partial x_t}\right|_{x_t,\theta_t} \cdot y_t + \left.\frac{\partial h_{t|t}}{\partial \theta_t}\right|_{x_t,\theta_t}$.
  Update and output $\theta_{t+1} \leftarrow \prod_\Theta(\theta_t - \eta G_t)$. */* Π_Θ is the Euclidean projection to Θ. */*
end

The standalone M-GAPS is modified from the M-GAPS algorithm but only with simpler dynamics and settings. Thus, in the setting where the online agent has exact knowledge of the time-varying dynamical models, simplified M-GAPS can achieve better regret guarantees than M-GAPS, and the same memory/computational complexities $O(1)$. This significantly improves computational efficiency as the Gradient-based Adaptive Policy Selection (GAPS) algorithm (Lin et al., 2023), which M-GAPS takes inspiration from, has memory/computational complexities of $O(\log T)$. To understand why this improvement is possible, note that, on the one hand, the M-GAPS algorithm considers the perturbation/biases which contributes additively to the local regret bound; on the other hand, to design an efficient estimation of $\nabla F_t(\theta_t)$, GAPS does two steps of approximations: first replacing the imaginary trajectory with the actual trajectory and then doing a bounded-memory truncation, but the (standalone) M-GAPS only keeps the first approximation step of GAPS by introducing the auxiliary internal state y_t that accumulates past partial derivatives. This greatly simplifies the computation. A more detailed comparison between standalone M-GAPS and GAPS can be found in Section 6.

3.2. System and Performance Definition/Analytical Framework

A key concept in this paper is the ideal trajectory obtained by rolling out a given policy–parameter sequence $\theta_{0:T-1}$ on the true dynamics. To formalize this, we introduce the multi-step dynamics/cost, which characterize how the system evolves under $\theta_{0:T-1}$ without any estimation module. This mirrors the construction in Lin et al. (2023) (which studies online policy selection with known dynamical systems) and differs only in notation. Concretely, throughout this work we use the multi-step mappings in Definition 1 to define the state and cost at time t resulting from applying the control inputs $u_\tau^*(\theta) := \pi_\tau(\tilde{x}_\tau^*(\theta), \theta)$ recursively from $\tau = 0$ to t on the true system. No estimated quantities or "actual" trajectories are needed in our setting; the ideal rollout under the true dynamics is the sole reference path used in both analysis and evaluation.

Definition 1 (Multi-Step Dynamics and Cost) *The multi-step dynamics $g_{t|\tau}$ between two time steps $\tau \leq t$ specifies the state x_t as a function of the previous state x_τ and previous policy parameters $\theta_{\tau:t-1}$. It is defined recursively, with the base case $g_{\tau|\tau}(x_\tau) := x_\tau$ and the recursive case*

$$g_{t+1|\tau}(x_\tau, \theta_{\tau:t}) = g_t(z_t,\ \pi_t(z_t, \theta_t)), \quad \forall\, t \geq \tau,$$

in which $z_t := g_{t|\tau}(x_\tau, \theta_{\tau:t-1})$.[2] *The multi-step cost $h_{t|\tau}$ specifies the cost c_t as function of x_τ and $\theta_{\tau:t}$. It is defined as*

$$h_{t|\tau}(x_\tau, \theta_{\tau:t}) := h_t(z_t, \pi_t(z_t, \theta_t), \theta_t).$$

It is worth emphasizing that, in our work, the concepts of multi-step dynamics/cost are only used for the theoretical analysis, because their definitions involve true model parameters that are unknown to the online agent. [Shengnan: Could you please go over the explanation part as you said in the meeting you have some ideas on how to formulate this one]

The multi-step dynamics and cost defined above describe the evolution of the system under an arbitrary policy parameter sequence $\theta_{\tau:t}$. In our analysis, we restrict attention to sequences that vary gradually over time, which plays a crucial role in ensuring contraction stability and controlling trajectory mismatch. We formalize this class below.

2. z_t is an auxiliary variable to denote the state at t under initial state x_τ and parameters $\theta_{\tau:t}$.

Definition 2 *We denote the set of policy parameter sequences with ϵ_θ-constrained step size by*

$$S_{\epsilon_\theta}(t_1 : t_2) \coloneqq \{\theta_{t_1:t_2} \in \Theta^{t_2-t_1+1} \mid \|\theta_{\tau+1} - \theta_\tau\| \le \epsilon_\theta, \forall \tau \in [t_1 : t_2 - 1]\}.$$

The above definition specifies the class of slowly-varying policy parameter sequences considered in our analysis. We next formalize the contraction framework of the induced multi-step dynamics under such admissible sequences.

Definition 3 (ϵ_θ-Time-varying Contractive Perturbation) *For $\epsilon_\theta \ge 0$, we say the ϵ_θ-time-varying contractive perturbation property holds for $R_C > 0, C \ge 1$, and $\rho \in (0, 1)$ if, for any parameter sequence $\theta_{\tau:t-1} \in S_{\epsilon_\theta}(\tau : t-1)$, the following inequality holds for arbitrary $x_\tau, x'_\tau \in B_n(0, R_C)$ and time steps $\tau \le t$:*

$$\left\|g_{t|\tau}(x_\tau, \theta_{\tau:t-1}) - g_{t|\tau}(x'_\tau, \theta_{\tau:t-1})\right\| \le C\rho^{t-\tau}\left\|x_\tau - x'_\tau\right\|$$

While the contractive perturbation property controls the relative distance between trajectories, we also require that the system remains uniformly bounded under admissible parameter sequences. This is formalized below.

Definition 4 (ϵ_θ-Time-varying Stability) *For $\epsilon_\theta \ge 0$, the ϵ_θ-time-varying stability property holds for $R_S > 0$ if, for any $\theta_{\tau:t-1} \in S_{\epsilon_\theta}(\tau : t-1)$, $\left\|g_{t|\tau}(0, \theta_{\tau:t-1})\right\| \le R_S$ holds for any $t \ge \tau$.*

Having specified the dynamical structure under which the system evolves, we now turn to the optimization perspective. Our goal is to evaluate the performance of the online policy update through appropriate regret notions.

In the literature of online optimization, regret is a common performance metric that directly compares the total cost $\sum_{t=0}^{T-1} c_t$ incurred by the online policy optimization algorithm against the optimal total cost one can achieve in hindsight.

However, in constrained and potentially nonconvex settings, first-order optimality cannot be characterized solely by the gradient; instead, first-order stationarity becomes the appropriate notion of optimality. Following Hazan et al. (2017), we therefore introduce the notion of the projected gradient, which enables a first-order characterization of optimality on constrained domains and serves as the basis for the projected-gradient-based local regret.

Definition 5 (Projected gradient) *Let $F : \Theta \to \mathbb{R}$ be a differentiable function on a closed convex set $\Theta \subseteq \mathbb{R}^d$. For $\eta > 0$, the (Θ, η)-projected gradient of F is defined as*

$$\nabla_{\Theta,\eta} F(\theta) \coloneqq \frac{1}{\eta}(\theta - \Pi_\Theta(\theta - \eta \nabla F(\theta))).$$

This concept of projected gradient is used to define the local regret introduced later in this section, and Section 5.1 that studies online gradient descent for online nonconvex optimization with constraints.

To cast the online control problem into an optimization framework, we introduce a time-dependent surrogate cost function that captures the cumulative effect of the policy parameter on the system trajectory.

Definition 6 (Surrogate Cost) *The surrogate cost function is*

$$F_t(\theta) := h_t(\tilde{x}_t(\theta), \tilde{u}_t(\theta), \theta),$$

where $\tilde{x}_t(\theta)$ and $\tilde{u}_t(\theta)$ are the system state and control input at time t if the agent is at the state recursively defined as

$$\tilde{x}_\tau(\theta) := g_\tau(\tilde{x}_{\tau-1}(\theta), \tilde{u}_{\tau-1}(\theta))$$

and applies the control inputs

$$\tilde{u}_\tau(\theta) := \pi_\tau(\tilde{x}_\tau(\theta), \theta)$$

at all previous time steps $\tau = 0, 1, \ldots, t$.

Intuitively, the surrogate cost $F_t(\theta)$ evaluates how good a policy parameter θ is at an intermediate time step t by eliminating the influence of any past policy parameters $\theta_{0:t-1}$. This concept is useful for defining different regret metrics. For example, the *static regret* is a widely-used metric in the literature of online policy optimization (Cesa-Bianchi and Lugosi (2006); Hazan and Seshadhri (2009)) that compares the total cost of an online agent with the best static policy parameter in hindsight can be written as

$$R^S(T) := \sum_{t=0}^{T-1} c_t - \min_{\theta \in \Theta} \sum_{t=0}^{T-1} F_t(\theta).$$

However, as noted in previous works (Lin et al., 2023; Hazan et al., 2017), regret metrics that directly compare the cost difference (like $R^S(T)$) are not always suitable for nonconvex cost functions because gradient-based online optimization algorithms may easily get stuck in local minima even when the cost functions are time-invariant. Thus, the metric of *local regret* is used. For a sequence of policy parameters $\theta_{0:T-1}$, the local regret is defined as

$$R^L_\eta(T) := \sum_{t=0}^{T-1} \|\nabla_{\eta,\Theta} F_t(\theta_t)\|^2.$$

Here, the projected gradient $\nabla_{\eta,\Theta} F_t(\theta_t)$ (parameterized by η) is a surrogate of the original gradient $\nabla F_t(\theta_t)$ that also considers the constraint set Θ. Specifically, an update step with the projected gradient is equivalent to projecting the output of the original gradient descent step back onto Θ, i.e.

$$\text{for any } \theta \in \Theta, \quad \theta - \eta \nabla_{\eta,\Theta} F_t(\theta) = \Pi_\Theta(\theta - \eta \nabla F_t(\theta)),$$

where Π_Θ is the Euclidean projection to Θ. This notion of local regret is first introduced by Hazan et al. (2017), and we provide a formal definition in Definition 5 in Appendix B. Intuitively, the local regret measures how well the policy parameter sequence $\theta_{0:T-1}$ tracks the changing stationary points of the surrogate cost functions $F_{0:T-1}$ (when $\Theta = \mathbb{R}^d$). In the time-invariant setting, sublinear local regret implies convergence to a stationary point.

Although local regret is useful for measuring the performance of an online policy optimization algorithm under nonconvex surrogate costs, applying it alone to our setting with time-varying dynamics has a limitation: the surrogate cost F_t is defined via an idealized multi-step rollout that fixes the policy parameter and thus serves as an analytical reference rather than a quantity directly optimized along the online parameter path.

To account for this, in addition to bounding the local regret of the policy parameters $\theta_{0:T-1}$, our analysis explicitly controls how the quantities used in the policy update— including the accumulated sensitivities—evolve along the single trajectory induced by the online updates under time-varying dynamics.

To summarize, the standalone M-GAPS minimizes the surrogate losses of a time-dependent sequence, and we use projected-gradient-based local regret to assess how well it is doing.

3.3. Regret Bound Guarantee

When applied to the dynamical system (5) and the policy class (6a), the joint dynamics induced by applying standalone M-GAPS are given by

$$x_{t+1} = q_t^x(x_t, y_t, \theta_t) = g_t(x_t, \pi_t(x_t, \theta_t)), \tag{7a}$$

$$y_{t+1} = q_t^y(x_t, y_t, \theta_t) = \left.\frac{\partial g_{t+1|t}}{\partial x_t}\right|_{x_t,\theta_t} \cdot y_t + \left.\frac{\partial g_{t+1|t}}{\partial \theta_t}\right|_{x_t,\theta_t}, \tag{7b}$$

$$\theta_{t+1} = q_t^\theta(x_t, y_t, \theta_t) = \Pi_\Theta\left(\theta_t - \eta\left(\left.\frac{\partial h_{t|t}}{\partial x_t}\right|_{x_t,\theta_t} \cdot y_t + \left.\frac{\partial h_{t|t}}{\partial \theta_t}\right|_{x_t,\theta_t}\right)\right). \tag{7c}$$

The following property captures the robustness of the policy update performed by M-GAPS. In the disturbance-free setting with known time-varying dynamics, the standalone M-GAPS algorithm achieves sublinear projected-gradient local regret.

Property 7 (Regret Bound) *Consider the dynamical system stated above in 7. Suppose the learning rate η satisfies $\eta < \min\left\{\frac{(1-\rho)\epsilon_\theta}{C_{L,h,\theta}}, \frac{1-\rho}{2C_{\ell,h,(\theta,\theta)}}\right\}$. Then, the parameter increments obey $\|\theta_t - \theta_{t-1}\| \leq \frac{C_{L,h,\theta}}{1-\rho}\eta \leq \epsilon_\theta$ for all t, so $\{\theta_t\}$ is slowly-time–varying. Further, the trajectory $\{\theta_t\}$ achieves the local regret guarantee*

$$R_\eta^L(T) \leq \frac{2}{\eta}(F_0(\theta_0) + S_0) + \frac{2}{1-\rho}(C_{L,h,\theta}S_1 + C_{\ell,h,(\theta,\theta)}\eta S_2), \text{ where}$$

$$\begin{aligned}
S_0 &\coloneqq \frac{2\bar{C}L_h(1 + L_{\pi,x} + L_{f,x})(1 + L_{g,u})}{(1-\rho)^2\rho} \cdot V_{sys} \\
&\quad + \frac{2\bar{C}L_h(1 + L_{\pi,x} + L_{f,x})}{1-\rho} \cdot \left(2\bar{C}\bar{R}_C + 2R_S\right), \\
S_1 &\coloneqq \left(\frac{\hat{C}_0}{1-\rho} + \frac{\hat{C}_1 + \hat{C}_2}{(1-\rho)^2} + \frac{\hat{C}_2}{(1-\rho)^3}\right)\eta T, \\
S_2 &\coloneqq \left(1 + \frac{\hat{C}_0}{1-\rho} + \frac{\hat{C}_1 + \hat{C}_2 + \hat{C}_3 + \hat{C}_5}{(1-\rho)^2} + \frac{\hat{C}_2 + \hat{C}_4}{(1-\rho)^3}\right) \cdot \\
&\quad \left(\frac{\hat{C}_0}{1-\rho} + \frac{\hat{C}_1 + \hat{C}_2}{(1-\rho)^2} + \frac{\hat{C}_2}{(1-\rho)^3}\right)\eta^2 T.
\end{aligned}$$

Here, the variation intensity is defined as

$$V_{sys} = \sum_{t=1}^{T-1}\left(\sup_{x\in\mathcal{X},u\in\mathcal{U}} \|g_t(x,u) - g_{t-1}(x,u)\| + \sup_{x\in\mathcal{X},\theta\in\Theta} \|\pi_t(x,\theta) - \pi_{t-1}(x,\theta)\|\right.$$

$$+ \sup_{x\in\mathcal{X},u\in\mathcal{U},\theta\in\Theta} |h_t(x,u,\theta) - h_{t-1}(x,u,\theta)| \Bigg).$$

The bound can be simplified to

$$R_\eta^L(T) = O\left(\frac{1}{\eta}(1+V_{sys}) + \eta T + \eta^3 T\right),$$

where the $O(\cdot)$ notation hides dependence on $\frac{1}{1-\rho}, R_x, R_y, \bar{C}$, and the Lipchitzness/smoothness coefficients defined in Assumption 11.

The definition of the coefficient $C_{L,h,\theta}$ can be found in Corollary 17.

To build intuition for Property 7, consider classical online gradient descent (OGD) in a setting without state evolution or system dynamics. With unbiased gradients, each update is a small step of size η, so the parameter trajectory is inherently slowly varying (i.e., $\|\theta_{t+1} - \theta_t\|$ remains small). Property 7 captures an analogous phenomenon in our dynamical setting: controlled per-step updates ensure gradual parameter variation, and when the stepsize is chosen within the admissible range $\eta \le \bar{\eta}$, the baseline rate $R_\eta^L(T)$ follows.

One thing to notice is that, for the standalone M-GAPS and the nonconvex local regret result in Lin et al. (2023), we have $\Theta = \mathbb{R}^d$. Since the projected gradients are identical to the original gradients when $\Theta = \mathbb{R}^d$, the local regret bound $R_\eta^L(T)$ of standalone M-GAPS given by Property 7 matches the local regret bound of GAPS in Lin et al. (2023).

4. Structural and Regularity Assumptions

4.1. Choice of Compact Domains and Radii

Having established the algorithmic and analytical framework, we now introduce the structural and regularity assumptions required for stability and regret analysis. The first main assumption (Assumption 8) is about regulating our domain and the choice of radius.

Assumption 8 (Auxiliary radii and compact domains) *Notice that when the ϵ_θ-time-varying contractive perturbation (Definition 3) holds globally, $R_C = \infty$. Therefore, like Lin et al. (2023), we assume there exists a real number $\bar{R}_C > 0$ such that*

$$R_C \;>\; \bar{R}_C \ge \bar{C}\big(R_S + \bar{C}\|x_0\|\big) + R_S,$$

and we fix any $\bar{R}_C > 0$ satisfying the strict inequality above when $R_C = \infty$. Similarly, to leverage the Lipschitzness/smoothness property, we require $\mathcal{X} \supseteq B_n(0, R_x)$ where $R_x \ge \bar{C}\bar{R}_C + R_S$ and $\mathcal{U} = \{\pi(x,\theta) \mid x \in \mathcal{X}, \theta \in \Theta, \pi \in \mathcal{G}\}$. But to pin down the compact domains on which the Lipschitz/smooth constants in Assumption 11 are evaluated, we choose

$$\mathcal{X} \;:=\; B_n(0, R_x), \qquad R_x = \bar{C}\bar{R}_C + R_S,$$

and define the control-value set

$$\mathcal{U} \;:=\; \{\, \pi_t(x,\theta) \mid x \in \mathcal{X},\ \theta \in \Theta,\ (g,\pi) \in \mathcal{G},\ t \in \mathcal{T} \,\}.$$

as the coefficients in Assumption 11 depend on $\mathcal{X}$ and $\mathcal{U}$. We also fix

$$\mathcal{Y} \;:=\; B_p(0, R_y), \qquad R_y \;:=\; \frac{\bar{C}\, L_{g,u}\, L_{\pi,\theta}}{\rho(1-\rho)},$$

so that the internal state y_t remains in $\mathcal{Y}$ along the trajectories considered. This assumption is purely notational/technical: it selects compact radii for evaluating Lipschitz/smoothness and implies the initial-state condition $\|x_0\| \le \frac{R_C-(\bar{C}+1)R_S}{\bar{C}^2}$ in Assumption 11 and Assumption 12.

4.2. Regularity Notions for Block-Structured Mappings

To formulate the regularity assumptions on the system dynamics, policy map, and cost functions in a structured manner, we introduce block-wise Lipschitz and smoothness notions. These definitions allow us to track the sensitivity of different argument blocks separately, which will be essential for deriving contraction and gradient bounds. Note that all norms are Euclidean unless otherwise specified.

Definition 9 (Block Lipschitzness) *Let $\mathcal{Z} \subseteq \mathbb{R}^d$ and let $z = (z^{(1)}, \dots, z^{(k)})$ with $z^{(i)} \in \mathbb{R}^{d_i}$, $\sum_i d_i = d$. A (possibly vector–valued) map $\phi : \mathcal{Z} \to \mathbb{R}^p$ is <u>$(L_1, \dots, L_k)$–Lipschitz on $\mathcal{Z}$</u> if there exist nonnegative constants $L_1, \dots, L_k$ such that, for all $z, z' \in \mathcal{Z}$,*

$$\big\|\phi(z) - \phi(z')\big\| \;\le\; \sum_{i=1}^{k} L_i \left\|z^{(i)} - z'^{(i)}\right\|.$$

When $k = 1$, this reduces to the usual L–Lipschitz condition. The constants L_i may depend on $\mathcal{Z}$.

Definition 10 (Block smoothness (Lipschitz Jacobian / gradient)) *With the same block decomposition, a differentiable map $\phi : \mathcal{Z} \to \mathbb{R}^p$ is <u>$(\ell_1, \dots, \ell_k)$–smooth on $\mathcal{Z}$</u> if its Jacobian $J\phi(z)$ exists for all $z \in \mathcal{Z}$ and is block–Lipschitz: for all $z, z' \in \mathcal{Z}$,*

$$\big\|J\phi(z) - J\phi(z')\big\| \;\le\; \sum_{i=1}^{k} \ell_i \left\|z^{(i)} - z'^{(i)}\right\|.$$

In the scalar case $f : \mathcal{Z} \to \mathbb{R}$, this is equivalent to the gradient being block–Lipschitz:

$$\big\|\nabla f(z) - \nabla f(z')\big\| \;\le\; \sum_{i=1}^{k} \ell_i \left\|z^{(i)} - z'^{(i)}\right\| \quad \textit{for all } z, z' \in \mathcal{Z}.$$

Remark (block vs. single constant). In finite dimensions, the block and single-constant formulations are equivalent up to a fixed multiplicative factor. We adopt the block form to explicitly track per-argument sensitivities, which leads to sharper contraction estimates and cleaner chain-rule bounds in the subsequent analysis.

4.3. Regularity of Dynamics, Policy, and Cost

Using the block Lipschitz and smoothness notions introduced above, we impose regularity conditions on the system dynamics, policy map, and cost functions, which is our second main assumption (Assumption 11).

Assumption 11 *The dynamics $g_{0:T-1}$, policies $\pi_{0:T-1}$, and costs $h_{0:T-1}$ are differentiable at every time step and satisfy that, for any compact sets $\mathcal{X} \subseteq \mathbb{R}^n$, $\mathcal{U} \subseteq \mathbb{R}^m$, one can find Lipschitzness/smoothness constants (can depend on $\mathcal{X}$ and $\mathcal{U}$) such that:*
1. $g_t(x, u)$ is $(L_{g,x}, L_{g,u})$-Lipschitz and $(\ell_{g,x}, \ell_{g,u})$-smooth in (x, u) on $\mathcal{X} \times \mathcal{U}$.
2. $\pi_t(x, \theta)$ is $(L_{\pi,x}, L_{\pi,\theta})$-Lipschitz and $(\ell_{\pi,x}, \ell_{\pi,\theta})$-smooth in (x, θ) on $\mathcal{X} \times \Theta$.
3. $h_t(x, u, \theta)$ is $(L_{h,x}, L_{h,u}, L_{h,\theta})$-Lipschitz and $(\ell_{h,x}, \ell_{h,u}, \ell_{h,\theta})$-smooth in (x, u, θ) on $\mathcal{X} \times \mathcal{U} \times \Theta$.

[Shengnan: I didn't see a clear definition of L_h, but probably it comes from the following?] For convenience, we also use the shorthand

$$L_h \coloneqq \max\{L_{h,x},\, L_{h,u},\, L_{h,\theta}\},$$

so that bounds involving the Lipschitz constants of $h_t(x, u, \theta)$ can be written compactly (e.g., $L_{h,x} + L_{h,u} L_{\pi,x} \leq L_h(1 + L_{\pi,x})$).

4.4. Contraction and Stability of Multi-Step Dynamics

In our third assumption (Assumption 12), we impose contraction and stability conditions on the multi-step dynamics induced by the environment. This assumption ensures that trajectory differences decay over time and that system states remain uniformly bounded under admissible parameter sequences.

Assumption 12 *Let $\mathcal{G}$ denote the set of all possible sequences $\{g_t, \pi_t\}_{t \in \mathcal{T}}$ the environment may provide. For a fixed $\epsilon_\theta \in \mathbb{R}_{\geq 0}$, the ϵ_θ-time-varying contractive perturbation holds with $(R_C, \bar{C}, \rho)$ for any sequence in $\mathcal{G}$. The ϵ_θ-time-varying stability holds with $R_S < R_C$ for any sequence in $\mathcal{G}$. We assume that the initial state satisfies $\|x_0\| \leq \frac{R_C - (\bar{C}+1)R_S}{\bar{C}^2}$. Further, we assume that if $\{g, \pi\}$ is the dynamics/disturbance/policy at an intermediate time step of a sequence in $\mathcal{G}$, then the time-invariant sequence $\{g, \pi\}_{\times T}$ is also in $\mathcal{G}$.*[3]

5. Technical Analysis and Regret Guarantees

In the following subsections, we summarize some useful existing results in Lin et al. (2023) that can help us in the proof of M-GAPS (Algorithm 1). We can build our proof upon some results shown in Lin et al. (2023) because of the similarity between our M-GAPS algorithm and the GAPS algorithm proposed by Lin et al. (2023) when applied to known dynamical systems: Both algorithms are designed to efficiently approximate the gradient $\nabla F_t(\theta_t)$ of the surrogate cost. Note that $\nabla F_t(\theta_t)$ can be expressed as

$$\nabla F_t(\theta_t) = \sum_{\tau=0}^{t} \frac{\partial h_{t|0}}{\partial \theta_{t-\tau}}\bigg|_{x_0, (\theta_t)_{\times(t+1)}}.$$

3. For $\{g, \pi\}_{\times T}$ to be in $\mathcal{G}$, it must satisfy other assumptions about contractive perturbation and stability that we impose on $\mathcal{G}$ but does not need to occur in real problem instances. This assumption can be made without the loss of generality for time-invariant dynamics and policy classes.

M-GAPS adopts the approximation G_t that is given by

$$G_t = \sum_{\tau=0}^{\min(B-1,t)} \frac{\partial h_{t|0}}{\partial \theta_{t-\tau}}\bigg|_{x_0,\theta_{0:t}}, \tag{8}$$

which simplifies $\nabla F_t(\theta_t)$ by replacing the imaginary trajectory achieved by using policy parameter θ_t repeatedly with the actual trajectory. The approximator of GAPS, which we denote as G'_t, takes an additional step to approximate G_t by truncating the summation from time 0 to t to at most B time steps, i.e.,

$$G'_t = \sum_{\tau=\max\{0,t-B\}}^{t} \frac{\partial h_{t|0}}{\partial \theta_{t-\tau}}\bigg|_{x_0,\theta_{0:t}},$$

where B is the buffer length parameter decided by the algorithm. Intuitively, the approximation G_t adopted by M-GAPS is closer to $\nabla F_t(\theta_t)$, which allows us to show the same guarantees for M-GAPS as GAPS when the true dynamics are known.

5.1. Local Regret of Inexact Projected Online Gradient Descent

[James: The following result controls the local regret of parameter sequences whose local steps have a bounded Euclidean error compared to online gradient descent.] We begin by establishing a general local regret bound for projected online gradient descent under inexact updates. This result applies to parameter sequences whose updates deviate from exact projected gradient descent by a bounded error, and will serve as a key tool for analyzing the regret of the M-GAPS algorithm.

Theorem 13 *Consider a parameter sequence (θ_t) that satisfies*

$$\|\theta_{t+1} - (\theta_t - \eta \nabla_{\Theta,\eta} F_t(\theta_t))\| \le \eta \beta_t, \text{ for all } t \ge 0.$$

Suppose at every time t, F_t is ℓ_F-smooth and L_F-Lipschitz in Θ. If the learning rate $\eta \le \frac{1}{\ell_F}$, then the local regret $\sum_{t=0}^{T-1} \|\nabla_{\Theta,\eta} F_t(\theta_t)\|^2$ is upper bounded by

$$\frac{1}{\eta(1-\eta\ell_F)}\left(F_0(\theta_0) + \sum_{t=1}^{T-1} \|F_t - F_{t-1}\|_\infty)\right) + \frac{L_F \sum_{t=0}^{T-1} \beta_t + \ell_F \eta \sum_{t=0}^{T-1} \beta_t^2}{1-\eta\ell_F},$$

where $\|F - F'\|_\infty := \sup_{\theta\in\Theta} |F(\theta) - F'(\theta)|$.

Proof To begin, we assert the nonexpansiveness of Euclidean projection onto the compact convex set $\Theta \in \mathbb{R}^d$. This is a classic result in convex optimization (see, for example, Theorem 1.2.1 in Schneider (2014)):

Lemma 14 *Let q and q' be arbitrary points in $\mathbb{R}^d$. Let $p = \Pi_\Theta(q)$ and $p' = \Pi_\Theta(q')$. Then, the following inequality holds:*

$$\|p - p'\| \le \|q - q'\|.$$

Now we return to the proof of Theorem 13. We define the quantity

$$\epsilon_t := \frac{1}{\eta}(\theta_{t+1} - (\theta_t - \eta \nabla_{\Theta,\eta} F_t(\theta_t))),$$

to name the deviation of the steps in our parameter sequence from those of ideal OGD:

$$\theta_{t+1} - \theta_t = -\eta \nabla_{\Theta,\eta} F_t(\theta_t) + \eta \epsilon_t. \tag{9}$$

By the smoothness of $F_t(\cdot)$, we have

$$\begin{aligned}
F_t(\theta_{t+1}) &\le F_t(\theta_t) + \langle \nabla F_t(\theta_t), \theta_{t+1} - \theta_t \rangle + \frac{\ell_F}{2} \|\theta_{t+1} - \theta_t\|^2 \\
&= F_t(\theta_t) - \eta \langle \nabla F_t(\theta_t), \nabla_{\Theta,\eta} F_t(\theta_t) - \epsilon_t \rangle + \frac{\ell_F \eta^2}{2} \|\nabla_{\Theta,\eta} F_t(\theta_t) - \epsilon_t\|^2 && \text{(10a)} \\
&= F_t(\theta_t) - \eta \langle \nabla F_t(\theta_t), \nabla_{\Theta,\eta} F_t(\theta_t) \rangle + \frac{\ell_F \eta^2}{2} \|\nabla_{\Theta,\eta} F_t(\theta_t)\|^2 \\
&\quad + \eta \langle \nabla F_t(\theta_t), \epsilon_t \rangle - \ell_F \eta^2 \langle \nabla_{\Theta,\eta} F_t(\theta_t), \epsilon_t \rangle + \frac{\ell_F \eta^2}{2} \|\epsilon_t\|^2, && \text{(10b)}
\end{aligned}$$

where we use (9) in (10a). Recall that Θ is a closed convex subset of $\mathbb{R}^d$. Let

$$y \coloneqq \theta_t - \eta \nabla F_t(\theta_t), \qquad p \coloneqq \Pi_\Theta(y) = \theta_t - \eta \nabla_{\Theta,\eta} F_t(\theta_t).$$

Note that $\theta_t - \eta \nabla_{\Theta,\eta} F_t(\theta_t)$ is the projection of $\theta_t - \eta \nabla F_t(\theta_t)$ onto Θ and $\theta_t \in \Theta$. By the projection variational inequality for projection onto a closed convex set,

$$\langle y - p,\ v - p \rangle \le 0 \qquad \forall\, v \in \Theta.$$

Taking $v = \theta_t \in \Theta$ gives

$$\langle (\theta_t - \eta \nabla F_t(\theta_t)) - (\theta_t - \eta \nabla_{\Theta,\eta} F_t(\theta_t)), \theta_t - (\theta_t - \eta \nabla_{\Theta,\eta} F_t(\theta_t)) \rangle \le 0.$$

Expanding $y - p = -\eta(\nabla F_t(\theta_t) - \nabla_{\Theta,\eta} F_t(\theta_t))$ and $\theta_t - p = \eta\, \nabla_{\Theta,\eta} F_t(\theta_t)$ and dividing by $-\eta^2$ with $\eta > 0$ yields

$$\langle \nabla F_t(\theta_t), \nabla_{\Theta,\eta} F_t(\theta_t) \rangle \ \ge\ \|\nabla_{\Theta,\eta} F_t(\theta_t)\|^2.$$

Substituting this inequality into (10) gives that

$$\begin{aligned}
F_t(\theta_{t+1}) &\le F_t(\theta_t) - \eta \big\|\nabla_{\Theta,\eta} F_t(\theta_t)\big\|^2 + \frac{\ell_F \eta^2}{2} \big\|\nabla_{\Theta,\eta} F_t(\theta_t)\big\|^2 \\
&\qquad + \eta \big\langle \nabla F_t(\theta_t), \epsilon_t \big\rangle - \ell_F \eta^2 \big\langle \nabla_{\Theta,\eta} F_t(\theta_t), \epsilon_t \big\rangle + \frac{\ell_F \eta^2}{2} \big\|\epsilon_t\big\|^2 \\
&\le\ F_t(\theta_t) - \eta \big\|\nabla_{\Theta,\eta} F_t(\theta_t)\big\|^2 + \frac{\ell_F \eta^2}{2} \big\|\nabla_{\Theta,\eta} F_t(\theta_t)\big\|^2 + \eta \big\|\nabla F_t(\theta_t)\big\| \, \big\|\epsilon_t\big\| \\
&\qquad - \ell_F \eta^2 \big\langle \nabla_{\Theta,\eta} F_t(\theta_t), \epsilon_t \big\rangle + \frac{\ell_F \eta^2}{2} \big\|\epsilon_t\big\|^2 && \text{(11a)} \\
&=\ F_t(\theta_t) - \eta \big\|\nabla_{\Theta,\eta} F_t(\theta_t)\big\|^2 - \frac{\ell_F \eta^2}{2} \big\|\nabla_{\Theta,\eta} F_t(\theta_t) + \epsilon_t\big\|^2 \\
&\qquad + \ell_F \eta^2 \big\|\nabla_{\Theta,\eta} F_t(\theta_t)\big\|^2 + \ell_F \eta^2 \big\|\epsilon_t\big\|^2 + \eta \big\|\nabla F_t(\theta_t)\big\| \, \big\|\epsilon_t\big\| && \text{(11b)} \\
&=\ F_t(\theta_t) - \eta(1 - \ell_F \eta) \big\|\nabla_{\Theta,\eta} F_t(\theta_t)\big\|^2 - \frac{\ell_F \eta^2}{2} \big\|\nabla_{\Theta,\eta} F_t(\theta_t) + \epsilon_t\big\|^2 \\
&\qquad + \ell_F \eta^2 \big\|\epsilon_t\big\|^2 + \eta \big\|\nabla F_t(\theta_t)\big\| \, \big\|\epsilon_t\big\| && \text{(11c)}
\end{aligned}$$

$$\leq\ F_t(\theta_t) - \eta(1-\ell_F\eta)\left\|\nabla_{\Theta,\eta}F_t(\theta_t)\right\|^2 + \eta L_F\beta_t + \ell_F\eta^2\beta_t^2, \tag{11d}$$

where in (11a) we apply Cauchy–Schwarz to the linear term $\eta\langle\nabla F_t(\theta_t),\epsilon_t\rangle \leq \eta\|\nabla F_t(\theta_t)\|\|\epsilon_t\|$, then in (11b) use the completion-of-squares identity to rewrite $\frac{\ell_F\eta^2}{2}\|\nabla_{\Theta,\eta}F_t(\theta_t)\|^2$, $\frac{\ell_F\eta^2}{2}\|\epsilon_t\|^2$, and $-\ell_F\eta^2\langle\nabla_{\Theta,\eta}F_t(\theta_t),\epsilon_t\rangle$ in (11a); (11c) factors out the common constant for $-\eta\|\nabla_{\Theta,\eta}F_t(\theta_t)\|^2$ and $+\ell_F\eta^2\|\nabla_{\Theta,\eta}F_t(\theta_t)\|^2$; lastly, (11d) uses the Lipschitz constant L_F of F_t, the assumption bound $\|\epsilon_t\| \leq \beta_t$, and drops the negative term $-\frac{\ell_F\eta^2}{2}\|\nabla_{\Theta,\eta}F_t(\theta_t)+\epsilon_t\|^2$.

Rearranging and summing (11) over $t = 0, 1, \ldots, T-1$ gives

$$\begin{aligned}
&\eta(1-\ell_F\eta)\sum_{t=0}^{T-1}\left\|\nabla_{\Theta,\eta}F_t(\theta_t)\right\|^2\\
\leq&\sum_{t=0}^{T-1}(F_t(\theta_t)-F_t(\theta_{t+1})) + \eta L_F\sum_{t=0}^{T-1}\beta_t + \ell_F\eta^2\sum_{t=0}^{T-1}\beta_t^2\\
=&F_0(\theta_0) + \sum_{t=1}^{T-1}(F_t(\theta_t)-F_{t-1}(\theta_t)) - F_{T-1}(\theta_T) + \eta L_F\sum_{t=0}^{T-1}\beta_t + \ell_F\eta^2\sum_{t=0}^{T-1}\beta_t^2 && \text{(12a)}\\
\leq&F_0(\theta_0) + \sum_{t=1}^{T-1}\|F_t-F_{t-1}\|_\infty - F_{T-1}(\theta_T) + \eta L_F\sum_{t=0}^{T-1}\beta_t + \ell_F\eta^2\sum_{t=0}^{T-1}\beta_t^2 && \text{(12b)}\\
\leq&F_0(\theta_0) + \sum_{t=1}^{T-1}\|F_t-F_{t-1}\|_\infty + \eta L_F\sum_{t=0}^{T-1}\beta_t + \ell_F\eta^2\sum_{t=0}^{T-1}\beta_t^2. && \text{(12c)}
\end{aligned}$$

where in (12a) we write out $t = 0$ case for the first sum and leave the rest in the summation format; in (12b) we apply the definition of $\|\cdot\|_\infty$ to $\sum_{t=1}^{T-1}(F_t(\theta_t)-F_{t-1}(\theta_t))$; in (12c) we use $F_{T-1}(\theta_T) \geq 0$ and thus drop this negative term[4]. ■

5.2. Regularity of Multi-Step Mappings

In this subsection, we translate some results from Lin et al. (2023) into the settings of matched-disturbance dynamics application discussed in Section 2. Lemma 16 is Lemma D.3 in Lin et al. (2023). We changed the condition $x_\tau, x'_\tau \in B_n(0, R_S + C\|x_0\|)$ to $x_\tau, x'_\tau \in B_n(0, \bar{R}_C)$, where $\bar{R}_C$ can be any positive number that satisfies $\bar{R}_C < R_C$ and $\bar{C}\bar{R}_C + R_S \leq R_x$. This minor change will not affect the proof provided in Lin et al. (2023). [Shengnan: Why keep the current R_x–$\bar{R}_C$ definition? In both GAPS and M-GAPS, the two radii serve different purposes: the contraction

4. This follows from our assumptions, which make the inputs of h_t uniformly bounded: $\theta \in \Theta$ with Θ compact, $\tilde{x}_t(\theta)$ remains in a compact ball by ϵ_θ–time-varying stability (hence $\tilde{x}_t(\theta) \in B_n(0, R_x)$), and $\tilde{u}_t(\theta) = \pi_t(\tilde{x}_t(\theta), \theta)$ belongs to a compact set $\mathcal{U}$ by continuity of π_t on a compact domain. Therefore $\mathcal{D}_t := \{(\tilde{x}_t(\theta), \tilde{u}_t(\theta), \theta) : \theta \in \Theta\}$ is compact. Since h_t is continuous, it attains a finite minimum on $\mathcal{D}_t$, so $I_t := \min_{(x,u,\theta)\in\mathcal{D}_t} h_t(x,u,\theta)$ is well-defined and finite. Note that I_t depends only on t (via $\mathcal{D}_t$), and not on the particular iterate θ at which we evaluate F_t. Define the shifted stage cost $\tilde{h}_t(x,u,\theta) := h_t(x,u,\theta) - I_t$. Then $\tilde{h}_t \geq 0$ on $\mathcal{D}_t$, and consequently for every $\theta \in \Theta$, $\tilde{F}_t(\theta) := \tilde{h}_t(\tilde{x}_t(\theta), \tilde{u}_t(\theta), \theta) = F_t(\theta) - I_t \geq 0$. Since I_t is independent of θ, we have $\nabla\tilde{F}_t(\theta) = \nabla F_t(\theta)$ and $\Pi_\Theta(\theta - \eta\nabla\tilde{F}_t(\theta)) = \Pi_\Theta(\theta - \eta\nabla F_t(\theta))$, so projected updates are unchanged. Moreover, in any regret expression of the form $\sum_t(F_t(\theta_t) - F_t(\theta))$, shifting by I_t leaves the value unchanged because the constants cancel termwise. Hence, w.l.o.g. we may work with the shifted costs but keep the notation h_t and F_t; under this convention $F_t(\theta) \geq 0$ for all θ, and in particular $F_{T-1}(\theta_T) \geq 0$ as used in (12c).

radius controls where the contractive perturbation property may be invoked, while the regularity radius specifies the compact domain on which the Lipschitz/smoothness constants are evaluated. The original GAPS presentation keeps these roles abstractly separate, which is why its arguments can be written in terms of conditions such as remaining in both admissible regions simultaneously. The standalone M-GAPS presentation keeps the same separation of roles, but makes the link between them explicit through the sufficient condition $R_x := \bar{C}\,\bar{R}_C + R_S$. With this choice, any trajectory that starts in $B(0, \bar{R}_C)$ is automatically guaranteed, by contraction and stability, to remain inside the compact regularity domain $B(0, R_x) \subseteq X$. Hence the change is only a bookkeeping change, not a conceptual one: it does not alter what the proof needs. This is why the text says that the change is minor and does not affect the proof from GAPS.]

Lemma 15 (Generalized Telescoping Identity for Matrix Products) *Let $n \in \mathbb{N}$, and let $A_1, \dots, A_n$ and $B_1, \dots, B_n$ be matrices of compatible dimensions so that all products below are well defined. Then*

$$\prod_{i=1}^{n} A_i - \prod_{i=1}^{n} B_i = \sum_{i=1}^{n} \left(\prod_{j=1}^{i-1} B_j \right) (A_i - B_i) \left(\prod_{j=i+1}^{n} A_j \right),$$

where the empty product is interpreted as the identity matrix of the appropriate dimension.

In particular, for $n = 2$, one obtains

$$A_1 A_2 - B_1 B_2 = (A_1 - B_1) A_2 + B_1 (A_2 - B_2).$$

Proof We prove the general identity first. For each $k = 0, 1, \dots, n$, define the mixed product

$$P_k := \left(\prod_{j=1}^{k} B_j \right) \left(\prod_{j=k+1}^{n} A_j \right),$$

with the convention that an empty product is the identity matrix. Then

$$P_0 = \prod_{i=1}^{n} A_i, \qquad P_n = \prod_{i=1}^{n} B_i.$$

Hence,

$$\prod_{i=1}^{n} A_i - \prod_{i=1}^{n} B_i = P_0 - P_n = \sum_{i=1}^{n} (P_{i-1} - P_i),$$

which is a telescoping sum.

Now, for each $i = 1, \dots, n$, we have

$$P_{i-1} = \left(\prod_{j=1}^{i-1} B_j \right) A_i \left(\prod_{j=i+1}^{n} A_j \right),$$

and

$$P_i = \left(\prod_{j=1}^{i-1} B_j \right) B_i \left(\prod_{j=i+1}^{n} A_j \right).$$

Subtracting gives

$$P_{i-1} - P_i = \left(\prod_{j=1}^{i-1} B_j\right)(A_i - B_i)\left(\prod_{j=i+1}^{n} A_j\right).$$

Substituting this back into the telescoping sum yields

$$\prod_{i=1}^{n} A_i - \prod_{i=1}^{n} B_i = \sum_{i=1}^{n}\left(\prod_{j=1}^{i-1} B_j\right)(A_i - B_i)\left(\prod_{j=i+1}^{n} A_j\right),$$

which proves the general formula.

For the special case $n = 2$, we simply add and subtract the mixed term $B_1 A_2$ to $A_1 A_2 - B_1 B_2$ to obtain the desired result. ■

Lemma 16 (Lipschitzness/Smoothness of the Multi-Step Dynamics) *Suppose Assumptions 11 and 12 hold. Given two time steps $t > \tau$, for any $x_\tau, x'_\tau \in B_n(0, \bar{R}_C)$ and $\theta_\tau, \theta'_\tau \in \Theta$, $\theta_{\tau+1:t-1} \in S_\varepsilon(\tau+1 : t-1)$, if $x'_{\tau+1} := g_{\tau+1|\tau}(x'_\tau, \theta'_\tau)$ is also in $B_n(0, \bar{R}_C)$, the multi-step dynamical function $g_{t|\tau}$ satisfies that [James: TODO: the use of Γ here was a mistake, we need to go back and clarify between C and $\bar{C}$] [Shengnan: Modified the use of Γ to $\bar{C}$ because the expressions are related to the time-varying contractive perturbation so the base contraction prefactor needs to be used instead of the summed decay gain.]*

$$\left\| \left.\frac{\partial g_{t|\tau}}{\partial x_\tau}\right|_{x_\tau, \theta_{\tau:t-1}} \right\| \le C_{L,g,x}\rho^{t-\tau}, \quad \left\| \left.\frac{\partial g_{t|\tau}}{\partial \theta_\tau}\right|_{x_\tau, \theta_{\tau:t-1}} \right\| \le C_{L,g,\theta}\rho^{t-\tau}, \forall \theta_{\tau+1:t-1} \in S_\varepsilon(\tau+1:t-1),\ \forall \theta_\tau \in \Theta,$$

$$\left\| \left.\frac{\partial g_{t|\tau}}{\partial x_\tau}\right|_{x_\tau, \theta_{\tau:t-1}} - \left.\frac{\partial g_{t|\tau}}{\partial x_\tau}\right|_{x'_\tau, \theta'_\tau, \theta_{\tau+1:t-1}} \right\| \le C_{\ell,g,(x,x)}\rho^{t-\tau}\|x_\tau - x'_\tau\| + C_{\ell,g,(x,\theta)}\rho^{t-\tau}\|\theta_\tau - \theta'_\tau\|,$$

$$\left\| \left.\frac{\partial g_{t|\tau}}{\partial \theta_\tau}\right|_{x_\tau, \theta_{\tau:t-1}} - \left.\frac{\partial g_{t|\tau}}{\partial \theta_\tau}\right|_{x'_\tau, \theta'_\tau, \theta_{\tau+1:t-1}} \right\| \le C_{\ell,g,(\theta,x)}\rho^{t-\tau}\|x_\tau - x'_\tau\| + C_{\ell,g,(\theta,\theta)}\rho^{t-\tau}\|\theta_\tau - \theta'_\tau\|,$$

where $C_{L,g,x} = \bar{C}$ as defined in Assumption 8 $C_{L,g,\theta} = \dfrac{\bar{C} L_{g,u} L_{\pi,\theta}}{\rho}$, [Shengnan: Given the changes, should we have $C_{L,g,x} := \frac{\bar{C}}{\rho}(L_{g,x} + L_{g,u}L_{\pi,x})$?] and

$$\begin{aligned}
C_{\ell,g,(x,x)} &= ((1 + L_{\pi,x})(\ell_{g,x} + \ell_{g,u}L_{\pi,x}) + L_{g,x}\ell_{\pi,x})\bar{C}^2(1 - \rho + \bar{C}\,(L_{g,x} + L_{g,u}L_{\pi,x}))\rho^{-1}(1-\rho)^{-1}, \\
C_{\ell,g,(x,\theta)} &= ((1 + L_{\pi,x})(\ell_{g,x} + \ell_{g,u}L_{\pi,x}) + L_{g,x}\ell_{\pi,x})\bar{C}^3 L_{g,u}L_{\pi,\theta}\rho^{-1}(1-\rho)^{-1} \\
&\quad + ((1 + L_{\pi,x})\ell_{g,u}L_{\pi,\theta} + L_{g,u}\ell_{\pi,\theta})\bar{C}\rho^{-1}(1-\rho)^{-1}, \\
C_{\ell,g,(\theta,x)} &= ((1 + L_{\pi,x})(\ell_{g,x} + \ell_{g,u}L_{\pi,x}) + L_{g,x}\ell_{\pi,x})(L_{g,x} + L_{g,u}L_{\pi,x}) \\
&\quad \bar{C}^3 L_{g,u}L_{\pi,\theta}\rho^{-2}(1-\rho)^{-1} + \bar{C}(L_{\pi,\theta}(\ell_{g,x} + \ell_{g,u}L_{\pi,x}) + L_{g,u}\ell_{\pi,x})\rho^{-1}, \\
C_{\ell,g,(\theta,\theta)} &= ((1 + L_{\pi,x})(\ell_{g,x} + \ell_{g,u}\cdot L_{\pi,x}) + L_{g,x}\cdot\ell_{\pi,x})L_{g,u}^2 L_{\pi,\theta}^2\bar{C}^3\rho^{-2}(1-\rho)^{-1} \\
&\quad + \left(L_{g,u}\ell_{\pi,\theta} + \ell_{g,u}L_{\pi,\theta}^2\right)\bar{C}\rho^{-1}.
\end{aligned}$$

[James: Possible to factor out any common constant from the list of $C_{\ell,g,*}$ above?]

$$A := (1 + L_{\pi,x})(\ell_{g,x} + \ell_{g,u} L_{\pi,x}) + L_{g,x}\, \ell_{\pi,x}.$$

$$\begin{aligned}
C_{\ell,g,(x,x)} &= A \cdot \bar{C}\, \rho^{-1} \big[\bar{C}\, (1 - \rho + \bar{C}\, (L_{g,x} + L_{g,u} L_{\pi,x}))(1-\rho)^{-1}\big], \\
C_{\ell,g,(x,\theta)} &= A \cdot \bar{C}\, \rho^{-1} \big[\bar{C}^2 L_{g,u} L_{\pi,\theta}\, (1-\rho)^{-1} + ((1 + L_{\pi,x})\, \ell_{g,u}\, L_{\pi,\theta} + L_{g,u}\, \ell_{\pi,\theta})\, (1-\rho)^{-1}\big], \\
C_{\ell,g,(\theta,x)} &= A \cdot \bar{C}\, \rho^{-1} \big[(L_{g,x} + L_{g,u} L_{\pi,x})\, \bar{C}^2 L_{g,u} L_{\pi,\theta}\, \rho^{-1} (1-\rho)^{-1} + (L_{\pi,\theta}\, (\ell_{g,x} + \ell_{g,u} L_{\pi,x}) + L_{g,u}\, \ell_{\pi,x})\big], \\
C_{\ell,g,(\theta,\theta)} &= A \cdot \bar{C}\, \rho^{-1} \big[L_{g,u}^2 L_{\pi,\theta}^2\, \bar{C}^2\, \rho^{-1} (1-\rho)^{-1} + \big(L_{g,u}\, \ell_{\pi,\theta} + \ell_{g,u}\, L_{\pi,\theta}^2\big)\big].
\end{aligned}$$

Intuitively, Lemma 16 shows that the dependence of the state x_t on the previous state x_τ and θ_τ decays exponentially with respect to their time distance $t - \tau$. Specifically, recall that the multi-step dynamics $g_{t|\tau}$ writes x_t as a function of x_τ and $\theta_{\tau:t-1}$. When other variables are fixed, the Lipschitzness and smoothness constants with respect to x_τ and θ_τ are both $O(\rho^{t-\tau})$. While the contractive Lipschitzness on x_τ is automatically guaranteed by ϵ-time-varying contractive perturbation (Definition 3), we use this property and the chain rule decomposition to show the Lipschitzness on θ_τ and the smoothness.

[James: remark: notice that θ_τ and θ'_τ are allowed to be arbitrarily far away, only $\theta_{\tau+1:t}$ needs to be slowly time varying.]

[Shengnan: The proof for the third inequality has been enhanced with more details. Could you please check the overall proof for Lemma 16 below?]

Proof For the first inequality, when $t = \tau + 1$, note that

$$\frac{\partial g_{\tau+1|\tau}}{\partial x_\tau}\Big|_{x_\tau,\theta_\tau} = \frac{\partial g_\tau}{\partial x_\tau}\Big|_{x_\tau,u_\tau} + \frac{\partial g_\tau}{\partial u_\tau}\Big|_{x_\tau,u_\tau} \cdot \frac{\partial \pi_\tau}{\partial x_\tau}\Big|_{x_\tau,\theta_\tau}, \text{where } u_\tau = \pi_\tau(x_\tau, \theta_\tau).$$

Taking norms on both sides gives

$$\left\| \frac{\partial g_{\tau+1|\tau}}{\partial x_\tau}\Big|_{x_\tau,\theta_\tau} \right\| \le \left\| \frac{\partial g_\tau}{\partial x_\tau}\Big|_{x_\tau,u_\tau} \right\| + \left\| \frac{\partial g_\tau}{\partial u_\tau}\Big|_{x_\tau,u_\tau} \right\| \cdot \left\| \frac{\partial \pi_\tau}{\partial x_\tau}\Big|_{x_\tau,\theta_\tau} \right\| \le L_{g,x} + L_{g,u} L_{\pi,x}.$$

Recall that

$$C_{L,g,x} = \frac{\bar{C}}{\rho}\, (L_{g,x} + L_{g,u} L_{\pi,x}).$$

Then, since $\bar{C} \ge 1$,

$$L_{g,x} + L_{g,u} L_{\pi,x} \le \frac{\bar{C}}{\rho} (L_{g,x} + L_{g,u} L_{\pi,x}) \cdot \rho = C_{L,g,x}\, \rho^{(\tau+1)-\tau}.$$

Hence the claim holds for $t = \tau + 1$.
When $t > \tau + 1$, by the chain rule,

$$\begin{aligned}
\left\| \frac{\partial g_{t|\tau}}{\partial x_\tau}\Big|_{x_\tau,\theta_{\tau:t-1}} \right\| &= \left\| \frac{\partial g_{t|\tau+1}}{\partial x_{\tau+1}}\Big|_{x_{\tau+1},\,\theta_{\tau+1:t-1}} \cdot \frac{\partial g_{\tau+1|\tau}}{\partial x_\tau}\Big|_{x_\tau,\,\theta_\tau} \right\| \\
&\le \left\| \frac{\partial g_{t|\tau+1}}{\partial x_{\tau+1}}\Big|_{x_{\tau+1},\theta_{\tau+1:t-1}} \right\| \cdot \left\| \frac{\partial g_{\tau+1|\tau}}{\partial x_\tau}\Big|_{x_\tau,\theta_\tau} \right\| \\
&\le \left(\bar{C}\, \rho^{t-(\tau+1)}\right) \cdot \left(L_{g,x} + L_{g,u} L_{\pi,x}\right) = \frac{\bar{C}}{\rho}\, (L_{g,x} + L_{g,u} L_{\pi,x})\, \rho^{t-\tau} = C_{L,g,x}\, \rho^{t-\tau},
\end{aligned}$$

where $x_{\tau+1} = g_{\tau+1|\tau}(x_\tau, \theta_\tau)$, and the bound $\left\|\partial g_{t|\tau+1}/\partial x_{\tau+1}\right\| \le \bar{C}\,\rho^{t-(\tau+1)}$ follows from Definition 3 applied to the tail, which only requires $\theta_{\tau+1:t-1} \in S_\varepsilon(\tau+1:t-1)$ and the intermediate states to lie in $B_n(0, \bar{R}_C)$.

For the second inequality, when $t = \tau+1$, note that

$$\frac{\partial g_{\tau+1|\tau}}{\partial \theta_\tau}\Big|_{x_\tau,\theta_\tau} = \frac{\partial g_\tau}{\partial u_\tau}\Big|_{x_\tau,u_\tau} \cdot \frac{\partial \pi_\tau}{\partial \theta_\tau}\Big|_{x_\tau,\theta_\tau}, \text{where } u_\tau = \pi_\tau(x_\tau, \theta_\tau).$$

Taking norms on both sides gives

$$\left\|\frac{\partial g_{\tau+1|\tau}}{\partial \theta_\tau}\Big|_{x_\tau,\theta_\tau}\right\| = \left\|\frac{\partial g_\tau}{\partial u_\tau}\Big|_{x_\tau,\,u_\tau} \cdot \frac{\partial \pi_\tau}{\partial \theta_\tau}\Big|_{x_\tau,\,\theta_\tau}\right\| \le \left\|\frac{\partial g_\tau}{\partial u_\tau}\Big|_{x_\tau,u_\tau}\right\| \cdot \left\|\frac{\partial \pi_\tau}{\partial \theta_\tau}\Big|_{x_\tau,\theta_\tau}\right\| \le L_{g,u}L_{\pi,\theta}.$$

Recall that

$$C_{L,g,\theta} = \frac{\bar{C}}{\rho}\,L_{g,u}L_{\pi,\theta}.$$

Then, since $\bar{C} \ge 1$,

$$L_{g,u}L_{\pi,\theta} \le \frac{\bar{C}}{\rho}L_{g,u}L_{\pi,\theta} \cdot \rho = C_{L,g,\theta}\,\rho^{(\tau+1)-\tau}.$$

Hence the claim holds for $t = \tau+1$.
When $t > \tau+1$, by the chain rule,

$$\begin{aligned}
\left\|\frac{\partial g_{t|\tau}}{\partial \theta_\tau}\Big|_{x_\tau,\theta_{\tau:t-1}}\right\| &= \left\|\frac{\partial g_{t|\tau+1}}{\partial x_{\tau+1}}\Big|_{x_{\tau+1},\,\theta_{\tau+1:t-1}} \cdot \frac{\partial g_{\tau+1|\tau}}{\partial \theta_\tau}\Big|_{x_\tau,\,\theta_\tau}\right\| \\
&\le \left\|\frac{\partial g_{t|\tau+1}}{\partial x_{\tau+1}}\Big|_{x_{\tau+1},\theta_{\tau+1:t-1}}\right\| \cdot \left\|\frac{\partial g_{\tau+1|\tau}}{\partial \theta_\tau}\Big|_{x_\tau,\theta_\tau}\right\| \\
&\le \left(\bar{C}\,\rho^{t-(\tau+1)}\right) \cdot \left(L_{g,u}L_{\pi,\theta}\right) = \frac{\bar{C}}{\rho}\,L_{g,u}L_{\pi,\theta}\,\rho^{t-\tau} = C_{L,g,\theta}\,\rho^{t-\tau},
\end{aligned}$$

where $x_{\tau+1} = g_{\tau+1|\tau}(x_\tau, \theta_\tau)$, and the bound $\left\|\partial g_{t|\tau+1}/\partial x_{\tau+1}\right\| \le \bar{C}\,\rho^{t-(\tau+1)}$ follows from Definition 3 applied to the tail which only requires $\theta_{\tau+1:t-1} \in S_\varepsilon(\tau+1:t-1)$ and the intermediate states to lie in $B_n(0, R_C)$.
For the third inequality, note that we have the chain rule decompositions

$$\begin{aligned}
\frac{\partial g_{t|\tau}}{\partial x_\tau}\Big|_{x_\tau,\theta_\tau,\theta_{\tau+1:t-1}} &= \frac{\partial g_{t|t-1}}{\partial x_{t-1}}\Big|_{x_{t-1},\theta_{t-1}} \cdot \frac{\partial g_{t-1|t-2}}{\partial x_{t-2}}\Big|_{x_{t-2},\theta_{t-2}} \cdot \ldots \cdot \frac{\partial g_{\tau+1|\tau}}{\partial x_\tau}\Big|_{x_\tau,\theta_\tau}, \\
\frac{\partial g_{t|\tau}}{\partial x_\tau}\Big|_{x'_\tau,\theta'_\tau,\theta_{\tau+1:t-1}} &= \frac{\partial g_{t|t-1}}{\partial x_{t-1}}\Big|_{x'_{t-1},\theta'_{t-1}} \cdot \frac{\partial g_{t-1|t-2}}{\partial x_{t-2}}\Big|_{x'_{t-2},\theta'_{t-2}} \cdot \ldots \cdot \frac{\partial g_{\tau+1|\tau}}{\partial x_\tau}\Big|_{x'_\tau,\theta'_\tau}.
\end{aligned} \tag{13}$$

where we use the notation

$$x_{\tau'} \coloneqq g_{\tau'|\tau}(x_\tau, \theta_{\tau:\tau'-1}) \quad \text{and} \quad x'_{\tau'} \coloneqq g_{\tau'|\tau}(x'_\tau, \theta'_\tau, \theta_{\tau+1:\tau'-1}) \quad \text{for } \tau' \in\ [\tau+1:t-1].$$

Note that for any $i \in [1:t-\tau]$ and any $\theta'_{t-i} \in \Theta$, letting

$$u_{t-i} \coloneqq \pi_{t-i}(x_{t-i}, \theta_{t-i}) \qquad \text{and} \qquad u'_{t-i} \coloneqq \pi_{t-i}(x'_{t-i}, \theta'_{t-i}),$$

we can write $g_{t-i+1|t-i}(x_{t-i}, \theta_{t-i}) = g_{t-i}(x_{t-i}, u_{t-i})$ and $g_{t-i+1|t-i}(x'_{t-i}, \theta'_{t-i}) = g_{t-i}(x'_{t-i}, u'_{t-i})$. Hence, by the chain rule,

$$
\begin{aligned}
&\frac{\partial g_{t-i+1|t-i}}{\partial x_{t-i}}\Big|_{x_{t-i},\theta_{t-i}} - \frac{\partial g_{t-i+1|t-i}}{\partial x_{t-i}}\Big|_{x'_{t-i},\theta'_{t-i}} \\
=&\left(\frac{\partial g_{t-i}}{\partial x_{t-i}}\Big|_{x_{t-i},u_{t-i}} + \frac{\partial g_{t-i}}{\partial u_{t-i}}\Big|_{x_{t-i},u_{t-i}} \frac{\partial \pi_{t-i}}{\partial x_{t-i}}\Big|_{x_{t-i},\theta_{t-i}}\right) - \left(\frac{\partial g_{t-i}}{\partial x_{t-i}}\Big|_{x'_{t-i},u'_{t-i}} + \frac{\partial g_{t-i}}{\partial u_{t-i}}\Big|_{x'_{t-i},u'_{t-i}} \frac{\partial \pi_{t-i}}{\partial x_{t-i}}\Big|_{x'_{t-i},\theta'_{t-i}}\right) && (14)\\
=&\left(\frac{\partial g_{t-i}}{\partial x_{t-i}}\Big|_{x_{t-i},u_{t-i}} - \frac{\partial g_{t-i}}{\partial x_{t-i}}\Big|_{x'_{t-i},u'_{t-i}}\right) + \left(\frac{\partial g_{t-i}}{\partial u_{t-i}}\Big|_{x_{t-i},u_{t-i}} \frac{\partial \pi_{t-i}}{\partial x_{t-i}}\Big|_{x_{t-i},\theta_{t-i}} - \frac{\partial g_{t-i}}{\partial u_{t-i}}\Big|_{x'_{t-i},u'_{t-i}} \frac{\partial \pi_{t-i}}{\partial x_{t-i}}\Big|_{x'_{t-i},\theta'_{t-i}}\right) && (15)\\
=&\frac{\partial g_{t-i}}{\partial x_{t-i}}\Big|_{x_{t-i},u_{t-i}} - \frac{\partial g_{t-i}}{\partial x_{t-i}}\Big|_{x'_{t-i},u'_{t-i}} + \left(\frac{\partial g_{t-i}}{\partial u_{t-i}}\Big|_{x_{t-i},u_{t-i}} - \frac{\partial g_{t-i}}{\partial u_{t-i}}\Big|_{x'_{t-i},u'_{t-i}}\right)\frac{\partial \pi_{t-i}}{\partial x_{t-i}}\Big|_{x_{t-i},\theta_{t-i}} \\
&+ \frac{\partial g_{t-i}}{\partial u_{t-i}}\Big|_{x'_{t-i},u'_{t-i}} \left(\frac{\partial \pi_{t-i}}{\partial x_{t-i}}\Big|_{x_{t-i},\theta_{t-i}} - \frac{\partial \pi_{t-i}}{\partial x_{t-i}}\Big|_{x'_{t-i},\theta'_{t-i}}\right). && (16)
\end{aligned}
$$

where in (14) we used the recursive representation $g_{t-i+1|t-i}(x, \theta) = g_{t-i}(x, \pi_{t-i}(x, \theta))$ and applied the chain rule, and we used the shorthand u_{t-i} and u'_{t-i} defined above to keep notation compact; in (15) we grouped terms after subtracting the two chain-rule expansions; and in (16) we applied Lemma 15. Taking norms on both sides and applying the triangle inequality gives

$$
\begin{aligned}
&\left\|\frac{\partial g_{t-i+1|t-i}}{\partial x_{t-i}}\Big|_{x_{t-i},\theta_{t-i}} - \frac{\partial g_{t-i+1|t-i}}{\partial x_{t-i}}\Big|_{x'_{t-i},\theta'_{t-i}}\right\| \\
\le&\ell_{g,x}\left\|x_{t-i} - x'_{t-i}\right\| + \ell_{g,u}\left\|\pi_{t-i}(x_{t-i}, \theta_{t-i}) - \pi_{t-i}(x'_{t-i}, \theta'_{t-i})\right\| \\
&+ L_{\pi,x}\big(\ell_{g,x}\left\|x_{t-i} - x'_{t-i}\right\| + \ell_{g,u}\left\|\pi_{t-i}(x_{t-i}, \theta_{t-i}) - \pi_{t-i}(x'_{t-i}, \theta'_{t-i})\right\|\big) \\
&+ L_{g,u}\big(\ell_{\pi,x}\left\|x_{t-i} - x'_{t-i}\right\| + \ell_{\pi,\theta}\left\|\theta_{t-i} - \theta'_{t-i}\right\|\big) && (17)\\
\le&\underbrace{\big((1 + L_{\pi,x})(\ell_{g,x} + \ell_{g,u}L_{\pi,x}) + L_{g,u}\,\ell_{\pi,x}\big)}_{\text{coeff. on }\left\|x_{t-i}-x'_{t-i}\right\|}\left\|x_{t-i} - x'_{t-i}\right\| \\
&+ \underbrace{\big((1 + L_{\pi,x})\,\ell_{g,u}\,L_{\pi,\theta} + L_{g,u}\,\ell_{\pi,\theta}\big)}_{\text{coeff. on}\left\|\theta_{t-i}-\theta'_{t-i}\right\|}\left\|\theta_{t-i} - \theta'_{t-i}\right\|, && (18)
\end{aligned}
$$

where we use Assumption 11 and definition of u_{t-i}, u'_{t-i} in (17), and Assumption 11 in (18). Therefore, by (13) and (18), we obtain

$$
\begin{aligned}
&\left\|\frac{\partial g_{t|\tau}}{\partial x_\tau}\Big|_{x_\tau,\theta_{\tau:t-1}} - \frac{\partial g_{t|\tau}}{\partial x_\tau}\Big|_{x'_\tau,\theta'_\tau,\theta_{\tau+1:t-1}}\right\| \\
=&\left\|\prod_{i=1}^{t-\tau}\frac{\partial g_{t-i+1|t-i}}{\partial x_{t-i}}\Big|_{x_{t-i},\theta_{t-i}} - \prod_{i=1}^{t-\tau}\frac{\partial g_{t-i+1|t-i}}{\partial x_{t-i}}\Big|_{x'_{t-i},\theta'_{t-i}}\right\| && (19)\\
\le&\sum_{i=1}^{t-\tau-1}\left(\left\|\prod_{\tau'=1}^{i-1}\frac{\partial g_{t-\tau'+1|t-\tau'}}{\partial x_{t-\tau'}}\Big|_{x'_{t-\tau'},\theta_{t-\tau'}}\right\| \cdot \left\|\frac{\partial g_{t-i+1|t-i}}{\partial x_{t-i}}\Big|_{x_{t-i},\theta_{t-i}} - \frac{\partial g_{t-i+1|t-i}}{\partial x_{t-i}}\Big|_{x'_{t-i},\theta_{t-i}}\right\|\right.
\end{aligned}
$$

$$\cdot \left\| \prod_{\tau'=i+1}^{t-\tau} \frac{\partial g_{t-\tau'+1|t-\tau'}}{\partial x_{t-\tau'}}\Big|_{x_{t-\tau'},\theta_{t-\tau'}} \right\| \Bigg)$$

$$+ \left\| \prod_{\tau'=1}^{t-\tau-1} \frac{\partial g_{t-\tau'+1|t-\tau'}}{\partial x_{t-\tau'}}\Big|_{x'_{t-\tau'},\theta_{t-\tau'}} \right\| \cdot \left\| \frac{\partial g_{\tau+1|\tau}}{\partial x_\tau}\Big|_{x_\tau,\theta_\tau} - \frac{\partial g_{\tau+1|\tau}}{\partial x_\tau}\Big|_{x'_\tau,\theta'_\tau} \right\| \tag{20}$$

$$\leq \sum_{i=1}^{t-\tau-1} \bar{C}\,\rho^{i-1} \cdot \left\| \frac{\partial g_{t-i+1|t-i}}{\partial x_{t-i}}\Big|_{x_{t-i},\theta_{t-i}} - \frac{\partial g_{t-i+1|t-i}}{\partial x_{t-i}}\Big|_{x'_{t-i},\theta_{t-i}} \right\| \cdot \bar{C}\,\rho^{t-\tau-i}$$

$$+ \bar{C}\,\rho^{t-\tau-1} \cdot \left\| \frac{\partial g_{\tau+1|\tau}}{\partial x_\tau}\Big|_{x_\tau,\theta_\tau} - \frac{\partial g_{\tau+1|\tau}}{\partial x_\tau}\Big|_{x'_\tau,\theta'_\tau} \right\| \tag{21}$$

where in (19) we rewrote the multi-step Jacobians using the chain-rule product decomposition in (13); in (20) we used Lemma 15 together with the triangle inequality and submultiplicativity of the operator norm; in (21) we applied the ϵ-time-varying contractive perturbation definition, and Assumption 11 to bound each prefix/suffix product by $\Gamma\rho^{(\cdot)}$[5]. Then, we can see that

$$\left\| \frac{\partial g_{t|\tau}}{\partial x_\tau}\Big|_{x_\tau,\theta_{\tau:t-1}} - \frac{\partial g_{t|\tau}}{\partial x_\tau}\Big|_{x'_\tau,\theta'_\tau,\theta_{\tau+1:t-1}} \right\|$$

$$\leq \sum_{i=1}^{t-\tau} \bar{C}\,\rho^{i-1} \cdot \Big((1+L_{\pi,x})(\ell_{g,x} + \ell_{g,u}\,L_{\pi,x}) + L_{g,u}\,\ell_{\pi,x} \Big) \cdot \|x_{t-i} - x'_{t-i}\| \cdot \bar{C}\,\rho^{t-\tau-i}$$

$$+ \bar{C}\,\rho^{t-\tau-1} \cdot \Big((1+L_{\pi,x})\,\ell_{g,u}\,L_{\pi,\theta} + L_{g,u}\,\ell_{\pi,\theta} \Big) \|\theta_\tau - \theta'_\tau\| \tag{22}$$

$$= \Big((1+L_{\pi,x})(\ell_{g,x} + \ell_{g,u}\,L_{\pi,x}) + L_{g,u}\,\ell_{\pi,x} \Big) \bar{C}^2 \cdot \rho^{t-\tau-1} \sum_{i=1}^{t-\tau} \|x_{t-i} - x'_{t-i}\|$$

$$+ \Gamma\,\rho^{t-\tau-1} \cdot \Big((1+L_{\pi,x})\,\ell_{g,u}\,L_{\pi,\theta} + L_{g,u}\,\ell_{\pi,\theta} \Big) \|\theta_\tau - \theta'_\tau\|$$

$$\leq C_{l,g,(x,x)} \cdot \rho^{t-\tau}\,\|x_\tau - x'_\tau\| \;+\; C_{l,g,(x,\theta)} \cdot \rho^{t-\tau}\,\|\theta_\tau - \theta'_\tau\| \tag{23}$$

where in (22) we applied (18) to bound the one-step Jacobian difference term, and we used Assumption 11 to bound the θ_τ-difference term; and in (23) we use the first two inequalities of Lemma 16 to bound $\|x_{t-i} - x'_{t-i}\| \leq \Gamma\,\rho^{t-i-\tau}\|x_\tau - x'_\tau\| + \frac{C\,L_{g,u}L_{\pi,\theta}}{\rho}\,\rho^{t-i-\tau}\|\theta_\tau - \theta'_\tau\|$[6].
Recall that

$$R_C \geq \bar{C}\big(R_S + \bar{C}\|x_0\|\big) + R_S.$$

We can apply the contractive perturbation because $x'_\tau \in B_n(0, R_S + C\|x_0\|)$, thus the trajectory $x'_{\tau:t}$ always stays within $B_n(0, \bar{R}_C)$, where the ϵ-time-varying contractive perturbation property holds.

5. By (13), each prefix/suffix product equals the Jacobian of a shorter multi-step map $g_{t|s}$ with time distance $t-s$ equal to the number of one-step state-transition Jacobians in the chain-rule product. The ϵ-time-varying contractive perturbation property the prefix $\prod_{\tau'=1}^{i-1} \frac{\partial g_{t-\tau'+1|t-\tau'}}{\partial x_{t-\tau'}}$ is exactly the Jacobian $\partial g_{t|t-i+1}/\partial x_{t-i+1}$ of an $(i-1)$-step map, so its norm is bounded by $\Gamma\rho^{i-1}$ and similarly the suffix has $t-\tau-i$ steps, giving $\Gamma\rho^{t-\tau-i}$.
6. The substitution is performed term-by-term inside $\sum_{i=1}^{t-\tau} \|x_{t-i} - x'_{t-i}\|$: plugging $\|x_{t-i} - x'_{t-i}\| \leq \Gamma\rho^{t-i-\tau}\|x_\tau - x'_\tau\| + \cdots$ yields sums of the form $\sum_{i=1}^{t-\tau} \rho^{t-i-\tau}$, which equals $\sum_{k=0}^{t-\tau-1} \rho^k$ after the change of variables $k = t-i-\tau$. We then use the geometric-series bound $\sum_{k=0}^{t-\tau-1} \rho^k \leq (1-\rho)^{-1}$ and absorb the resulting prefactors into $C_{l,g,(x,x)}$ and $C_{l,g,(x,\theta)}$.

In (23), we note that

$$\begin{aligned}\|x_{\tau+1} - x'_{\tau+1}\| &\le L_{g,x}\|x_\tau - x'_\tau\| + L_{g,u}\|u_\tau - u'_\tau\| \\ &\le L_{g,x}\|x_\tau - x'_\tau\| + L_{g,u}\big(L_{\pi,x}\|x_\tau - x'_\tau\| + L_{\pi,\theta}\|\theta_\tau - \theta'_\tau\|\big) \\ &\le (L_{g,x} + L_{g,u}L_{\pi,x})\|x_\tau - x'_\tau\| + L_{g,u}L_{\pi,\theta}\|\theta_\tau - \theta'_\tau\|. \end{aligned} \tag{24}$$

Therefore, using the ϵ-time-varying contractive perturbation property, we obtain that

$$\|x_{\tau'} - x'_{\tau'}\| \le \bar{C}\,\rho^{\tau'-\tau-1}\Big((L_{g,x} + L_{g,u}L_{\pi,x})\|x_\tau - x'_\tau\| + L_{g,u}L_{\pi,\theta}\|\theta_\tau - \theta'_\tau\|\Big)$$

holds for all $\tau' \ge \tau + 1$.

For the last inequality, when $t = \tau + 1$, we see that

$$\begin{aligned}&\left\|\frac{\partial g_{\tau+1|\tau}}{\partial \theta_\tau}\Big|_{x_\tau,\theta_\tau} - \frac{\partial g_{\tau+1|\tau}}{\partial \theta_\tau}\Big|_{x'_\tau,\theta'_\tau}\right\| \\ =&\left\|\frac{\partial g_\tau}{\partial u_\tau}\Big|_{x_\tau,u_\tau}\cdot\frac{\partial \pi_\tau}{\partial \theta_\tau}\Big|_{x_\tau,\theta_\tau} - \frac{\partial g_\tau}{\partial u_\tau}\Big|_{x'_\tau,u'_\tau}\cdot\frac{\partial \pi_\tau}{\partial \theta_\tau}\Big|_{x'_\tau,\theta'_\tau}\right\| \\ \le&\left\|\left(\frac{\partial g_\tau}{\partial u_\tau}\Big|_{x_\tau,u_\tau} - \frac{\partial g_\tau}{\partial u_\tau}\Big|_{x'_\tau,u'_\tau}\right)\cdot\frac{\partial \pi_\tau}{\partial \theta_\tau}\Big|_{x_\tau,\theta_\tau}\right\| + \left\|\frac{\partial g_\tau}{\partial u_\tau}\Big|_{x'_\tau,u'_\tau}\cdot\left(\frac{\partial \pi_\tau}{\partial \theta_\tau}\Big|_{x_\tau,\theta_\tau} - \frac{\partial \pi_\tau}{\partial \theta_\tau}\Big|_{x'_\tau,\theta'_\tau}\right)\right\| && (25)\\ \le& L_{\pi,\theta}\big(\ell_{g,x}\|x_\tau - x'_\tau\| + \ell_{g,u}\|u_\tau - u'_\tau\|\big) + L_{g,u}\big(\ell_{\pi,x}\|x_\tau - x'_\tau\| + \ell_{\pi,\theta}\|\theta_\tau - \theta'_\tau\|\big) && (26)\\ \le&(L_{\pi,\theta}(\ell_{g,x} + \ell_{g,u}L_{\pi,x}) + L_{g,u}\ell_{\pi,x})\|x_\tau - x'_\tau\| + \big(L^2_{\pi,\theta}\ell_{g,u} + L_{g,u}\ell_{\pi,\theta}\big)\|\theta_\tau - \theta'_\tau\|, && (27)\end{aligned}$$

where we use the notations $u_\tau = \pi_\tau(x_\tau, \theta_\tau)$, $u'_\tau = \pi_\tau(x_\tau, \theta'_\tau)$. We use the triangle inequality in (25); we use Assumption 11 in both (26) and (27).
Meanwhile, when $t > \tau + 1$, we also see that

$$\begin{aligned}&\left\|\frac{\partial g_{t|\tau}}{\partial \theta_\tau}\Big|_{x_\tau,\theta_\tau,\theta_{\tau+1:t-1}} - \frac{\partial g_{t|\tau}}{\partial \theta_\tau}\Big|_{x'_\tau,\theta'_\tau,\theta_{\tau+1:t-1}}\right\| \\ \le&\left\|\frac{\partial g_{t|\tau+1}}{\partial x_{\tau+1}}\Big|_{x_{\tau+1},\theta_{\tau+1:t-1}}\cdot\frac{\partial g_{\tau+1|\tau}}{\partial \theta_\tau}\Big|_{x_\tau,\theta_\tau} - \frac{\partial g_{t|\tau+1}}{\partial x_{\tau+1}}\Big|_{x'_{\tau+1},\theta_{\tau+1:t-1}}\cdot\frac{\partial g_{\tau+1|\tau}}{\partial \theta_\tau}\Big|_{x'_\tau,\theta'_\tau}\right\| && (28)\\ \le&\left\|\left(\frac{\partial g_{t|\tau+1}}{\partial x_{\tau+1}}\Big|_{x_{\tau+1},\theta_{\tau+1:t-1}} - \frac{\partial g_{t|\tau+1}}{\partial x_{\tau+1}}\Big|_{x'_{\tau+1},\theta_{\tau+1:t-1}}\right)\frac{\partial g_{\tau+1|\tau}}{\partial \theta_\tau}\Big|_{x_\tau,\theta_\tau}\right\| \\ &\quad+\left\|\frac{\partial g_{t|\tau+1}}{\partial x_{\tau+1}}\Big|_{x'_{\tau+1},\theta_{\tau+1:t-1}}\left(\frac{\partial g_{\tau+1|\tau}}{\partial \theta_\tau}\Big|_{x_\tau,\theta_\tau} - \frac{\partial g_{\tau+1|\tau}}{\partial \theta_\tau}\Big|_{x'_\tau,\theta'_\tau}\right)\right\| && (29)\\ \le&\left\|\frac{\partial g_{t|\tau+1}}{\partial x_{\tau+1}}\Big|_{x_{\tau+1},\theta_{\tau+1:t-1}} - \frac{\partial g_{t|\tau+1}}{\partial x_{\tau+1}}\Big|_{x'_{\tau+1},\theta_{\tau+1:t-1}}\right\|\cdot\left\|\frac{\partial g_{\tau+1|\tau}}{\partial \theta_\tau}\Big|_{x_\tau,\theta_\tau}\right\| \\ &\quad+\left\|\frac{\partial g_{t|\tau+1}}{\partial x_{\tau+1}}\Big|_{x'_{\tau+1},\theta_{\tau+1:t-1}}\right\|\cdot\left\|\frac{\partial g_{\tau+1|\tau}}{\partial \theta_\tau}\Big|_{x_\tau,\theta_\tau} - \frac{\partial g_{\tau+1|\tau}}{\partial \theta_\tau}\Big|_{x'_\tau,\theta'_\tau}\right\| \\ \le&\frac{((1 + L_{\pi,x})(\ell_{g,x} + \ell_{g,u}\,L_{\pi,x}) + L_{g,x}\,\ell_{\pi,x})C_0^3}{\rho(1-\rho)}\cdot\rho^{t-\tau-1}\cdot\big\|x_{\tau+1} - x'_{\tau+1}\big\|\cdot L_{g,u}L_{\pi,\theta} \\ &\quad+ C_0\cdot\rho^{t-\tau-1}\cdot\big(L_{\pi,\theta}(\ell_{g,x} + \ell_{g,u}L_{\pi,x}) + L_{g,u}\ell_{\pi,x}\big)\big\|x_\tau - x'_\tau\big\| \\ &\quad+ C_0\cdot\rho^{t-\tau-1}\cdot\big(L^2_{\pi,\theta}\ell_{g,u} + L_{g,u}\ell_{\pi,\theta}\big)\big\|\theta_\tau - \theta'_\tau\big\| && (30)\end{aligned}$$

$$\leq C_{\ell,g,(\theta,x)}\,\rho^{\,t-\tau}\,\big\|x_\tau - x'_\tau\big\| \;+\; C_{\ell,g,(\theta,\theta)}\,\rho^{\,t-\tau}\,\big\|\theta_\tau - \theta'_\tau\big\|, \tag{31}$$

where we use the notations $x_{\tau+1} = g_{\tau+1|\tau}(x_\tau, \theta_\tau)$, $x'_{\tau+1} = g_{\tau+1|\tau}(x_\tau, \theta'_\tau)$. We use the chain rule decomposition in (28); we use the triangle inequality in (29); we use the first and the third inequality of Lemma 16 as well as (25) - (27) in (30); we use the first two inequalities of Lemma 16 in (31). ■

Corollary 17 is implied by Lemma 16 and corresponds to Corollary D.4 in Lin et al. (2023).

Corollary 17 (Lipschitzness/Smoothness of the Multi-Step Costs) *Under the same assumptions as Lemma 16, the multi-step cost function $h_{t|\tau}$ satisfies that*

$$\begin{aligned}
\left\| \left.\frac{\partial h_{t|\tau}}{\partial x_\tau}\right|_{x_\tau,\theta_{\tau:t}} \right\| &\leq C_{L,h,x}\rho^{t-\tau}, \\
\left\| \left.\frac{\partial h_{t|\tau}}{\partial \theta_\tau}\right|_{x_\tau,\theta_{\tau:t}} \right\| &\leq C_{L,h,\theta}\rho^{t-\tau}, \\
\left\| \left.\frac{\partial h_{t|\tau}}{\partial x_\tau}\right|_{x_\tau,\theta_\tau,\theta_{\tau+1:t}} - \left.\frac{\partial h_{t|\tau}}{\partial x_\tau}\right|_{x'_\tau,\theta'_\tau,\theta_{\tau+1:t}} \right\| &\leq C_{\ell,h,(x,x)}\rho^{t-\tau}\big\|x_\tau - x'_\tau\big\| + C_{\ell,h,(x,\theta)}\rho^{t-\tau}\big\|\theta_\tau - \theta'_\tau\big\|, \\
\left\| \left.\frac{\partial h_{t|\tau}}{\partial \theta_\tau}\right|_{x_\tau,\theta_\tau,\theta_{\tau+1:t}} - \left.\frac{\partial h_{t|\tau}}{\partial \theta_\tau}\right|_{x'_\tau,\theta'_\tau,\theta_{\tau+1:t}} \right\| &\leq C_{\ell,h,(\theta,x)}\rho^{t-\tau}\big\|x_\tau - x'_\tau\big\| + C_{\ell,h,(\theta,\theta)}\rho^{t-\tau}\big\|\theta_\tau - \theta'_\tau\big\|,
\end{aligned}$$

[James: TODO: check that we are accounting for the θ_t input to h_t properly. This proof was copied from the original GAPS paper where h_t depends only on x_t, u_t.] where $C_{L,h,x} = L_h\bar{C}(1 + L_{\pi,x})$, $C_{L,h,\theta} = L_h \max\{C_{L,g,\theta}(1+L_{\pi,x}), L_{\pi,\theta}\}$, and [Shengnan: $C_{L,h,\theta} = L_h \max\{C_{L,g,\theta}(1+L_{\pi,x}), 1+L_{\pi,\theta}\}$]

$$\begin{aligned}
C_{\ell,h,(x,x)} &= L_h(1 + L_{\pi,x})\,C_{\ell,g,(x,x)} + ((\ell_{h,x} + \ell_{h,u}L_{\pi,x})(1 + L_{\pi,x}) + L_h\,\ell_{\pi,x})\,C_{L,g,x} \\
&\quad \cdot \max\big\{\bar{C}\big(L_{g,x} + L_{g,u}L_{\pi,x}\big),\, 1\big\}, \\
C_{\ell,h,(x,\theta)} &= L_h(1 + L_{\pi,x})\,C_{\ell,g,(x,\theta)} + ((\ell_{h,x} + \ell_{h,u}L_{\pi,x})(1 + L_{\pi,x}) + L_h\,\ell_{\pi,x})\,C_{L,g,x} \\
&\quad \cdot \max\big\{\bar{C}\,L_{g,u}L_{\pi,\theta},\, 1\big\}, \\
C_{\ell,h,(\theta,x)} &= L_h(1 + L_{\pi,x})\,C_{\ell,g,(\theta,x)} + ((\ell_{h,x} + \ell_{h,u}L_{\pi,x})(1 + L_{\pi,x}) + L_h\,\ell_{\pi,x})\,C_{L,g,\theta} \\
&\quad \cdot \max\big\{\bar{C}\big(L_{g,x} + L_{g,u}L_{\pi,x}\big),\, 1\big\}, \\
C_{\ell,h,(\theta,\theta)} &= L_h(1 + L_{\pi,x})\,C_{\ell,g,(\theta,\theta)} + ((\ell_{h,x} + \ell_{h,u}L_{\pi,x})(1 + L_{\pi,x}) + L_h\,\ell_{\pi,x})\,C_{L,g,\theta} \\
&\quad \cdot \max\big\{\bar{C}\,L_{g,u}L_{\pi,\theta},\, 1\big\}.
\end{aligned}$$

[Shengnan:

$$\begin{aligned}
C_{\ell,h,(x,x)} &= L_h(1 + L_{\pi,x})\,C_{\ell,g,(x,x)} + \big((\ell_{h,x} + \ell_{h,u}L_{\pi,x})(1 + L_{\pi,x}) + L_h\,\ell_{\pi,x}\big)\,C_{L,g,x} \\
&\quad \cdot \max\big\{\bar{C}\big(L_{g,x} + L_{g,u}L_{\pi,x}\big),\, 1\big\}, \\
C_{\ell,h,(x,\theta)} &= L_h(1 + L_{\pi,x})\,C_{\ell,g,(x,\theta)} + \big((\ell_{h,x} + \ell_{h,u}L_{\pi,x})(1 + L_{\pi,x}) + L_h\,\ell_{\pi,x}\big)\,C_{L,g,x} \\
&\quad \cdot \max\big\{\bar{C}\,L_{g,u}L_{\pi,\theta},\, 1\big\} + \big((\ell_{h,u}L_{\pi,\theta} + \ell_{h,\theta})(1 + L_{\pi,x}) + L_h\,\ell_{\pi,\theta}\big), \\
C_{\ell,h,(\theta,x)} &= L_h(1 + L_{\pi,x})\,C_{\ell,g,(\theta,x)} + \big((\ell_{h,x} + \ell_{h,u}L_{\pi,x})(1 + L_{\pi,x}) + L_h\,\ell_{\pi,x}\big)\,C_{L,g,\theta}
\end{aligned}$$

$$\cdot \max\{\bar{C}\big(L_{g,x} + L_{g,u}L_{\pi,x}\big),\, 1\} + \big(L_h\,\ell_{\pi,x} + (\ell_{h,x} + \ell_{h,u}L_{\pi,x})(1 + L_{\pi,\theta})\big),$$
$$C_{\ell,h,(\theta,\theta)} = L_h(1 + L_{\pi,x})\,C_{\ell,g,(\theta,\theta)} + \big((\ell_{h,x} + \ell_{h,u}L_{\pi,x})(1 + L_{\pi,x}) + L_h\,\ell_{\pi,x}\big)\,C_{L,g,\theta}$$
$$\cdot \max\{\bar{C}\,L_{g,u}L_{\pi,\theta},\, 1\} + \big(L_h\,\ell_{\pi,\theta} + (\ell_{h,u}L_{\pi,\theta} + \ell_{h,\theta})(1 + L_{\pi,\theta})\big).$$

] [James: Copy proof from GAPS paper – or if it is nothing more than a product of Lipschitz constants for function composition, then say that.]

[Shengnan: could you please check the proof for Corollary 17 below?]

Proof For the first inequality in Corollary 17, note that

$$\frac{\partial h_{t|\tau}}{\partial x_\tau} = \left(\frac{\partial h_t}{\partial x_t}\Big|_{x_t,u_t} + \frac{\partial h_t}{\partial u_t}\Big|_{x_t,u_t} \cdot \frac{\partial \pi_t}{\partial x_t}\Big|_{x_t,\theta_t}\right) \cdot \frac{\partial g_{t|\tau}}{\partial x_\tau}\Big|_{x_\tau,\theta_{\tau:t-1}}, \tag{32}$$

where $x_t = g_{t|\tau}(x_\tau, \theta_{\tau:t-1})$, $u_t = \pi_t(x_t, \theta_t)$. Thus, by ϵ-time-varying contractive perturbation, we see that

$$\left\|\frac{\partial h_{t|\tau}}{\partial x_\tau}\Big|_{x_\tau,\theta_{\tau:t}}\right\| \le \left(\left\|\frac{\partial h_t}{\partial x_t}\Big|_{x_t,u_t}\right\| + \left\|\frac{\partial h_t}{\partial u_t}\Big|_{x_t,u_t}\right\| \cdot \left\|\frac{\partial \pi_t}{\partial x_t}\Big|_{x_t,\theta_t}\right\|\right)\left\|\frac{\partial g_{t|\tau}}{\partial x_\tau}\Big|_{x_\tau,\theta_{\tau:t-1}}\right\| \tag{33}$$
$$\le L_h(1 + L_{\pi,x})\,\bar{C}\rho^{t-\tau}. \tag{34}$$

[Shengnan: For the first inequality in Corollary 17, note that

$$\frac{\partial h_{t|\tau}}{\partial x_\tau} = \left(\frac{\partial h_t}{\partial x_t}\Big|_{x_t,u_t,\theta_t} + \frac{\partial h_t}{\partial u_t}\Big|_{x_t,u_t,\theta_t} \cdot \frac{\partial \pi_t}{\partial x_t}\Big|_{x_t,\theta_t}\right) \cdot \frac{\partial g_{t|\tau}}{\partial x_\tau}\Big|_{x_\tau,\theta_{\tau:t-1}}, \tag{35}$$

where $x_t = g_{t|\tau}(x_\tau, \theta_{\tau:t-1})$, $u_t = \pi_t(x_t, \theta_t)$. Since θ_t is held fixed when differentiating with respect to x_τ, the direct θ_t-dependence of h_t does not produce an additional term. Thus, by ϵ-time-varying contractive perturbation, we see that

$$\left\|\frac{\partial h_{t|\tau}}{\partial x_\tau}\Big|_{x_\tau,\theta_{\tau:t}}\right\| \le \left(\left\|\frac{\partial h_t}{\partial x_t}\Big|_{x_t,u_t,\theta_t}\right\| + \left\|\frac{\partial h_t}{\partial u_t}\Big|_{x_t,u_t,\theta_t}\right\| \cdot \left\|\frac{\partial \pi_t}{\partial x_t}\Big|_{x_t,\theta_t}\right\|\right)\left\|\frac{\partial g_{t|\tau}}{\partial x_\tau}\Big|_{x_\tau,\theta_{\tau:t-1}}\right\| \tag{36}$$
$$\le L_h(1 + L_{\pi,x})\,\bar{C}\,\rho^{t-\tau} = C_{L,h,x}\,\rho^{t-\tau}. \tag{37}$$

] [Ryan: What is this inequality when $\tau = t$ used for?] [Shengnan: The $\tau = t$ case is included mainly so that the statement holds uniformly for all $\tau \le t$; the main downstream uses are for $\tau < t$. When $\tau = t$, Definition 1 gives $g_{t|t}(x_t) = x_t$, so the bound reduces to the one-step closed-loop cost sensitivity bound.]

For the second inequality, when $\tau = t$, since $x_t \in B_n(0, \bar{R}_C)$ and $u_t \in \mathcal{U}$, we see that

$$\left\|\frac{\partial h_{t|t}}{\partial \theta_t}\Big|_{x_t,\theta_t}\right\| = \left\|\frac{\partial h_t}{\partial u_t}\Big|_{x_t,u_t} \cdot \frac{\partial \pi_t}{\partial \theta_t}\Big|_{x_t,\theta_t}\right\| \le \left\|\frac{\partial h_t}{\partial u_t}\Big|_{x_t,u_t}\right\|\left\|\frac{\partial \pi_t}{\partial \theta_t}\Big|_{x_t,\theta_t}\right\| \le L_h L_{\pi,\theta}.$$

[Shengnan: For the second inequality, when $\tau = t$, since $x_t \in B_n(0, \bar{R}_C)$ and $u_t \in \mathcal{U}$, we see that

$$h_{t|t}(x_t, \theta_t) = h_t(x_t, \pi_t(x_t, \theta_t), \theta_t), \qquad u_t = \pi_t(x_t, \theta_t),$$

and hence

$$\frac{\partial h_{t|t}}{\partial \theta_t}\Big|_{x_t,\theta_t} = \frac{\partial h_t}{\partial u_t}\Big|_{x_t,u_t,\theta_t} \cdot \frac{\partial \pi_t}{\partial \theta_t}\Big|_{x_t,\theta_t} + \frac{\partial h_t}{\partial \theta_t}\Big|_{x_t,u_t,\theta_t}.$$

Therefore,

$$\left\| \frac{\partial h_{t|t}}{\partial \theta_t}\Big|_{x_t,\theta_t} \right\| \le \left\| \frac{\partial h_t}{\partial u_t}\Big|_{x_t,u_t,\theta_t} \right\| \left\| \frac{\partial \pi_t}{\partial \theta_t}\Big|_{x_t,\theta_t} \right\| + \left\| \frac{\partial h_t}{\partial \theta_t}\Big|_{x_t,u_t,\theta_t} \right\|$$
$$\le L_h L_{\pi,\theta} + L_h \;=\; L_h(1 + L_{\pi,\theta}).$$

]

When $\tau < t$, the second inequality can be shown similarly with the first inequality in Corollary 17 because we have the chain-rule decomposition

$$\frac{\partial h_{t|\tau}}{\partial \theta_\tau}\Big|_{x_\tau,\theta_{\tau:t}} = \left(\frac{\partial h_t}{\partial x_t}\Big|_{x_t,u_t} + \frac{\partial h_t}{\partial u_t}\Big|_{x_t,u_t} \cdot \frac{\partial \pi_t}{\partial x_t}\Big|_{x_t,\theta_t} \right) \cdot \frac{\partial g_{t|\tau}}{\partial \theta_\tau}\Big|_{x_\tau,\theta_{\tau:t-1}}. \tag{38}$$

Applying Lemma 16 gives that $C_{L,h,\theta} \;=\; L_h\, C_{L,g,\theta}\,(1 + L_{\pi,x})$.

[Shengnan: When $\tau < t$, the second inequality can be shown similarly with the first inequality in Corollary 17. Since

$$h_{t|\tau}(x_\tau, \theta_{\tau:t}) = h_t(x_t, \pi_t(x_t, \theta_t), \theta_t), \qquad x_t = g_{t|\tau}(x_\tau, \theta_{\tau:t-1}),$$

and θ_t belongs to the fixed tail $\theta_{\tau+1:t}$, we have

$$\frac{\partial \theta_t}{\partial \theta_\tau} = 0.$$

Hence the direct θ_t-dependence of h_t does not contribute when differentiating with respect to θ_τ, and we obtain the chain-rule decomposition

$$\frac{\partial h_{t|\tau}}{\partial \theta_\tau}\Big|_{x_\tau,\theta_{\tau:t}} = \left(\frac{\partial h_t}{\partial x_t}\Big|_{x_t,u_t,\theta_t} + \frac{\partial h_t}{\partial u_t}\Big|_{x_t,u_t,\theta_t} \cdot \frac{\partial \pi_t}{\partial x_t}\Big|_{x_t,\theta_t} \right) \cdot \frac{\partial g_{t|\tau}}{\partial \theta_\tau}\Big|_{x_\tau,\theta_{\tau:t-1}}. \tag{39}$$

Applying Lemma 16 gives that $C_{L,h,\theta} \;=\; L_h\, C_{L,g,\theta}\,(1 + L_{\pi,x})$.]

For the third inequality, using (35), we see that

$$\begin{aligned}
&\left\| \frac{\partial h_{t|\tau}}{\partial x_\tau}\Big|_{x_\tau,\theta_\tau,\theta_{\tau+1:t}} - \frac{\partial h_{t|\tau}}{\partial x_\tau}\Big|_{x'_\tau,\theta'_\tau,\theta_{\tau+1:t}} \right\| \\
\le &\left\| \frac{\partial h_t}{\partial x_t}\Big|_{x_t,u_t} \cdot \frac{\partial g_{t|\tau}}{\partial x_\tau}\Big|_{x_\tau,\theta_\tau,\theta_{\tau+1:t-1}} - \frac{\partial h_t}{\partial x_t}\Big|_{x_t,u_t} \cdot \frac{\partial g_{t|\tau}}{\partial x_\tau}\Big|_{x'_\tau,\theta'_\tau,\theta_{\tau+1:t-1}} \right\| \\
&+ \left\| \left(\frac{\partial h_t}{\partial x_t}\Big|_{x_t,u_t} - \frac{\partial h_t}{\partial x_t}\Big|_{x'_t,u'_t} \right) \cdot \frac{\partial g_{t|\tau}}{\partial x_\tau}\Big|_{x'_\tau,\theta'_\tau,\theta_{\tau+1:t-1}} \right\| \\
&+ \left\| \frac{\partial h_t}{\partial u_t}\Big|_{x_t,u_t} \cdot \frac{\partial \pi_t}{\partial x_t}\Big|_{x_t,\theta_t} \cdot \left(\frac{\partial g_{t|\tau}}{\partial x_\tau}\Big|_{x_\tau,\theta_\tau,\theta_{\tau+1:t-1}} - \frac{\partial g_{t|\tau}}{\partial x_\tau}\Big|_{x'_\tau,\theta'_\tau,\theta_{\tau+1:t-1}} \right) \right\| \\
&+ \left\| \frac{\partial h_t}{\partial u_t}\Big|_{x_t,u_t} \cdot \left(\frac{\partial \pi_t}{\partial x_t}\Big|_{x_t,\theta_t} - \frac{\partial \pi_t}{\partial x_t}\Big|_{x'_t,\theta_t} \right) \cdot \frac{\partial g_{t|\tau}}{\partial x_\tau}\Big|_{x'_\tau,\theta'_\tau,\theta_{\tau+1:t-1}} \right\| \\
&+ \left\| \left(\frac{\partial h_t}{\partial u_t}\Big|_{x_t,u_t} - \frac{\partial h_t}{\partial u_t}\Big|_{x'_t,u'_t} \right) \cdot \frac{\partial \pi_t}{\partial x_t}\Big|_{x'_t,\theta_t} \cdot \frac{\partial g_{t|\tau}}{\partial x_\tau}\Big|_{x'_\tau,\theta'_\tau,\theta_{\tau+1:t-1}} \right\| \\
\le &L_h\, \rho^{t-\tau} \big(C_{\ell,g,(x,x)} \|x_\tau - x'_\tau\| + C_{\ell,g,(x,\theta)} \|\theta_\tau - \theta'_\tau\| \big) \\
&+ \big(\ell_{h,x} \|x_t - x'_t\| + \ell_{h,u} \|u_t - u'_t\| \big)\, C_{L,g,x}\, \rho^{t-\tau}
\end{aligned} \tag{40}$$

$$
\begin{aligned}
&+ L_h L_{\pi,x}\, \rho^{t-\tau} \big(C_{\ell,g,(x,x)} \|x_\tau - x'_\tau\| + C_{\ell,g,(x,\theta)} \|\theta_\tau - \theta'_\tau\| \big) \\
&+ L_h \ell_{\pi,x} \|x_t - x'_t\|\, C_{L,g,x}\, \rho^{t-\tau} + \big(\ell_{h,x} \|x_t - x'_t\| + \ell_{h,u} \|u_t - u'_t\| \big)\, L_{\pi,x} C_{L,g,x}\, \rho^{t-\tau}, \quad (41)
\end{aligned}
$$

where we use the notations $x_t = g_{t|\tau}(x_\tau, \theta_{\tau:t-1})$, $x'_t = g_{t|\tau}(x'_\tau, \theta_{\tau:t-1})$, $u_t = \pi_t(x_t, \theta_t)$, $u'_t = \pi_t(x'_t, \theta_t)$. We use (35) and the triangle inequality in (42); we use Lemma 16 in (43).
Note that by (24), we have, for $t > \tau$,

$$
\begin{aligned}
&\|x_t - x'_t\| \le \bar{C}\, \rho^{t-\tau-1} ((L_{g,x} + L_{g,u} L_{\pi,x}) \|x_\tau - x'_\tau\| + L_{g,u} L_{\pi,\theta} \|\theta_\tau - \theta'_\tau\|), \\
&\|u_t - u'_t\| \le L_{\pi,x}\, \bar{C}\, \rho^{t-\tau-1} ((L_{g,x} + L_{g,u} L_{\pi,x}) \|x_\tau - x'_\tau\| + L_{g,u} L_{\pi,\theta} \|\theta_\tau - \theta'_\tau\|).
\end{aligned}
$$

Substituting these two inequalities into (43) finishes the proof of the third inequality.

[Shengnan: For the third inequality, when $\tau = t$, we have

$$
\frac{\partial h_{t|t}}{\partial x_t}\Big|_{x_t,\theta_t} = \frac{\partial h_t}{\partial x_t}\Big|_{x_t,u_t,\theta_t} + \frac{\partial h_t}{\partial u_t}\Big|_{x_t,u_t,\theta_t} \cdot \frac{\partial \pi_t}{\partial x_t}\Big|_{x_t,\theta_t},
$$

where $u_t = \pi_t(x_t, \theta_t)$. Hence,

$$
\begin{aligned}
&\Big\| \frac{\partial h_{t|t}}{\partial x_t}\Big|_{x_t,\theta_t} - \frac{\partial h_{t|t}}{\partial x_t}\Big|_{x'_t,\theta'_t} \Big\| \\
&\le \Big((\ell_{h,x} + \ell_{h,u} L_{\pi,x})(1 + L_{\pi,x}) + L_h\, \ell_{\pi,x} \Big) \|x_t - x'_t\| \\
&\qquad + \Big((\ell_{h,u} L_{\pi,\theta} + \ell_{h,\theta})(1 + L_{\pi,x}) + L_h\, \ell_{\pi,\theta} \Big) \|\theta_t - \theta'_t\|.
\end{aligned}
$$

For $\tau < t$, using (35), we see that

$$
\begin{aligned}
&\Big\| \frac{\partial h_{t|\tau}}{\partial x_\tau}\Big|_{x_\tau,\theta_\tau,\theta_{\tau+1:t}} - \frac{\partial h_{t|\tau}}{\partial x_\tau}\Big|_{x'_\tau,\theta'_\tau,\theta_{\tau+1:t}} \Big\| \\
\le &\Big\| \frac{\partial h_t}{\partial x_t}\Big|_{x_t,u_t,\theta_t} \cdot \frac{\partial g_{t|\tau}}{\partial x_\tau}\Big|_{x_\tau,\theta_\tau,\theta_{\tau+1:t-1}} - \frac{\partial h_t}{\partial x_t}\Big|_{x_t,u_t,\theta_t} \cdot \frac{\partial g_{t|\tau}}{\partial x_\tau}\Big|_{x'_\tau,\theta'_\tau,\theta_{\tau+1:t-1}} \Big\| \\
&+ \Big\| \Big(\frac{\partial h_t}{\partial x_t}\Big|_{x_t,u_t,\theta_t} - \frac{\partial h_t}{\partial x_t}\Big|_{x'_t,u'_t,\theta_t} \Big) \cdot \frac{\partial g_{t|\tau}}{\partial x_\tau}\Big|_{x'_\tau,\theta'_\tau,\theta_{\tau+1:t-1}} \Big\| \\
&+ \Big\| \frac{\partial h_t}{\partial u_t}\Big|_{x_t,u_t,\theta_t} \cdot \frac{\partial \pi_t}{\partial x_t}\Big|_{x_t,\theta_t} \cdot \Big(\frac{\partial g_{t|\tau}}{\partial x_\tau}\Big|_{x_\tau,\theta_\tau,\theta_{\tau+1:t-1}} - \frac{\partial g_{t|\tau}}{\partial x_\tau}\Big|_{x'_\tau,\theta'_\tau,\theta_{\tau+1:t-1}} \Big) \Big\| \\
&+ \Big\| \frac{\partial h_t}{\partial u_t}\Big|_{x_t,u_t,\theta_t} \cdot \Big(\frac{\partial \pi_t}{\partial x_t}\Big|_{x_t,\theta_t} - \frac{\partial \pi_t}{\partial x_t}\Big|_{x'_t,\theta_t} \Big) \cdot \frac{\partial g_{t|\tau}}{\partial x_\tau}\Big|_{x'_\tau,\theta'_\tau,\theta_{\tau+1:t-1}} \Big\| \\
&+ \Big\| \Big(\frac{\partial h_t}{\partial u_t}\Big|_{x_t,u_t,\theta_t} - \frac{\partial h_t}{\partial u_t}\Big|_{x'_t,u'_t,\theta_t} \Big) \cdot \frac{\partial \pi_t}{\partial x_t}\Big|_{x'_t,\theta_t} \cdot \frac{\partial g_{t|\tau}}{\partial x_\tau}\Big|_{x'_\tau,\theta'_\tau,\theta_{\tau+1:t-1}} \Big\| \quad (42) \\
\le &L_h\, \rho^{t-\tau} \big(C_{\ell,g,(x,x)} \|x_\tau - x'_\tau\| + C_{\ell,g,(x,\theta)} \|\theta_\tau - \theta'_\tau\| \big) \\
&+ \big(\ell_{h,x} \|x_t - x'_t\| + \ell_{h,u} \|u_t - u'_t\| \big)\, C_{L,g,x}\, \rho^{t-\tau} \\
&+ L_h L_{\pi,x}\, \rho^{t-\tau} \big(C_{\ell,g,(x,x)} \|x_\tau - x'_\tau\| + C_{\ell,g,(x,\theta)} \|\theta_\tau - \theta'_\tau\| \big) \\
&+ L_h \ell_{\pi,x} \|x_t - x'_t\|\, C_{L,g,x}\, \rho^{t-\tau} + \big(\ell_{h,x} \|x_t - x'_t\| + \ell_{h,u} \|u_t - u'_t\| \big)\, L_{\pi,x} C_{L,g,x}\, \rho^{t-\tau}, \quad (43)
\end{aligned}
$$

where we use the notations $x_t = g_{t|\tau}(x_\tau, \theta_{\tau:t-1})$, $x'_t = g_{t|\tau}(x'_\tau, \theta'_\tau, \theta_{\tau+1:t-1})$, $u_t = \pi_t(x_t, \theta_t)$, $u'_t = \pi_t(x'_t, \theta_t)$. We use (35) and the triangle inequality in (42); we use Lemma 16 in (43). Since

$\tau < t$, the variable θ_t is the same fixed tail input on both sides, so the direct θ_t-dependence of h_t does not produce any additional term when differentiating with respect to x_τ. Note that by (24), we have, for $t > \tau$,

$$\|x_t - x_t'\| \le \bar{C}\,\rho^{t-\tau-1}((L_{g,x} + L_{g,u}L_{\pi,x})\|x_\tau - x_\tau'\| + L_{g,u}L_{\pi,\theta}\|\theta_\tau - \theta_\tau'\|),$$
$$\|u_t - u_t'\| \le L_{\pi,x}\,\bar{C}\,\rho^{t-\tau-1}((L_{g,x} + L_{g,u}L_{\pi,x})\|x_\tau - x_\tau'\| + L_{g,u}L_{\pi,\theta}\|\theta_\tau - \theta_\tau'\|).$$

Substituting these two inequalities into (43) finishes the proof of the third inequality for $\tau < t$.]
For the last inequality, note that when $\tau = t$, we have that

$$\begin{aligned}
&\Big\|\frac{\partial h_{t,t}}{\partial \theta_t}\Big|_{x_t,\theta_t} - \frac{\partial h_{t,t}}{\partial \theta_t}\Big|_{x_t',\theta_t'}\Big\| && (44)\\
=&\Big\|\frac{\partial h_t}{\partial u_t}\Big|_{x_t,u_t}\cdot\frac{\partial \pi_t}{\partial \theta_t}\Big|_{x_t,\theta_t} - \frac{\partial h_t}{\partial u_t}\Big|_{x_t',u_t'}\cdot\frac{\partial \pi_t}{\partial \theta_t}\Big|_{x_t',\theta_t'}\Big\| && (45)\\
\le&\Big\|\frac{\partial h_t}{\partial u_t}\Big|_{x_t,u_t}\cdot\Big(\frac{\partial \pi_t}{\partial \theta_t}\Big|_{x_t,\theta_t} - \frac{\partial \pi_t}{\partial \theta_t}\Big|_{x_t',\theta_t'}\Big)\Big\| + \Big\|\Big(\frac{\partial h_t}{\partial u_t}\Big|_{x_t,u_t} - \frac{\partial h_t}{\partial u_t}\Big|_{x_t',u_t'}\Big)\cdot\frac{\partial \pi_t}{\partial \theta_t}\Big|_{x_t',\theta_t'}\Big\| && (46)\\
\le& L_h\big(\ell_{\pi,\theta}\|\theta_t - \theta_t'\| + \ell_{\pi,x}\|x_t - x_t'\|\big) + \big(\ell_{h,x}\|x_t - x_t'\| + \ell_{h,u}\|u_t - u_t'\|\big)L_{\pi,\theta} && (47)\\
\le&(L_h\ell_{\pi,\theta} + (\ell_{h,x} + \ell_{h,u}L_{\pi,x})L_{\pi,\theta})\|x_t - x_t'\| + \big(L_h\ell_{\pi,\theta} + \ell_{h,u}L_{\pi,\theta}^2\big)\|\theta_t - \theta_t'\|, && (48)
\end{aligned}$$

where we use the chain rule decomposition in (49); we use the triangle inequality in (50); we use Assumption 11 in both (51) and (52).

[Shengnan: For the last inequality, note that when $\tau = t$, we have that

$$\begin{aligned}
&\Big\|\frac{\partial h_{t,t}}{\partial \theta_t}\Big|_{x_t,\theta_t} - \frac{\partial h_{t,t}}{\partial \theta_t}\Big|_{x_t',\theta_t'}\Big\|\\
=&\Big\|\Big(\frac{\partial h_t}{\partial u_t}\Big|_{x_t,u_t,\theta_t}\cdot\frac{\partial \pi_t}{\partial \theta_t}\Big|_{x_t,\theta_t} + \frac{\partial h_t}{\partial \theta_t}\Big|_{x_t,u_t,\theta_t}\Big) - \Big(\frac{\partial h_t}{\partial u_t}\Big|_{x_t',u_t',\theta_t'}\cdot\frac{\partial \pi_t}{\partial \theta_t}\Big|_{x_t',\theta_t'} + \frac{\partial h_t}{\partial \theta_t}\Big|_{x_t',u_t',\theta_t'}\Big)\Big\| && (49)\\
\le&\Big\|\frac{\partial h_t}{\partial u_t}\Big|_{x_t,u_t,\theta_t}\cdot\Big(\frac{\partial \pi_t}{\partial \theta_t}\Big|_{x_t,\theta_t} - \frac{\partial \pi_t}{\partial \theta_t}\Big|_{x_t',\theta_t'}\Big)\Big\|\\
&+\Big\|\Big(\frac{\partial h_t}{\partial u_t}\Big|_{x_t,u_t,\theta_t} - \frac{\partial h_t}{\partial u_t}\Big|_{x_t',u_t',\theta_t'}\Big)\cdot\frac{\partial \pi_t}{\partial \theta_t}\Big|_{x_t',\theta_t'}\Big\|\\
&+\Big\|\frac{\partial h_t}{\partial \theta_t}\Big|_{x_t,u_t,\theta_t} - \frac{\partial h_t}{\partial \theta_t}\Big|_{x_t',u_t',\theta_t'}\Big\| && (50)\\
\le& L_h\big(\ell_{\pi,\theta}\|\theta_t - \theta_t'\| + \ell_{\pi,x}\|x_t - x_t'\|\big)\\
&+\big(\ell_{h,x}\|x_t - x_t'\| + \ell_{h,u}\|u_t - u_t'\| + \ell_{h,\theta}\|\theta_t - \theta_t'\|\big)L_{\pi,\theta}\\
&+\ell_{h,x}\|x_t - x_t'\| + \ell_{h,u}\|u_t - u_t'\| + \ell_{h,\theta}\|\theta_t - \theta_t'\| && (51)\\
\le&(L_h\ell_{\pi,x} + (\ell_{h,x} + \ell_{h,u}L_{\pi,x})L_{\pi,\theta} + \ell_{h,x} + \ell_{h,u}L_{\pi,x})\|x_t - x_t'\|\\
&+(L_h\ell_{\pi,\theta} + (\ell_{h,u}L_{\pi,\theta} + \ell_{h,\theta})L_{\pi,\theta} + \ell_{h,u}L_{\pi,\theta} + \ell_{h,\theta})\|\theta_t - \theta_t'\| && (52)
\end{aligned}$$

where we use the chain rule decomposition in (49); we use the triangle inequality in (50); we use Assumption 11 in both (51) and (52).]
When $\tau < t$, by (39), we have that

$$\Big\|\frac{\partial h_{t|\tau}}{\partial \theta_\tau}\Big|_{x_\tau,\theta_\tau,\theta_{\tau+1:t}} - \frac{\partial h_{t|\tau}}{\partial \theta_\tau}\Big|_{x_\tau',\theta_\tau',\theta_{\tau+1:t}}\Big\|$$

$$
\begin{aligned}
\leq & \Big\| \frac{\partial h_t}{\partial x_t}\Big|_{x_t,u_t} \cdot \left(\frac{\partial g_{t|\tau}}{\partial \theta_\tau}\Big|_{x_\tau,\theta_\tau,\theta_{\tau+1:t-1}} - \frac{\partial g_{t|\tau}}{\partial \theta_\tau}\Big|_{x'_\tau,\theta'_\tau,\theta_{\tau+1:t-1}} \right) \Big\| \\
& + \Big\| \left(\frac{\partial h_t}{\partial x_t}\Big|_{x_t,u_t} - \frac{\partial h_t}{\partial x_t}\Big|_{x'_t,u'_t} \right) \cdot \frac{\partial g_{t|\tau}}{\partial \theta_\tau}\Big|_{x'_\tau,\theta'_\tau,\theta_{\tau+1:t-1}} \Big\| \\
& + \Big\| \frac{\partial h_t}{\partial u_t}\Big|_{x_t,u_t} \cdot \frac{\partial \pi_t}{\partial x_t}\Big|_{x_t,\theta_t} \cdot \left(\frac{\partial g_{t|\tau}}{\partial \theta_\tau}\Big|_{x_\tau,\theta_\tau,\theta_{\tau+1:t-1}} - \frac{\partial g_{t|\tau}}{\partial \theta_\tau}\Big|_{x'_\tau,\theta'_\tau,\theta_{\tau+1:t-1}} \right) \Big\| \\
& + \Big\| \frac{\partial h_t}{\partial u_t}\Big|_{x_t,u_t} \cdot \left(\frac{\partial \pi_t}{\partial x_t}\Big|_{x_t,\theta_t} - \frac{\partial \pi_t}{\partial x_t}\Big|_{x'_t,\theta_t} \right) \cdot \frac{\partial g_{t|\tau}}{\partial \theta_\tau}\Big|_{x'_\tau,\theta'_\tau,\theta_{\tau+1:t-1}} \Big\| \\
& + \Big\| \left(\frac{\partial h_t}{\partial u_t}\Big|_{x_t,u_t} - \frac{\partial h_t}{\partial u_t}\Big|_{x'_t,u'_t} \right) \cdot \frac{\partial \pi_t}{\partial x_t}\Big|_{x'_t,\theta_t} \cdot \frac{\partial g_{t|\tau}}{\partial \theta_\tau}\Big|_{x'_\tau,\theta'_\tau,\theta_{\tau+1:t-1}} \Big\| \qquad (53) \\
\leq & L_h \, \rho^{t-\tau} \big(C_{\ell,g,(\theta,x)} \|x_\tau - x'_\tau\| + C_{\ell,g,(\theta,\theta)} \|\theta_\tau - \theta'_\tau\| \big) \\
& + \big(\ell_{h,x} \|x_t - x'_t\| + \ell_{h,u} \|u_t - u'_t\| \big) \, C_{L,g,\theta} \, \rho^{t-\tau} \\
& + L_h L_{\pi,x} \, \rho^{t-\tau} \big(C_{\ell,g,(\theta,x)} \|x_\tau - x'_\tau\| + C_{\ell,g,(\theta,\theta)} \|\theta_\tau - \theta'_\tau\| \big) \\
& + L_h \ell_{\pi,x} \|x_t - x'_t\| \, C_{L,g,\theta} \, \rho^{t-\tau} + \big(\ell_{h,x} \|x_t - x'_t\| + \ell_{h,u} \|u_t - u'_t\| \big) \, L_{\pi,x} \, C_{L,g,\theta} \, \rho^{t-\tau}, \qquad (54)
\end{aligned}
$$

where we use (39) and the triangle inequality in (55); we use Assumption 11 in (56). Note that by (24), we have, for $t > \tau$,

$$
\begin{aligned}
\|x_t - x'_t\| &\leq \bar{C} \, \rho^{t-\tau-1} ((L_{g,x} + L_{g,u} L_{\pi,x}) \|x_\tau - x'_\tau\| + L_{g,u} L_{\pi,\theta} \|\theta_\tau - \theta'_\tau\|), \\
\|u_t - u'_t\| &\leq L_{\pi,x} \, \bar{C} \, \rho^{t-\tau-1} ((L_{g,x} + L_{g,u} L_{\pi,x}) \|x_\tau - x'_\tau\| + L_{g,u} L_{\pi,\theta} \|\theta_\tau - \theta'_\tau\|).
\end{aligned}
$$

Substituting these into (56) finishes the proof of the fourth inequality.

[Shengnan: When $\tau < t$, by (39), we have that

$$
\begin{aligned}
& \Big\| \frac{\partial h_{t|\tau}}{\partial \theta_\tau}\Big|_{x_\tau,\theta_\tau,\theta_{\tau+1:t}} - \frac{\partial h_{t|\tau}}{\partial \theta_\tau}\Big|_{x'_\tau,\theta'_\tau,\theta_{\tau+1:t}} \Big\| \\
\leq & \Big\| \frac{\partial h_t}{\partial x_t}\Big|_{x_t,u_t,\theta_t} \cdot \left(\frac{\partial g_{t|\tau}}{\partial \theta_\tau}\Big|_{x_\tau,\theta_\tau,\theta_{\tau+1:t-1}} - \frac{\partial g_{t|\tau}}{\partial \theta_\tau}\Big|_{x'_\tau,\theta'_\tau,\theta_{\tau+1:t-1}} \right) \Big\| \\
& + \Big\| \left(\frac{\partial h_t}{\partial x_t}\Big|_{x_t,u_t,\theta_t} - \frac{\partial h_t}{\partial x_t}\Big|_{x'_t,u'_t,\theta_t} \right) \cdot \frac{\partial g_{t|\tau}}{\partial \theta_\tau}\Big|_{x'_\tau,\theta'_\tau,\theta_{\tau+1:t-1}} \Big\| \\
& + \Big\| \frac{\partial h_t}{\partial u_t}\Big|_{x_t,u_t,\theta_t} \cdot \frac{\partial \pi_t}{\partial x_t}\Big|_{x_t,\theta_t} \cdot \left(\frac{\partial g_{t|\tau}}{\partial \theta_\tau}\Big|_{x_\tau,\theta_\tau,\theta_{\tau+1:t-1}} - \frac{\partial g_{t|\tau}}{\partial \theta_\tau}\Big|_{x'_\tau,\theta'_\tau,\theta_{\tau+1:t-1}} \right) \Big\| \\
& + \Big\| \frac{\partial h_t}{\partial u_t}\Big|_{x_t,u_t,\theta_t} \cdot \left(\frac{\partial \pi_t}{\partial x_t}\Big|_{x_t,\theta_t} - \frac{\partial \pi_t}{\partial x_t}\Big|_{x'_t,\theta_t} \right) \cdot \frac{\partial g_{t|\tau}}{\partial \theta_\tau}\Big|_{x'_\tau,\theta'_\tau,\theta_{\tau+1:t-1}} \Big\| \\
& + \Big\| \left(\frac{\partial h_t}{\partial u_t}\Big|_{x_t,u_t,\theta_t} - \frac{\partial h_t}{\partial u_t}\Big|_{x'_t,u'_t,\theta_t} \right) \cdot \frac{\partial \pi_t}{\partial x_t}\Big|_{x'_t,\theta_t} \cdot \frac{\partial g_{t|\tau}}{\partial \theta_\tau}\Big|_{x'_\tau,\theta'_\tau,\theta_{\tau+1:t-1}} \Big\| \qquad (55) \\
\leq & L_h \, \rho^{t-\tau} \big(C_{\ell,g,(\theta,x)} \|x_\tau - x'_\tau\| + C_{\ell,g,(\theta,\theta)} \|\theta_\tau - \theta'_\tau\| \big) \\
& + \big(\ell_{h,x} \|x_t - x'_t\| + \ell_{h,u} \|u_t - u'_t\| \big) \, C_{L,g,\theta} \, \rho^{t-\tau} \\
& + L_h L_{\pi,x} \, \rho^{t-\tau} \big(C_{\ell,g,(\theta,x)} \|x_\tau - x'_\tau\| + C_{\ell,g,(\theta,\theta)} \|\theta_\tau - \theta'_\tau\| \big) \\
& + L_h \ell_{\pi,x} \|x_t - x'_t\| \, C_{L,g,\theta} \, \rho^{t-\tau} + \big(\ell_{h,x} \|x_t - x'_t\| + \ell_{h,u} \|u_t - u'_t\| \big) \, L_{\pi,x} \, C_{L,g,\theta} \, \rho^{t-\tau}, \qquad (56)
\end{aligned}
$$

where we use the notations

$$
x_t = g_{t|\tau}(x_\tau, \theta_{\tau:t-1}), \qquad x'_t = g_{t|\tau}(x'_\tau, \theta'_\tau, \theta_{\tau+1:t-1}),
$$

$$u_t = \pi_t(x_t, \theta_t), \qquad u'_t = \pi_t(x'_t, \theta_t).$$

We use (39) and the triangle inequality in (55); we use Assumption 11 in (56). Since $\tau < t$, the variable θ_t is the same fixed tail input on both sides, so the direct θ_t-dependence of h_t does not produce any additional term when differentiating with respect to θ_τ. Note that by (24), we have, for $t > \tau$,

$$\|x_t - x'_t\| \le \bar{C}\,\rho^{t-\tau-1}((L_{g,x} + L_{g,u}L_{\pi,x})\|x_\tau - x'_\tau\| + L_{g,u}L_{\pi,\theta}\|\theta_\tau - \theta'_\tau\|),$$
$$\|u_t - u'_t\| \le L_{\pi,x}\,\bar{C}\,\rho^{t-\tau-1}((L_{g,x} + L_{g,u}L_{\pi,x})\|x_\tau - x'_\tau\| + L_{g,u}L_{\pi,\theta}\|\theta_\tau - \theta'_\tau\|).$$

Substituting these into (56) finishes the proof of the fourth inequality for $\tau < t$.]

■

5.3. Contraction and Stability of the M-GAPS Joint Dynamics

Building on the regularity properties established above, we now show that the M-GAPS joint dynamics exhibit contraction stability as a consequence of the contractive perturbation structure of the closed-loop dynamics. In particular, the coupled evolution of the system state x_t and internal sensitivity y_t state remains bounded and contractive under slowly varying policy parameters.

Lemma 18 (Contraction Stability) *[James: I believe this should be fairly straightforward, maybe a few more $(AB - CD)$ instances to point out, and some extra hand-holding, but hopefully no big changes in explanation needed.] Suppose the sequence $\theta_{0:T-1}$ is given and it satisfies the constraint that $\|\theta_t - \theta_{t-1}\| \le \epsilon_\theta$ for all time step t. Consider the dynamical system*

$$x_{t+1} = q_t^x(x_t, y_t, \theta_t) = g_t(x_t, \pi_t(x_t, \theta_t)),$$
$$y_{t+1} = q_t^y(x_t, y_t, \theta_t) = \left.\frac{\partial g_{t+1|t}}{\partial x_t}\right|_{x_t,\theta_t} \cdot y_t + \left.\frac{\partial g_{t+1|t}}{\partial \theta_t}\right|_{x_t,\theta_t}.$$

We have that $\|x_t\| \le R_S < R_x$, $\|y_t\| \le R_y$ always hold if the system starts from $(x_\tau, y_\tau) = (0, 0)$. Here,

$$R_y = \frac{\bar{C}\,L_{g,u}\,L_{\pi,\theta}}{\rho(1-\rho)} = \frac{C_{L,g,\theta}}{1-\rho},$$

where recall that ρ is the decay factor defined in Assumption 12. Further, from any initial states $(x_\tau, y_\tau), (x'_\tau, y'_\tau)$ that satisfy $\|x_\tau\|, \|x'_\tau\| \le R_x$ and $\|y_\tau\|, \|y'_\tau\| \le R_y$, the trajectory satisfies

$$\big\|(x_t, y_t) - (x'_t, y'_t)\big\| \le \gamma(t-\tau) \cdot \big\|(x_\tau, y_\tau) - (x'_\tau, y'_\tau)\big\|,$$

where

$$\gamma(\tau) = \big(\bar{C} + C_{\ell,g,(x,x)}R_y + C_{\ell,g,(\theta,x)}\bar{C}\tau\big)\rho^\tau.$$

Note that γ satisfies

$$\sum_{\tau=0}^{\infty}\gamma(\tau) \le \Gamma, \text{ where } \Gamma = \frac{\bar{C} + C_{\ell,g,(x,x)}R_y}{1-\rho} + \frac{C_{\ell,g,(\theta,x)}\bar{C}}{(1-\rho)^2}.$$

The definitions of the coefficients $C_{L,g,\theta}, C_{\ell,g,(x,x)}, C_{\ell,g,(\theta,x)}$ can be found in Lemma 16 in Section 5.2.

Note that Lemma 18 is different from the contraction assumption of Lin et al. (2023) because it also considers the internal state y_t of ALG in addition to the system state x_t. The requirement that $\sum_{t=0}^{\infty} \gamma(t) \leq \Gamma$ is also weaker than the exponential decay rate in Lin et al. (2023).

Proof Consider two trajectories $\{x_{t_1:t_2}, y_{t_1:t_2}\}$ and $\{x'_{t_1:t_2}, y'_{t_1:t_2}\}$ given by

$$\begin{aligned} x_{\tau+1} &= g_\tau(x_\tau, \pi_t(x_\tau, \theta_\tau)), \\ y_{\tau+1} &= \left.\frac{\partial g_{\tau+1|\tau}}{\partial x_\tau}\right|_{x_\tau,\theta_\tau} \cdot y_\tau + \left.\frac{\partial g_{\tau+1|\tau}}{\partial \theta_\tau}\right|_{x_\tau,\theta_\tau}, \end{aligned}$$

and

$$\begin{aligned} x'_{\tau+1} &= g_\tau(x'_\tau, \pi_t(x'_\tau, \theta_\tau)), \\ y'_{\tau+1} &= \left.\frac{\partial g_{\tau+1|\tau}}{\partial x_\tau}\right|_{x'_\tau,\theta_\tau} \cdot y'_\tau + \left.\frac{\partial g_{\tau+1|\tau}}{\partial \theta_\tau}\right|_{x'_\tau,\theta_\tau}, \end{aligned}$$

where $\tau = t_1, t_1+1, \ldots, t_2$.

Note that by Assumption 12, we have that $\|x_{t_2}\| \leq R_S$ and for any x_{t_1}, x'_{t_1} whose norms are upper bounded by R_C

$$\left\|x_{t_2} - x'_{t_2}\right\| \leq \bar{C}\rho^{t_2-t_1}\left\|x_{t_1} - x'_{t_1}\right\|. \tag{57}$$

where ρ is the decay factor of the contractive perturbation property defined in Assumption 12. For the y sequence, note that y_{t_2} and y'_{t_2} can be expressed equivalently as

$$y_{t_2} = \left.\frac{\partial g_{t_2|t_1}}{\partial x_{t_1}}\right|_{x_{t_1},\theta_{t_1:t_2-1}} \cdot y_{t_1} + \sum_{\tau=t_1}^{t_2-1} \left.\frac{\partial g_{t_2|\tau}}{\partial \theta_\tau}\right|_{x_\tau,\theta_{\tau:t_2-1}}, \tag{58a}$$

$$y'_{t_2} = \left.\frac{\partial g_{t_2|t_1}}{\partial x_{t_1}}\right|_{x'_{t_1},\theta_{t_1:t_2-1}} \cdot y'_{t_1} + \sum_{\tau=t_1}^{t_2-1} \left.\frac{\partial g_{t_2|\tau}}{\partial \theta_\tau}\right|_{x'_\tau,\theta_{\tau:t_2-1}}. \tag{58b}$$

By Lemma 16, we see that if $y_{t_1} = 0$, then

$$\|y_{t_2}\| = \left\|\sum_{\tau=t_1}^{t_2-1} \left.\frac{\partial g_{t_2|\tau}}{\partial \theta_\tau}\right|_{x_\tau,\theta_{\tau:t_2-1}}\right\| \leq \sum_{\tau=t_1}^{t_2-1} \left\|\left.\frac{\partial g_{t_2|\tau}}{\partial \theta_\tau}\right|_{x_\tau,\theta_{\tau:t_2-1}}\right\| \leq \sum_{\tau=t_1}^{t_2-1} C_{L,g,\theta}\rho^{t_2-\tau} = \frac{C_{L,g,\theta}}{1-\rho}. \tag{59}$$

[James: The block below needs some discussion...] We also see that

$$\begin{aligned} y_{t_2} - y'_{t_2} = &\left.\frac{\partial g_{t_2|t_1}}{\partial x_{t_1}}\right|_{x_{t_1},\theta_{t_1:t_2-1}} \cdot y_{t_1} - \left.\frac{\partial g_{t_2|t_1}}{\partial x_{t_1}}\right|_{x'_{t_1},\theta_{t_1:t_2-1}} \cdot y'_{t_1} \\ &+ \sum_{\tau=t_1}^{t_2-1}\left(\left.\frac{\partial g_{t_2|\tau}}{\partial \theta_\tau}\right|_{x_\tau,\theta_{\tau:t_2-1}} - \left.\frac{\partial g_{t_2|\tau}}{\partial \theta_\tau}\right|_{x'_\tau,\theta_{\tau:t_2-1}}\right) \end{aligned}$$

$$= \left(\frac{\partial g_{t_2|t_1}}{\partial x_{t_1}}\bigg|_{x_{t_1},\theta_{t_1:t_2-1}} - \frac{\partial g_{t_2|t_1}}{\partial x_{t_1}}\bigg|_{x'_{t_1},\theta_{t_1:t_2-1}} \right) \cdot y_{t_1} + \frac{\partial g_{t_2|t_1}}{\partial x_{t_1}}\bigg|_{x'_{t_1},\theta_{t_1:t_2-1}} \cdot (y_{t_1} - y'_{t_1})$$
$$+ \sum_{\tau=t_1}^{t_2-1} \left(\frac{\partial g_{t_2|\tau}}{\partial \theta_\tau}\bigg|_{x_\tau,\theta_{\tau:t_2-1}} - \frac{\partial g_{t_2|\tau}}{\partial \theta_\tau}\bigg|_{x'_\tau,\theta_{\tau:t_2-1}} \right), \tag{60a}$$

Therefore,

$$\begin{aligned}
&\left\| y_{t_2} - y'_{t_2} \right\| \\
&\leq \left\| \left(\frac{\partial g_{t_2|t_1}}{\partial x_{t_1}}\bigg|_{x_{t_1},\theta_{t_1:t_2-1}} - \frac{\partial g_{t_2|t_1}}{\partial x_{t_1}}\bigg|_{x'_{t_1},\theta_{t_1:t_2-1}} \right) \cdot y_{t_1} \right\| + \left\| \frac{\partial g_{t_2|t_1}}{\partial x_{t_1}}\bigg|_{x'_{t_1},\theta_{t_1:t_2-1}} \cdot (y_{t_1} - y'_{t_1}) \right\| \\
&\quad + \sum_{\tau=t_1}^{t_2-1} \left\| \frac{\partial g_{t_2|\tau}}{\partial \theta_\tau}\bigg|_{x_\tau,\theta_{\tau:t_2-1}} - \frac{\partial g_{t_2|\tau}}{\partial \theta_\tau}\bigg|_{x'_\tau,\theta_{\tau:t_2-1}} \right\| && \text{(60b)} \\
&\leq C_{\ell,g,(x,x)} \rho^{t_2-t_1} \left\| x_{t_1} - x'_{t_1} \right\| \cdot R_y + C_{L,g,x} \rho^{t_2-t_1} \left\| y_{t_1} - y'_{t_1} \right\| \\
&\quad + C_{\ell,g,(\theta,x)} \sum_{\tau=t_1}^{t_2-1} \rho^{t_2-\tau} \left\| x_\tau - x'_\tau \right\| && \text{(60c)} \\
&\leq C_{\ell,g,(x,x)} \rho^{t_2-t_1} \left\| x_{t_1} - x'_{t_1} \right\| \cdot R_y + C_{L,g,x} \rho^{t_2-t_1} \left\| y_{t_1} - y'_{t_1} \right\| \\
&\quad + C_{\ell,g,(\theta,x)} \sum_{\tau=t_1}^{t_2-1} \rho^{t_2-\tau} \cdot \bar{C} \rho^{\tau-t_1} \left\| x_{t_1} - x'_{t_1} \right\| && \text{(60d)} \\
&\leq \left(C_{\ell,g,(x,x)} R_y + C_{\ell,g,(\theta,x)} \bar{C}(t_2 - t_1) \right) \rho^{t_2-t_1} \left\| x_{t_1} - x'_{t_1} \right\| + C_{L,g,x} \rho^{t_2-t_1} \left\| y_{t_1} - y'_{t_1} \right\|. && \text{(60e)}
\end{aligned}$$

where (60a) is obtained by subtracting (58a) and (58b) and then applying the identity $AB - CD = (A - C)B + C(B - D)$ to the term $\left(\partial g_{t_2|t_1}/\partial x_{t_1}\right) y_{t_1} - \left(\partial g_{t_2|t_1}/\partial x_{t_1}\right) y'_{t_1}$ with $A = \frac{\partial g_{t_2|t_1}}{\partial x_{t_1}}\Big|_{x_{t_1},\theta_{t_1:t_2-1}}$, $B = y_{t_1}$, $C = \frac{\partial g_{t_2|t_1}}{\partial x_{t_1}}\Big|_{x'_{t_1},\theta_{t_1:t_2-1}}$, and $D = y'_{t_1}$); (60b) follows by taking norms, using the triangle inequality, and using submultiplicativity of the operator norm; in (60c), we applied Lemma 16 to bound each term in (60b) and used $\|y_{t_1}\| \leq R_y$ for admissible initial conditions; in (60d), we substituted the contraction bound (57) to obtain $\|x_\tau - x'_\tau\| \leq \bar{C}\rho^{\tau-t_1}\|x_{t_1} - x'_{t_1}\|$; (60e) follows since $\rho^{t_2-\tau}\rho^{\tau-t_1} = \rho^{t_2-t_1}$ for each τ, so $\sum_{\tau=t_1}^{t_2-1} \rho^{t_2-\tau}\bar{C}\rho^{\tau-t_1} = (t_2 - t_1)\bar{C}\rho^{t_2-t_1}$. Therefore, we see that

$$\begin{aligned}
&\left\| (x_{t_2}, y_{t_2}) - (x'_{t_2}, y'_{t_2}) \right\| \\
&\leq \left\| x_{t_2} - x'_{t_2} \right\| + \left\| y_{t_2} - y'_{t_2} \right\| && \text{(61a)} \\
&\leq \bar{C}\rho^{t_2-t_1} \left\| x_{t_1} - x'_{t_1} \right\| + \left(C_{\ell,g,(x,x)} R_y + C_{\ell,g,(\theta,x)} \bar{C}(t_2 - t_1) \right) \rho^{t_2-t_1} \left\| x_{t_1} - x'_{t_1} \right\| \\
&\quad + \bar{C}\rho^{t_2-t_1} \left\| y_{t_1} - y'_{t_1} \right\| && \text{(61b)} \\
&\leq \gamma(t_2 - t_1) \left\| (x_{t_1}, y_{t_1}) - (x'_{t_1}, y'_{t_1}) \right\|, && \text{(61c)}
\end{aligned}$$

where we use the triangle inequality in (61a); we use (57), (60e), and $\bar{C} = C_{L,g,x}$ in (61b); we use the inequality that

$$\left\| x_{t_1} - x'_{t_1} \right\| + \left\| y_{t_1} - y'_{t_1} \right\| \leq \sqrt{2} \left\| (x_{t_1}, y_{t_1}) - (x'_{t_1}, y'_{t_1}) \right\|$$

and the definition of $\gamma(\cdot)$ in (61c). ■

5.4. Trajectory Mismatch Control

We now turn to the control of trajectory mismatch. Before deriving the main mismatch bound, we record a simple state-domain inclusion result under slow parameter variation. This lemma ensures that the state trajectory remains inside the compact analysis domain $\mathcal{X}$, so that the Lipschitz and smoothness constants introduced earlier are applicable throughout the argument.

Lemma 19 (State Bound and Domain Inclusion Under Slow Parameter Variation) *Suppose Assumptions 8, 11, and 12 hold. Let $\mathcal{X} = B_n(0, R_x)$, where R_x is defined in Assumption 8. Fix $\tau, t \in \mathcal{T}$ with $\tau \le t$. If $x_\tau \in B_n(0, \bar{R}_C)$ and $\theta_{\tau:t-1} \in S_{\epsilon_\theta}(\tau : t-1)$, then the propagated state $x_t := g_{t|\tau}(x_\tau, \theta_{\tau:t-1})$ satisfies $\|x_t\| \le \bar{C}\,\rho^{t-\tau}\,\|x_\tau\| + R_S \le \bar{C}\,\bar{R}_C + R_S = R_x$. In particular, $x_t \in \mathcal{X}$.*

Proof By the ϵ_θ-time-varying contractive perturbation property (Definition 3),

$$\|x_t - g_{t|\tau}(0, \theta_{\tau:t-1})\| \le \bar{C}\,\rho^{t-\tau}\,\|x_\tau\|.$$

Moreover, by the ϵ_θ-time-varying stability property in Assumption 12,

$$\|g_{t|\tau}(0, \theta_{\tau:t-1})\| \le R_S.$$

Therefore, by the triangle inequality,

$$\|x_t\| \le \|x_t - g_{t|\tau}(0, \theta_{\tau:t-1})\| + \|g_{t|\tau}(0, \theta_{\tau:t-1})\| \le \bar{C}\,\rho^{t-\tau}\,\|x_\tau\| + R_S.$$

Since $\|x_\tau\| \le \bar{R}_C$, we further obtain

$$\|x_t\| \le \bar{C}\,\rho^{t-\tau}\,\bar{R}_C + R_S \le \bar{C}\,\bar{R}_C + R_S = R_x,$$

where the last equality is the definition of R_x in Assumption 8. Hence $x_t \in B_n(0, R_x) = \mathcal{X}$. ■

Now, we state and prove the main theorem (Theorem 20) that bounds the distances between the trajectory of M-GAPS and the imaginary trajectory achieved by using θ_t repeatedly from time step 0.

Theorem 20 *Suppose Assumptions 8, 11, and 12 hold. Let $\{x_t, u_t, \theta_t\}_{t\in\mathcal{T}}$ denote the trajectory of*

$$x_{t+1} = q_t^x(x_t, y_t, \theta_t) = g_t(x_t, \pi_t(x_t, \theta_t)), \tag{62a}$$

$$y_{t+1} = q_t^y(x_t, y_t, \theta_t) = \left.\frac{\partial g_{t+1|t}}{\partial x_t}\right|_{x_t,\theta_t} \cdot y_t + \left.\frac{\partial g_{t+1|t}}{\partial \theta_t}\right|_{x_t,\theta_t}, \tag{62b}$$

$$\theta_{t+1} = q_t^\theta(x_t, y_t, \theta_t) = \Pi_\Theta\left(\theta_t - \eta\left(\left.\frac{\partial h_{t|t}}{\partial x_t}\right|_{x_t,\theta_t} \cdot y_t + \left.\frac{\partial h_{t|t}}{\partial \theta_t}\right|_{x_t,\theta_t}\right)\right). \tag{62c}$$

Suppose η satisfies the constraint that $\bar{\varepsilon} := \frac{C_{L,h,\theta}\eta}{1-\rho} \le \epsilon_\theta$, where ϵ_θ is the per-step parameter-variation budget specified in Assumption 12. Under this condition, the M-GAPS update satisfies $\|\theta_{t+1} - \theta_t\| \le \bar{\varepsilon} \le \epsilon_\theta$, and hence the parameter sequence is slowly time-varying, i.e., $\theta_{0:t-1} \in$

$S_{\epsilon_\theta}(0:t-1)$. Then, both $\|G_t\|$ and $\|\nabla F_t(\theta_t)\|$ are upper bounded by $\frac{C_{L,h,\theta}}{1-\rho}$ (as defined in Corollary 17), and the following inequalities holds for any two time steps $\tau \le t$:

$$\|\theta_t - \theta_\tau\| \le \frac{C_{L,h,\theta}}{1-\rho} \cdot (t-\tau)\eta, \text{ and } \|x_\tau - \hat{x}_\tau(\theta_t)\| \le \frac{C_{L,h,\theta} C_{L,g,\theta}}{(1-\rho)^2}\left((t-\tau) + \frac{1}{1-\rho}\right) \cdot \rho \cdot \eta,$$

where we use the notation $\hat{x}_\tau(\theta) := g_{\tau|w}(x_0, \theta_{\times(\tau+1)}), \forall \theta \in \Theta$. Further, we have that

$$|h_t(x_t, u_t, \theta_t) - F_t(\theta_t)| \le \frac{C_{L,h,\theta} C_{L,g,\theta} L_h (1 + L_{\pi,x} + L_{f,x})\rho}{(1-\rho)^3} \cdot \eta.$$

Proof We first use induction to show that for all time step $t \in \mathcal{T}$,

$$\|G_t\| \le \frac{C_{L,h,\theta}}{1-\rho},\ x_t \in B_n(0, \bar{C}(R_S + \bar{C}\|x_0\|) + R_S),\ u_t \in \mathcal{U}, \text{ and } \|\theta_{t+1} - \theta_t\| \le \epsilon_\theta, \tag{63}$$

where $\mathcal{U}$ is defined in Assumption 8. Note that $\|G_0\| \le C_{L,h,\theta} \le \frac{C_{L,h,\theta}}{1-\rho}$ by Corollary 17. Since $\bar{C} \ge 1$ by Definition 3,

$$\bar{C}(R_S + \bar{C}\|x_0\|) + R_S \ \ge\ \bar{C}^2\|x_0\| \ \ge\ \bar{C}\|x_0\| \ \ge\ \|x_0\|.$$

Thus, we have $x_0 \in B_n(0, \bar{C}(R_S + \bar{C}\|x_0\|) + R_S)$. We also have $u_0 \in \mathcal{U}$.

Suppose $\|G_{t-1}\| \le \frac{C_{L,h,\theta}}{1-\rho}$ for some $t \ge 1$. Then, since the projection onto Θ is nonexpansive (Lemma 14) and $\eta \le \frac{(1-\rho)\epsilon_\theta}{C_{L,h,\theta}}$, we see that

$$\|\theta_t - \theta_{t-1}\| = \|\Pi_\Theta(\theta_{t-1} - \eta G_{t-1}) - \Pi_\Theta(\theta_{t-1})\| \le \|\eta G_{t-1}\| \le \epsilon_\theta.$$

Suppose $\|\theta_\tau - \theta_{\tau-1}\| \le \epsilon_\theta$ holds for all $\tau \le t$, i.e., $\theta_{0:t} \in S_{\epsilon_\theta}(0:t)$. Applying Lemma 19 with $x_\tau = x_0$ yields $x_t \in B_n(0, \bar{C}(R_S + \bar{C}\|x_0\|) + R_S)$ for any t. By the definition of $\mathcal{U}$ (constructed over a state domain that contains this ball), we also have $u_t \in \mathcal{U}$. Therefore, from the definition of G_t in M-GAPS (8), we see that

$$\begin{aligned}
\|G_t\| &= \left\| \sum_{\tau=0}^{t} \frac{\partial h_{t|t-\tau}}{\partial \theta_{t-\tau}}\bigg|_{x_{t-\tau}, \theta_{t-\tau:t}} \right\| \\
&\le \sum_{\tau=0}^{t} \left\| \frac{\partial h_{t|t-\tau}}{\partial \theta_{t-\tau}}\bigg|_{x_{t-\tau}, \theta_{t-\tau:t}} \right\| && \text{(64a)} \\
&\le \sum_{\tau=0}^{t} C_{L,h,\theta}\rho^\tau && \text{(64b)} \\
&\le \frac{C_{L,h,\theta}}{1-\rho},
\end{aligned}$$

where we use the triangle inequality in (64a) and Corollary 17 in (64b). Note that we can apply Corollary 17 because $x_t \in B_n(0, \bar{R}_C)$. Therefore, we have shown (63) by induction. One can use

the same technique as (64) to show $\|\nabla F_t(\theta_t)\| \le \frac{C_{L,h,\theta}}{1-\rho}$. Since the projection onto Θ is nonexpansive according to Lemma 14, we have for every k,

$$\begin{aligned}\|\theta_{k+1} - \theta_k\| &= \|\Pi_\Theta(\theta_k - \eta G_k) - \Pi_\Theta(\theta_k)\| \\ &\le \|(\theta_k - \eta G_k) - \theta_k\| = \eta \|G_k\| \;\le\; \frac{C_{L,h,\theta}}{1-\rho}\eta,\end{aligned}$$

where the last step uses the uniform bound on $\|G_k\|$ established above. Summing from $k = \tau$ to $t-1$ (triangle inequality) yields

$$\|\theta_t - \theta_\tau\| \le \sum_{k=\tau}^{t-1} \|\theta_{k+1} - \theta_k\| \le \frac{C_{L,h,\theta}}{1-\rho} \cdot (t-\tau)\eta. \tag{65}$$

Now we bound the distance between x_τ and $\hat{x}_\tau(\theta_t)$ for $\tau \le t$. We see that

$$\begin{aligned}\big\|x_\tau - \hat{x}_\tau(\theta_t)\big\| &= \big\|g_{\tau|0}(x_0, \theta_{0:\tau-1}) - g_{\tau|0}(x_0, (\theta_t)_{\times\tau})\big\| \\ &\le \sum_{\tau'=0}^{\tau-1} \big\|g_{\tau|0}(x_0, \theta_{0:\tau'}, (\theta_t)_{\times(\tau-\tau'-1)}) - g_{\tau|0}(x_0, \theta_{0:\tau'-1}, (\theta_t)_{\times(\tau-\tau')})\big\| \qquad (66a) \\ &\le \sum_{\tau'=0}^{\tau-1} \big\|g_{\tau|\tau'}(x_{\tau'}, \theta_{\tau'}, (\theta_t)_{\times(\tau-\tau'-1)}) - g_{\tau|\tau'}(x_{\tau'}, (\theta_t)_{\times(\tau-\tau')})\big\| \qquad (66b)\end{aligned}$$

where we use the triangle inequality in (66a) as follows: we bridge the two parameter strings $\theta_{0:\tau-1}$ and $(\theta_t)_{\times\tau}$ by a sequence that changes one coordinate at a time, from left to right. Concretely, start from $g_{\tau|0}(x_0, \theta_{0:\tau-1})$ and, for each $\tau' = 0, 1, \ldots, \tau-1$, replace the τ'-th entry $\theta_{\tau'}$ by θ_t, keeping all other entries fixed; after τ such single-coordinate replacements we reach $g_{\tau|0}(x_0, (\theta_t)_{\times\tau})$ (here $(\theta_t)_{\times m}$ means m copies of θ_t). Writing the difference between the endpoints as the sum of these τ one-step differences and applying $\|\sum_i v_i\| \le \sum_i \|v_i\|$ yields (66a).
We use the definition of the multi-step dynamics in (66b) by factoring the horizon at each τ'. It is equality by the semigroup (composition) property of the multi-step map. By definition,

$$g_{t|\tau}(x, \theta_{\tau:t-1}) := g_{t-1}(g_{t-2}(\cdots g_\tau(x, \theta_\tau) \cdots, \theta_{t-2}),\, \theta_{t-1}).$$

Hence, concatenating the parameter string as $(\theta_{0:\tau'},\, (\theta_t)_{\times(\tau-\tau'-1)}) = (\theta_{0:\tau'-1},\, \theta_{\tau'},\, (\theta_t)_{\times(\tau-\tau'-1)})$, we can factor the $0 \to \tau$ evolution at time τ':

$$\begin{aligned}g_{\tau|0}(x_0, \theta_{0:\tau'}, (\theta_t)_{\times(\tau-\tau'-1)}) &= g_{\tau-1}\big(\cdots g_{\tau'}\big(g_{\tau'|0}(x_0, \theta_{0:\tau'-1}),\, \theta_{\tau'}\big) \cdots,\, \underbrace{\theta_t, \ldots, \theta_t}_{\tau-\tau'-1 \text{ times}}\big) \\ &= g_{\tau|\tau'}\big(g_{\tau'|0}(x_0, \theta_{0:\tau'-1}),\, \theta_{\tau'},\, (\theta_t)_{\times(\tau-\tau'-1)}\big).\end{aligned}$$

Similarly, for the string $(\theta_{0:\tau'-1},\, (\theta_t)_{\times(\tau-\tau')})$ we obtain

$$g_{\tau|0}(x_0, \theta_{0:\tau'-1}, (\theta_t)_{\times(\tau-\tau')}) = g_{\tau|\tau'}\big(g_{\tau'|0}(x_0, \theta_{0:\tau'-1}),\, (\theta_t)_{\times(\tau-\tau')}\big).$$

Defining the intermediate state $x_{\tau'} := g_{\tau'|0}(x_0, \theta_{0:\tau'-1})$, both identities become

$$g_{\tau|0}(x_0, \theta_{0:\tau'}, (\theta_t)_{\times(\tau-\tau'-1)}) = g_{\tau|\tau'}\big(x_{\tau'},\, \theta_{\tau'},\, (\theta_t)_{\times(\tau-\tau'-1)}\big),$$

$$g_{\tau|0}(x_0, \theta_{0:\tau'-1}, (\theta_t)_{\times(\tau-\tau')}) = g_{\tau|\tau'}\big(x_{\tau'},\ (\theta_t)_{\times(\tau-\tau')}\big),$$

so each "one–coordinate change" term in (66a) is exactly a difference of two $\tau' \to \tau$ evolutions that share the same intermediate state $x_{\tau'}$ and the same suffix length $\tau - \tau'$, which is the form in (66b).

Then, we see that

$$\begin{aligned}
\big\|x_\tau - \hat{x}_\tau(\theta_t)\big\| &\le \sum_{\tau'=0}^{\tau-1} C_{L,g,\theta}\, \rho^{\tau-\tau'}\, \big\|\theta_t - \theta_{\tau'}\big\| && \text{(67a)}\\
&\le \sum_{\tau'=0}^{\tau-1} C_{L,g,\theta}\, \rho^{\tau-\tau'} \left(\frac{C_{L,h,\theta}}{1-\rho}\,(t-\tau')\,\eta\right) && \text{(67b)}\\
&= \frac{C_{L,h,\theta}\, C_{L,g,\theta}}{1-\rho}\,\eta \sum_{\tau'=0}^{\tau-1} (t-\tau')\,\rho^{\tau-\tau'} \\
&= \frac{C_{L,h,\theta}\, C_{L,g,\theta}}{1-\rho}\,\eta \left((t\tau)\sum_{j=1}^{\tau}\rho^j \ + \ \sum_{j=1}^{\tau} j\,\rho^j\right) && \text{(67c)}
\end{aligned}$$

where we use Lemma 16 in (67a); we use (65) in (67b); to pass from (67b) to (67c), we rewrite and reindex the sum: $\sum_{\tau'=0}^{\tau-1}(t-\tau')\,\rho^{\tau-\tau'} = \sum_{j=1}^{\tau}[(t-\tau)+j]\,\rho^j = (t-\tau)\sum_{j=1}^{\tau}\rho^j \ + \ \sum_{j=1}^{\tau} j\,\rho^j$, where we set $j = \tau - \tau'$ so that as τ' runs $0 \to \tau - 1$, j runs $1 \to \tau$, and $t - \tau' = t - (\tau - j) = (t - \tau) + j$.

Then, the derivation follows as:

$$\begin{aligned}
\big\|x_\tau - \hat{x}_\tau(\theta_t)\big\| &\le \frac{C_{L,h,\theta}\, C_{L,g,\theta}}{1-\rho}\,\eta\left((t-\tau)\frac{\rho}{1-\rho} \ + \ \frac{\rho}{(1-\rho)^2}\right) && \text{(68a)}\\
&= \frac{C_{L,h,\theta}\, C_{L,g,\theta}}{(1-\rho)^2}\left((t-\tau) + \frac{1}{1-\rho}\right)\rho\,\eta, && \text{(68b)}
\end{aligned}$$

where we upper bound the finite sums by their infinite geometric counterparts: $\sum_{j=1}^{\tau}\rho^j \le \sum_{j=1}^{\infty}\rho^j = \frac{\rho}{1-\rho}$, $\sum_{j=1}^{\tau} j\,\rho^j \le \sum_{j=1}^{\infty} j\,\rho^j = \frac{\rho}{(1-\rho)^2}$ in (68a); we rearranged and factored out common terms in (68b). Similarly, given that $\hat{x}_t(\theta_t) = g_{t|0}(x_0, (\theta_t)_{\times t})$, since $(\theta_t)_{\times t} \in S_{\epsilon_\theta}(0 : t-1)$, and consecutive differences are $0 \le \epsilon_\theta$, by Lemma 19 with $\tau = 0$ and ϵ_θ-time-varying stability, we have $\|\hat{x}_t(\theta_t) - g_{t|0}(0, (\theta_t)_{\times t})\| \le \bar{C}\rho^t\|x_0\|$ and $\|g_{t|0}(0, (\theta_t)_{\times t})\| \le R_S$, hence $\|\hat{x}_t(\theta_t)\| \le R_S + \bar{C}\|x_0\| + \le \bar{C}(R_S + \bar{C}\|x_0\|) + R_S)$, which means $\hat{x}_t(\theta_t) \in B_n(0, \bar{C}(R_S + \bar{C}\|x_0\|) + R_S)$. Together with $x_t \in B_n(0, \bar{C}(R_S + \bar{C}\|x_0\|) + R_S)$ derived earlier, we obtain that

$$\begin{aligned}
|h_t(x_t, u_t, \theta_t) - F_t(\theta_t)| &= |h_t(x_t, u_t, \theta_t) - h_t(\hat{x}_t(\theta_t), \hat{u}_t(\theta_t), \theta_t)| \\
&\le L_h(\|x_t - \hat{x}_t(\theta_t)\| + \|u_t - \hat{u}_t(\theta_t)\|) && \text{(69a)}\\
&\le L_h(1 + L_{\pi,x} + L_{f,x})\|x_t - \hat{x}_t(\theta_t)\| && \text{(69b)}\\
&\le \frac{C_{L,h,\theta} C_{L,g,\theta} L_h(1 + L_{\pi,x} + L_{h,x})\rho}{(1-\rho)^3}\cdot\eta, && \text{(69c)}
\end{aligned}$$

where we use Theorem 11 in (69a) and (69b); we use (68) in (69c). [James: Where was L_h defined? I thought Theorem 11 only gives us separate Lipschitz constants for each input of h.] ■

Recall that we define the gradient approximation G_t for M-GAPS in (8). Using this notation, the update rule of $\theta_{0:T-1}$ in joint dynamics (62) can be simplified as

$$\theta_{t+1} = \Pi_\Theta(\theta_t - \eta G_t).$$

To compare the trajectory of M-GAPS with the trajectory achieved by the online gradient descent trajectory $\theta_{t+1} = \Pi_\Theta(\theta_t - \eta \nabla F_t(\theta_t))$, we bound the difference between the gradient approximation G_t and the true surrogate gradient $\nabla F_t(\theta_t)$ in Theorem 21. This difference/discrepancy, referred to as gradient bias, arises from the mismatch between the surrogate gradient—defined under fixed parameters—and the recursive sensitivity propagation under slowly varying parameters. Leveraging the contraction stability and trajectory mismatch bounds established above, we show that this gradient bias remains uniformly controlled.

Theorem 21 (Gradient Bias) *Suppose Assumptions 11 and 12 hold. Let $\{x_t, u_t, \theta_t\}_{t\in\mathcal{T}}$ denote the trajectory of (62). Suppose η satisfies the constraint that $\bar{\varepsilon} := \frac{C_{L,h,\theta}\eta}{1-\rho} \le \varepsilon$. Then, the following holds for all $\tau \le t$:*

$$\left\| \left.\frac{\partial h_{t|0}}{\partial \theta_\tau}\right|_{x_0,\theta_{0:t}} - \left.\frac{\partial h_{t|0}}{\partial \theta_\tau}\right|_{x_0,(\theta_t)_{\times(t+1)}} \right\| \le \left(\hat{C}_0 + \hat{C}_1(t-\tau) + \hat{C}_2(t-\tau)^2\right)\rho^{t-\tau}\cdot\eta.$$

for

$$\hat{C}_0 = \frac{\rho C_{L,h,\theta} C_{L,g,\theta} C_{\ell,h,(\theta,x)}}{(1-\rho)^3},\ \hat{C}_1 = \frac{(1-\rho)C_{L,h,\theta}C_{\ell,h,(\theta,x)} + \rho C_{L,h,\theta}C_{L,g,\theta}C_{\ell,h,(\theta,\theta)}}{(1-\rho)^2},$$
$$\hat{C}_2 = \frac{C_{L,h,\theta}C_{\ell,h,(x,\theta)}C_{L,g,\theta}}{1-\rho}.$$

Next,

$$\|G_t - \nabla F_t(\theta_t)\| \le \left(\frac{\hat{C}_0}{1-\rho} + \frac{\hat{C}_1 + \hat{C}_2}{(1-\rho)^2} + \frac{2\hat{C}_2}{(1-\rho)^3}\right)\eta.$$

Proof To simplify the notation, we adopt the shorthand notations $\hat{x}_\tau(\theta) := g_{\tau|0}(x_0, \theta_{\times\tau})$ and $\hat{u}_\tau(\theta) := \pi_\tau(\hat{x}_\tau(\theta), \theta)$ throughout the proof.

We use the triangle inequality to do the decomposition

$$\begin{aligned}
&\left\| \left.\frac{\partial h_{t|0}}{\partial \theta_\tau}\right|_{x_0,\theta_{0:t}} - \left.\frac{\partial h_{t|0}}{\partial \theta_\tau}\right|_{x_0,(\theta_t)_{\times(t+1)}} \right\| \\
&= \left\| \left.\frac{\partial h_{t|\tau}}{\partial \theta_\tau}\right|_{x_\tau,\theta_{\tau:t}} - \left.\frac{\partial h_{t|\tau}}{\partial \theta_\tau}\right|_{\hat{x}_\tau(\theta_t),(\theta_t)_{\times(t-\tau+1)}} \right\| \\
&\le \left\| \left.\frac{\partial h_{t|\tau}}{\partial \theta_\tau}\right|_{x_\tau,\theta_\tau,(\theta_t)_{\times(t-\tau)}} - \left.\frac{\partial h_{t|\tau}}{\partial \theta_\tau}\right|_{\hat{x}_\tau(\theta_t),(\theta_t)_{\times(t-\tau+1)}} \right\| \\
&\quad + \sum_{\tau'=\tau+1}^{t-1} \left\| \left.\frac{\partial h_{t|\tau}}{\partial \theta_\tau}\right|_{x_\tau,\theta_{\tau:\tau'},(\theta_t)_{\times(t-\tau')}} - \left.\frac{\partial h_{t|\tau}}{\partial \theta_\tau}\right|_{x_\tau,\theta_{\tau:\tau'-1},(\theta_t)_{\times(t-\tau'+1)}} \right\|.
\end{aligned} \tag{70}$$

Note that we can apply Corollary 17 to bound each term in (70). For the first term in (70), since $x_\tau, \hat{x}_\tau(\theta_t), x_{\tau+1} \in B_n(0, \bar{R}_C)$, we see that

$$\begin{aligned}
&\left\| \frac{\partial h_{t|\tau}}{\partial \theta_\tau}\bigg|_{x_\tau, \theta_\tau, (\theta_t)_{\times(t-\tau)}} - \frac{\partial h_{t|\tau}}{\partial \theta_\tau}\bigg|_{\hat{x}_\tau(\theta_t), (\theta_t)_{\times(t-\tau+1)}} \right\| \\
&\le \rho^{t-\tau}\big(C_{\ell,h,(\theta,x)}\|x_\tau - \hat{x}_\tau(\theta_t)\| + C_{\ell,h,(\theta,\theta)}\|\theta_t - \theta_\tau\|\big) && \text{(71a)} \\
&\le \frac{(1-\rho)C_{L,h,\theta}C_{\ell,h,(\theta,x)} + \rho C_{L,h,\theta}C_{L,g,\theta}C_{\ell,h,(\theta,\theta)}}{(1-\rho)^2} \cdot (t-\tau)\rho^{t-\tau} \cdot \eta \\
&\quad + \frac{\rho C_{L,h,\theta}C_{L,g,\theta}C_{\ell,h,(\theta,x)}}{(1-\rho)^3} \cdot \rho^{t-\tau} \cdot \eta, && \text{(71b)}
\end{aligned}$$

where we invoke Corollary 17 in (71a) because both terms there evaluate $\partial h_{t|\tau}/\partial\theta_\tau$ along "frozen-tail" θ sequences in which the future parameters $\theta_{\tau+1:t}$ are held fixed at θ_t (i.e., $(\theta_t)_{\times(t-\tau)}$ or $(\theta_t)_{\times(t-\tau+1)}$), and Theorem 20 in (71b).

For any $\tau' \in [\tau+1 : t-1]$, since $x_{\tau'}, x_{\tau'+1} \in B_n(0, \bar{R}_C)$, we see that

$$\begin{aligned}
&\left\| \frac{\partial h_{t|\tau}}{\partial \theta_\tau}\bigg|_{x_\tau, \theta_{\tau:\tau'}, (\theta_t)_{\times(t-\tau')}} - \frac{\partial h_{t|\tau}}{\partial \theta_\tau}\bigg|_{x_\tau, \theta_{\tau:\tau'-1}, (\theta_t)_{\times(t-\tau'+1)}} \right\| \\
&= \left\| \left(\frac{\partial h_{t|\tau'}}{\partial x_{\tau'}}\bigg|_{x_{\tau'}, \theta_{\tau'}, (\theta_t)_{\times(t-\tau')}} - \frac{\partial h_{t|\tau'}}{\partial x_{\tau'}}\bigg|_{x_{\tau'}, (\theta_t)_{\times(t-\tau'+1)}} \right) \frac{\partial g_{\tau'|\tau}}{\partial \theta_\tau}\bigg|_{x_\tau, \theta_{\tau:\tau'-1}} \right\| \\
&\le \left\| \frac{\partial h_{t|\tau'}}{\partial x_{\tau'}}\bigg|_{x_{\tau'}, \theta_{\tau'}, (\theta_t)_{\times(t-\tau')}} - \frac{\partial h_{t|\tau'}}{\partial x_{\tau'}}\bigg|_{x_{\tau'}, (\theta_t)_{\times(t-\tau'+1)}} \right\| \cdot \left\| \frac{\partial g_{\tau'|\tau}}{\partial \theta_\tau}\bigg|_{x_\tau, \theta_{\tau:\tau'-1}} \right\| \\
&\le C_{\ell,h,(x,\theta)}\rho^{t-\tau'}\|\theta_t - \theta_{\tau'}\| \cdot C_{L,g,\theta}\rho^{\tau'-\tau} && \text{(72a)} \\
&\le C_{\ell,h,(x,\theta)}C_{L,g,\theta} \cdot \rho^{t-\tau} \cdot \left(\frac{C_{L,h,\theta}}{1-\rho} \cdot (t-\tau')\eta \right), && \text{(72b)}
\end{aligned}$$

where we use Lemma 16 and Corollary 17 in (72a); we use Theorem 20 in (72b). Substituting (71) and (72) into (70) finishes the proof of the first inequality.

For the second inequality, recall that G_t and $\nabla F_t(\theta_t)$ are given by

$$G_t \coloneqq \sum_{\tau=0}^{t} \frac{\partial h_{t|0}}{\partial \theta_{t-\tau}}\bigg|_{x_0, \theta_{0:t}}, \nabla F_t(\theta_t) = \sum_{\tau=0}^{t} \frac{\partial h_{t|0}}{\partial \theta_{t-\tau}}\bigg|_{x_0, (\theta_t)_{\times(t+1)}}.$$

Therefore, we see that

$$\begin{aligned}
\|G_t - \nabla F_t(\theta_t)\| &= \left\| \sum_{\tau=0}^{t} \frac{\partial h_{t|0}}{\partial \theta_{t-\tau}}\bigg|_{x_0, \theta_{0:t}} - \sum_{\tau=0}^{t} \frac{\partial h_{t|0}}{\partial \theta_{t-\tau}}\bigg|_{x_0, (\theta_t)_{\times(t+1)}} \right\| \\
&\le \sum_{\tau=0}^{t} \left\| \frac{\partial h_{t|0}}{\partial \theta_{t-\tau}}\bigg|_{x_0, \theta_{0:t}} - \frac{\partial h_{t|0}}{\partial \theta_{t-\tau}}\bigg|_{x_0, (\theta_t)_{\times(t+1)}} \right\| && \text{(73a)} \\
&\le \sum_{\tau=0}^{t} \Big(\hat{C}_0 + \hat{C}_1\tau + \hat{C}_2\tau^2 \Big)\rho^\tau \eta && \text{(73b)}
\end{aligned}$$

$$\leq \left(\frac{\hat{C}_0}{1-\rho} + \frac{\hat{C}_1 + \hat{C}_2}{(1-\rho)^2} + \frac{2\hat{C}_2}{(1-\rho)^3} \right) \eta, \tag{73c}$$

where we use the triangle inequality in (73a); we use the first inequality in Theorem 21 that we have shown and Corollary 17 in (73b). To pass from (73b) to (73c), we bound the finite sum by its infinite counterpart and apply standard geometric–series identities. Specifically, starting from

$$\sum_{\tau=0}^{t} \left(\hat{C}_0 + \hat{C}_1 \tau + \hat{C}_2 \tau^2 \right) \rho^\tau \;\leq\; \sum_{\tau=0}^{\infty} \left(\hat{C}_0 + \hat{C}_1 \tau + \hat{C}_2 \tau^2 \right) \rho^\tau,$$

we use, for $|\rho| < 1$,

$$\sum_{\tau=0}^{\infty} \rho^\tau = \frac{1}{1-\rho}, \quad \sum_{\tau=0}^{\infty} \tau\, \rho^\tau = \frac{\rho}{(1-\rho)^2}, \quad \sum_{\tau=0}^{\infty} \tau(\tau-1)\, \rho^\tau = \frac{2\rho^2}{(1-\rho)^3}.$$

Since $\tau^2 = \tau(\tau-1) + \tau$, it follows that

$$\sum_{\tau=0}^{\infty} \tau^2\, \rho^\tau = \frac{\rho}{(1-\rho)^2} + \frac{2\rho^2}{(1-\rho)^3}.$$

Substituting into the infinite sum gives

$$\sum_{\tau=0}^{\infty} (\hat{C}_0 + \hat{C}_1 \tau + \hat{C}_2 \tau^2)\, \rho^\tau = \frac{\hat{C}_0}{1-\rho} + \hat{C}_1 \frac{\rho}{(1-\rho)^2} + \hat{C}_2 \left(\frac{\rho}{(1-\rho)^2} + \frac{2\rho^2}{(1-\rho)^3} \right).$$

Finally, since $|\rho| < 1$, we can loose the bound to obtain

$$\frac{\hat{C}_0}{1-\rho} + \frac{\hat{C}_1 + \hat{C}_2}{(1-\rho)^2} + \frac{2\hat{C}_2}{(1-\rho)^3},$$

which is exactly the coefficient stated in (73c). ■

[James: Full proof was carefully checked by Shengnan and James in person]

5.5. Cumulative Variation of the Surrogate Objectives

Before proving Property 7, we collect a technical bound on the cumulative variation of the surrogate objectives $\{F_t\}_{t=0}^{T-1}$ induced by the time-varying dynamics. In particular, the next lemma controls the path-length of $\{F_t\}$ in terms of the total variation budget V and the diameter of the state domain, and will be used to upper bound the "environment drift" term that appears in the regret decomposition.

Lemma 22 (Cumulative Variation of the Surrogate Objectives) *[James: TODO: I need to check this carefully] [James: TODO: Check if the θ_t contribution to h_t has been handled correctly when moving from GAPS to M-GAPS – similar to Corollary 17.] Suppose Assumptions 11 and 12. Then, we have*

$$\sum_{t=1}^{T-1} \|F_t - F_{t-1}\|_{\infty,s} \leq \frac{2\bar{C} L_h (1 + L_{\pi,x})(1 + L_{g,u})}{(1-\rho)^2 \rho} V \;+\; \frac{2\bar{C} L_h (1 + L_{\pi,x})}{1-\rho} \mathsf{diam}(\mathcal{X}),$$

where $\|F_t - F_{t-1}\|_{\infty,s} := \sup_{\theta \in \Theta} |F_t(\theta) - F_{t-1}(\theta)|$ *and* $\mathsf{diam}(\mathcal{X}) := 2\bar{C}\big(R_S + \bar{C}\|x_0\|\big) + 2R_S$.

Proof To simplify the notation, we introduce the notations

$$\|g - g'\|_{\infty,d} \coloneqq \sup_{x\in\mathcal{X},\, u\in\mathcal{U}} \|g(x,u) - g'(x,u)\|,$$
$$\|\pi - \pi'\|_{\infty,p} \coloneqq \sup_{x\in\mathcal{X},\, \theta\in\Theta} \|\pi(x,\theta) - \pi'(x,\theta)\|,$$
$$\|h - h'\|_{\infty,c} \coloneqq \sup_{x\in\mathcal{X},\, u\in\mathcal{U}} |h(x,u) - h'(x,u)|.$$

For any $\theta \in \Theta$, we see that

$$\begin{aligned}
&|F_t(\theta) - F_{t-1}(\theta)| \\
=&\Big|h_t\big(\tilde{x}_t(\theta),\, \pi_t(\tilde{x}_t(\theta),\theta)\big) - h_{t-1}\big(\tilde{x}_{t-1}(\theta),\, \pi_{t-1}(\tilde{x}_{t-1}(\theta),\theta)\big)\Big| && \text{(74a)}\\
\le&\Big|h_t\big(\tilde{x}_t(\theta),\, \pi_t(\tilde{x}_t(\theta),\theta)\big) - h_t\big(\tilde{x}_{t-1}(\theta),\, \pi_{t-1}(\tilde{x}_{t-1}(\theta),\theta)\big)\Big| \\
&\quad + \Big|h_t\big(\tilde{x}_{t-1}(\theta),\, \pi_{t-1}(\tilde{x}_{t-1}(\theta),\theta)\big) - h_{t-1}\big(\tilde{x}_{t-1}(\theta),\, \pi_{t-1}(\tilde{x}_{t-1}(\theta),\theta)\big)\Big| && \text{(74b)}\\
\le&\Big|h_t\big(\tilde{x}_t(\theta),\, \pi_t(\tilde{x}_t(\theta),\theta)\big) - h_t\big(\tilde{x}_{t-1}(\theta),\, \pi_{t-1}(\tilde{x}_{t-1}(\theta),\theta)\big)\Big| + \|h_t - h_{t-1}\|_{\infty,c} && \text{(74c)}
\end{aligned}$$

where (74a) follows from the definition of the surrogate cost (Definition 6); in (74b), we add and subtract the intermediate term $h_t\big(\tilde{x}_{t-1}(\theta),\, \pi_{t-1}(\tilde{x}_{t-1}(\theta),\theta)\big)$, and apply the triangle inequality; in (74c), we bound the second term by $\|h_t, h_{t-1}\|_{\infty,c}$ by definition. Then, by Lipschitzness, we can deduct

$$\begin{aligned}
&|F_t(\theta) - F_{t-1}(\theta)| \\
\le& L_h\Big(\|\tilde{x}_t(\theta) - \tilde{x}_{t-1}(\theta)\| + \|\pi_t(\tilde{x}_t(\theta),\theta) - \pi_{t-1}(\tilde{x}_{t-1}(\theta),\theta)\|\Big) + \|h_t - h_{t-1}\|_{\infty,c} && \text{(74d)}\\
\le& L_h\Big(\|\tilde{x}_t(\theta) - \tilde{x}_{t-1}(\theta)\| + \|\pi_t(\tilde{x}_t(\theta),\theta) - \pi_t(\tilde{x}_{t-1}(\theta),\theta)\|\Big) + L_h\|\pi_t - \pi_{t-1}\|_{\infty,p} + \|h_t - h_{t-1}\|_{\infty,c} && \text{(74e)}\\
\le& L_h(1 + L_{\pi,x})\,\|\tilde{x}_t(\theta) - \tilde{x}_{t-1}(\theta)\| + \Big(L_h\|\pi_t - \pi_{t-1}\|_{\infty,p} + \|h_t - h_{t-1}\|_{\infty,c}\Big) && \text{(74f)}
\end{aligned}$$

where in (74d), we use h_t is L_h-Lipschitz in (x,u) uniformly over $\theta \in \Theta$, so $\big|h_t(x,u,\theta) - h_t(x',u',\theta)\big| \le L_h\big(\|x - x'\| + \|u - u'\|\big)$; in (74e), we further add and subtract $\pi_t(\tilde{x}_{t-1}(\theta),\theta)$, and bound $\|\pi_t(\tilde{x}_{t-1}(\theta),\theta) - \pi_{t-1}(\tilde{x}_{t-1}(\theta),\theta)\|$ by the definition of $\|\pi - \pi'\|_{\infty,p}$; (74f) follows from the Lipschitzness of π_t in its state argument, $\|\pi_t(x,\theta) - \pi_t(x',\theta)\| \le L_{\pi,x}\|x - x'\|$. To bound $\|\tilde{x}_t(\theta) - \tilde{x}_{t-1}(\theta)\|$, we first bound the difference between $\tilde{x}_t(\theta)$ and $x_t^{(g,\pi)}(\theta)$ for arbitrary $(g,\pi) \in \mathcal{G}$. Note that $\tilde{x}_t(\theta) = g_{t|0}\big(x_0^{(g,\pi)},\theta\big)$, thus

$$\begin{aligned}
&\|\tilde{x}_t(\theta) - x_t^{(g,\pi)}(\theta)\| \\
\le& \sum_{\tau=0}^{t-1}\Big\|g_{t|\tau+1}\big(x_{\tau+1}^{(g,\pi)}(\theta),\theta,x_{(\tau+1:t-1)}\big) - g_{t|\tau}\big(x_\tau^{(g,\pi)}(\theta),\theta,x_{(\tau:t-1)}\big)\Big\| && \text{(75a)}\\
=& \sum_{\tau=0}^{t-1}\Big\|g_{t|\tau+1}\big(x_{\tau+1}^{(g,\pi)}(\theta),\theta,x_{(\tau+1:t-1)}\big) - g_{t|\tau+1}\Big(g_{\tau+1|\tau}\big(x_\tau^{(g,\pi)}(\theta),\theta\big),\theta,x_{(\tau+1:t-1)}\Big)\Big\| && \text{(75b)}
\end{aligned}$$

$$\leq \bar{C} \sum_{\tau=0}^{t-1} \rho^{t-\tau-1} \Big\| x_{\tau+1}^{(g,\pi)}(\theta) - g_{\tau+1|\tau}\big(x_\tau^{(g,\pi)}(\theta), \theta\big) \Big\| \tag{75c}$$

where (75a) is obtained by a telescoping decomposition of $\tilde{x}_t(\theta) - x_t^{(g,\pi)}(\theta)$ across the intermediate times τ and the triangle inequality; (75b) uses the multi-step composition identity $g_{t|\tau}(\cdot) = g_{t|\tau+1}\big(g_{\tau+1|\tau}(\cdot)\big)$ and the definition of the one-step map $g_{\tau+1|\tau}$ to rewrite the second term inside the norm; (75c) follows from the ϵ-time-varying contractive perturbation (Definition 3), which bounds the Lipschitz constant of $g_{t|\tau+1}$ in its initial state by $\bar{C}\rho^{t-\tau-1}$. Then, we can apply frozen trajectory and time-varying update definitions:

$$\begin{aligned}
&\|\tilde{x}_t(\theta) - x_t^{(g,\pi)}(\theta)\| \\
=&\bar{C} \sum_{\tau=0}^{t-1} \rho^{t-\tau-1} \Big\| g\big(x_\tau^{(g,\pi)}(\theta),\, \pi(x_\tau^{(g,\pi)}(\theta), \theta)\big) - g_\tau\big(x_\tau^{(g,\pi)}(\theta),\, \pi_\tau(x_\tau^{(g,\pi)}(\theta), \theta)\big) \Big\| && \tag{75d} \\
\leq&\bar{C} \sum_{\tau=0}^{t-1} \rho^{t-\tau-1} \Big(\big\| g\big(x_\tau^{(g,\pi)}(\theta),\, \pi(x_\tau^{(g,\pi)}(\theta), \theta)\big) - g\big(x_\tau^{(g,\pi)}(\theta),\, \pi_\tau(x_\tau^{(g,\pi)}(\theta), \theta)\big) \big\| \\
&\qquad + \big\| g\big(x_\tau^{(g,\pi)}(\theta),\, \pi_\tau(x_\tau^{(g,\pi)}(\theta), \theta)\big) - g_\tau\big(x_\tau^{(g,\pi)}(\theta),\, \pi_\tau(x_\tau^{(g,\pi)}(\theta), \theta)\big) \big\| \Big) && \tag{75e} \\
\leq&\bar{C} \sum_{\tau=0}^{t-1} \rho^{t-\tau-1} \big(\|g_\tau - g\|_{\infty,d} + L_{g,u} \|\pi_\tau, \pi\|_{\infty,p} \big) && \tag{75f}
\end{aligned}$$

where (75d) uses the definition of the frozen trajectory $x_{\tau+1}^{(g,\pi)}(\theta) = g\big(x_\tau^{(g,\pi)}(\theta), \pi(x_\tau^{(g,\pi)}(\theta), \theta)\big)$ and the time-varying update $g_{\tau+1|\tau}\big(x_\tau^{(g,\pi)}(\theta), \theta\big) = g_\tau\big(x_\tau^{(g,\pi)}(\theta), \pi_\tau(x_\tau^{(g,\pi)}(\theta), \theta)\big)$; (75e) is obtained by adding and subtracting $g\big(x_\tau^{(g,\pi)}(\theta), \pi_\tau(x_\tau^{(g,\pi)}(\theta), \theta)\big)$ and applying the triangle inequality; and (75f) follows from the definitions of dist_d and dist_p together with the Lipschitzness of g in u, i.e., $\|g(x,u) - g(x,u')\| \leq L_{g,u}\|u - u'\|$.

Therefore, we see that

$$\begin{aligned}
&\|\tilde{x}_t(\theta) - \tilde{x}_{t-1}(\theta)\| \\
\leq&\|\tilde{x}_t(\theta) - x_t^{(g_t,\pi_t)}(\theta)\| + \|\tilde{x}_{t-1}(\theta) - x_{t-1}^{(g_t,\pi_t)}(\theta)\| \\
&\quad + \|x_t^{(g_t,\pi_t)}(\theta) - x_\infty^{(g_t,\pi_t)}(\theta)\| + \|x_{t-1}^{(g_t,\pi_t)}(\theta) - x_\infty^{(g_t,\pi_t)}(\theta)\| && \tag{76a} \\
\leq&\bar{C} \sum_{\tau=0}^{t-1} \rho^{t-\tau-1} \big(\|g_\tau, g_t\|_{\infty,d}) + L_{g,u} \|\pi_\tau, \pi_t\|_{\infty,p} \big) \\
&\quad + \bar{C} \sum_{\tau=0}^{t-2} \rho^{t-\tau-2} \big(\|g_\tau, g_t\|_{\infty,d}) + L_{g,u} \|\pi_\tau, \pi_t\|_{\infty,p} \big) \\
&\quad + \bar{C}\rho^t \mathsf{diam}(\mathcal{X}) + C\rho^{t-1}\mathsf{diam}(\mathcal{X}) && \tag{76b} \\
\leq&\frac{2\bar{C}}{1-\rho} \sum_{\tau=0}^{t-1} \rho^{t-\tau-2} \big(\|g_\tau, g_t\|_{\infty,d} + L_{g,u} \|\pi_\tau, \pi_t\|_{\infty,p} \big) + 2\bar{C}\rho^{t-1}\mathsf{diam}(\mathcal{X}) && \tag{76c}
\end{aligned}$$

where we use the triangle inequality in (76a); in (76b), we use the bound we derived in (75); in (76c), we use the triangle inequality decomposition

$$\|g_\tau, g_t\|_{\infty,d} \leq \sum_{j=\tau}^{t-1} \|g_{j+1}, g_j\|_{\infty,d}, \qquad \|\pi_\tau, \pi_t\|_{\infty,p} \leq \sum_{j=\tau}^{t-1} \|\pi_{j+1}, \pi_j\|_{\infty,p}.$$

Substituting this into (74) gives that

$$\begin{aligned}
&\|F_t, F_{t-1}\|_{\infty,s} \\
\leq &\frac{2\bar{C}L_h(1+L_{\pi,x})}{1-\rho} \cdot \sum_{\tau=0}^{t-1} \rho^{t-\tau-2}\big(\|g_\tau, g_{\tau+1}\|_{\infty,d} + L_{g,u}\|\pi_\tau, \pi_{\tau+1}\|_{\infty,p}\big) \\
&+ 2\bar{C}L_h(1+L_{\pi,x})\rho^{t-1}\mathsf{diam}(\mathcal{X}) + \big(L_h\|\pi_t, \pi_{t-1}\|_{\infty,p} + \|h_t, h_{t-1}\|_{\infty,c}\big),
\end{aligned} \tag{77}$$

because (76) holds for arbitrary $\theta \in \Theta$. Summing (77) over $t = 1, \ldots, T-1$ and rearranging the terms give that

$$\begin{aligned}
&\sum_{t=1}^{T-1} \|F_t, F_{t-1}\|_{\infty,s} \\
\leq &\frac{2\bar{C}L_h(1+L_{\pi,x})}{1-\rho} \cdot \sum_{t=1}^{T-1}\sum_{\tau=0}^{t-1} \rho^{t-\tau-2}\big(\|g_\tau, g_{\tau+1}\|_{\infty,d} + L_{g,u}\|\pi_\tau, \pi_{\tau+1}\|_{\infty,p}\big) \\
&+ 2\bar{C}L_f(1+L_{\pi,x})\sum_{t=1}^{T-1}\rho^{t-1}\mathsf{diam}(\mathcal{X}) + \sum_{t=1}^{T-1}\big(L_h\|\pi_t, \pi_{t-1}\|_{\infty,p} + \|h_t, h_{t-1}\|_{\infty,c}\big) \\
\leq &\frac{2\bar{C}L_h(1+L_{\pi,x})(1+L_{g,u})}{(1-\rho)^2\rho} P_T + \frac{2\bar{C}L_h(1+L_{\pi,x})}{1-\rho}\mathsf{diam}(\mathcal{X}).
\end{aligned}$$

■

5.6. Proof of the Regret Bound and Robustness Guarantee (Property 7)

We now complete the regret analysis by combining the trajectory mismatch and gradient bias bounds established above. These results show that the M-GAPS update closely approximates projected gradient descent with a controlled error at each step. By invoking the local regret bound for inexact projected online gradient descent (Theorem 13), we obtain the desired regret guarantee and robustness property for M-GAPS under slowly time-varying dynamics.

Proof We compare the dynamical system (7) with the Ideal OGD update rule:

$$\theta_{t+1} = \Pi_\Theta(\theta_t - \eta\nabla F_t(\theta_t)). \tag{78}$$

Note that the update on θ_t that the dynamical system (7) performs can be written equivalently as

$$\theta_{t+1} = \Pi_\Theta(\theta_t - \eta G_t), \tag{79}$$

where

$$G_t := \sum_{\tau=0}^{t} \frac{\partial h_{t|0}}{\partial \theta_{t-\tau}}\bigg|_{x_0,\theta_{0:t}}. \tag{80}$$

By Theorem 21, we know that

$$\|G_t - \nabla F_t(\theta_t)\| \leq \left(\frac{\hat{C}_0}{1-\rho} + \frac{\hat{C}_1 + \hat{C}_2}{(1-\rho)^2} + \frac{\hat{C}_2}{(1-\rho)^3}\right)\eta,$$

where the constants $\hat{C}_{0:5}$ are given in Theorem 21. Let θ_{t+1} be the actual next policy parameter (following the update rule (79)). By Lemma 14, we see that

$$\begin{aligned}\|\theta_{t+1} - \Pi_\Theta(\theta_t - \eta\nabla F_t(\theta_t))\| &\leq \eta\|G_t - \nabla F_t(\theta_t)\| \\ &\leq \left(\frac{\hat{C}_0}{1-\rho} + \frac{\hat{C}_1 + \hat{C}_2}{(1-\rho)^2} + \frac{\hat{C}_2}{(1-\rho)^3}\right)\eta^2.\end{aligned}$$

Then, we can apply Theorem 13 to obtain that

$$\sum_{t=0}^{T-1} \|\nabla_{\Theta,\eta} F_t(\theta_t)\|^2 \leq \frac{1}{\eta(1-\eta\ell_F)}\left(F_0(\theta_0) + \sum_{t=1}^{T-1}\|F_t - F_{t-1}\|_\infty\right) + \frac{L_F S_1 + \ell_F \eta S_2}{1-\eta\ell_F}, \tag{81}$$

where $\|\cdot\|_\infty$ is a metric that measures the distance between two surrogate cost functions (see Theorem 13 for definition), and S_1 and S_2 are given by

$$\begin{aligned}S_1 &:= (\left(\frac{\hat{C}_0}{1-\rho} + \frac{\hat{C}_1 + \hat{C}_2}{(1-\rho)^2} + \frac{\hat{C}_2}{(1-\rho)^3}\right)\eta T, \\ S_2 &:= \left(1 + \frac{\hat{C}_0}{1-\rho} + \frac{\hat{C}_1 + \hat{C}_2 + \hat{C}_3 + \hat{C}_5}{(1-\rho)^2} + \frac{\hat{C}_2 + \hat{C}_4}{(1-\rho)^3}\right)\cdot \\ &\quad \left[\left(\frac{\hat{C}_0}{1-\rho} + \frac{\hat{C}_1 + \hat{C}_2}{(1-\rho)^2} + \frac{\hat{C}_2}{(1-\rho)^3}\right)\eta^2 T\right].\end{aligned}$$

By applying Lemma 22, we can bound the total variational intensity on the surrogate costs by [James: We should reproduce this lemma in this tutorial]

$$\sum_{t=1}^{T-1}\|F_t - F_{t-1}\|_\infty \leq \frac{2\bar{C}L_h(1 + L_{\pi,x} + L_{f,x})(1 + L_{g,u})}{(1-\rho)^2\rho}\cdot V_{sys}$$

$$+\frac{2\bar{C}L_h(1+L_{\pi,x}+L_{f,x})}{1-\rho}\cdot\big(2\bar{C}\bar{R}_C+2R_S\big).$$

Substituting the above inequality and $L_F = \frac{C_{L,f,\theta}}{1-\rho}, \ell_F = \frac{C_{\ell,h,(\theta,\theta)}}{1-\rho}$ into (81) finishes the proof. [James: If we do this substitution and simplify into an expression involving η, can the motivation for the choice of η become clear? Maybe this would be easier than the discussions below.] ■

5.7. Discussion on the Choice of Learning Rate

Let the one–step surrogate be

$$F_t(\theta) \;=\; h_t(\tilde{x}_t(\theta),\ \pi_t(\tilde{x}_t(\theta),\theta),\ \theta),$$

and denote the (exact) sensitivity of the state to the policy parameter by

$$y_t(\theta) \;:=\; \frac{\partial \tilde{x}_t(\theta)}{\partial \theta}\,.$$

5.7.1. Slow–variation requirement

The projected update is $\theta_{t+1} = \Pi_\Theta(\theta_t - \eta\nabla F_t(\theta_t))$. By nonexpansiveness of Euclidean projection,

$$\|\theta_{t+1}-\theta_t\| \;=\; \|\Pi_\Theta(\theta_t-\eta\nabla F_t(\theta_t)) - \Pi_\Theta(\theta_t)\| \;\le\; \eta\,\|\nabla F_t(\theta_t)\|.$$

By the chain rule,

$$\nabla F_t(\theta) \;=\; \underbrace{\partial_\theta h_t(\cdot)}_{\text{direct }\theta\text{-channel}} \;+\; \underbrace{\partial_x h_t(\cdot)\,y_t(\theta)}_{\text{through state}} \;+\; \underbrace{\partial_u h_t(\cdot)\,(\partial_\theta \pi_t(\cdot)+\partial_x\pi_t(\cdot)\,y_t(\theta))}_{\text{through policy}}\,.$$

Property 18 (contraction) implies that perturbing θ affects the state with a gain bounded by the geometric series of decay factors, hence $\|y_t(\theta)\| \le \frac{\bar{C}_y}{1-\rho}$ for a time–uniform constant $\bar{C}_y$. Bounding $\|\partial_x h_t\|,\ \|\partial_u h_t\|,\ \|\partial_\theta h_t\|$ and $\|\partial_\theta \pi_t\|$ and $\|\partial_x \pi_t\|$ by their Lipschitz constants and folding $\bar{C}_y$ into a single coefficient gives

$$\|\nabla F_t(\theta)\| \;\le\; \frac{C_{L,h,\theta}}{1-\rho}.$$

Therefore

$$\|\theta_{t+1}-\theta_t\| \;\le\; \eta\,\frac{C_{L,h,\theta}}{1-\rho}.$$

To enforce the slowly–time–varying constraint $\|\theta_{t+1}-\theta_t\| \le \epsilon_\theta$, it suffices to choose

$$\eta \;<\; \frac{(1-\rho)\,\epsilon_\theta}{C_{L,h,\theta}}\,.$$

5.7.2. Smoothness requirement (stability of the gradient step)

We also need a stepsize that is small enough for the projected gradient analysis on a smooth objective. We seek a Lipschitz bound of the form

$$\big\|\nabla F_t(\theta) - \nabla F_t(\theta')\big\| \;\le\; \ell_F \,\big\|\theta - \theta'\big\|.$$

Split the difference using the chain rule and bound each term:

$$\begin{aligned}
&\big\|\nabla F_t(\theta) - \nabla F_t(\theta')\big\| \\
&= \Big\|\partial_\theta h_t(\theta) - \partial_\theta h_t(\theta') + \partial_x h_t(\theta)\, y_t(\theta) - \partial_x h_t(\theta')\, y_t(\theta') \\
&\qquad + \partial_u h_t(\theta)\Big(\partial_\theta \pi_t(\theta) + \partial_x \pi_t(\theta)\, y_t(\theta)\Big) - \partial_u h_t(\theta')\Big(\partial_\theta \pi_t(\theta') + \partial_x \pi_t(\theta')\, y_t(\theta')\Big)\Big\| \\
&\le \big\|\partial_\theta h_t(\theta) - \partial_\theta h_t(\theta')\big\| + \big\|\partial_x h_t(\theta)\, y_t(\theta) - \partial_x h_t(\theta')\, y_t(\theta')\big\| \\
&\qquad + \Big\|\partial_u h_t(\theta)\Big(\partial_\theta \pi_t(\theta) + \partial_x \pi_t(\theta)\, y_t(\theta)\Big) - \partial_u h_t(\theta')\Big(\partial_\theta \pi_t(\theta') + \partial_x \pi_t(\theta')\, y_t(\theta')\Big)\Big\| \\
&\le C_{\ell,h,(\theta,\theta)}\, \|\theta - \theta'\| + \big\|\partial_x h_t(\theta) - \partial_x h_t(\theta')\big\|\, \|y_t(\theta)\| \\
&\qquad + \|\partial_x h_t(\theta')\|\, \|y_t(\theta) - y_t(\theta')\| + C_{\ell,h,(\theta,\theta)}\, \|\theta - \theta'\| \\
&\le C_{\ell,h,(\theta,\theta)}\, \|\theta - \theta'\| + C_{\ell,h,(\theta,x)}\, \|\theta - \theta'\|\, \frac{C_y}{1-\rho} + L_{h,x}\, \frac{\tilde{C}_y}{1-\rho}\, \|\theta - \theta'\| + C_{\ell,h,(\theta,\theta)}\, \|\theta - \theta'\| \\
&\le \Big(C_{\ell,h,(\theta,\theta)} + \frac{C_{\ell,h,(\theta,x)}\, C_y + L_{h,x}\, \tilde{C}_y}{1-\rho}\Big)\, \|\theta - \theta'\|. \qquad (82)
\end{aligned}$$

The sensitivity bounds

$$\|y_t(\theta)\| \;\le\; \frac{C_y}{1-\rho}, \qquad \|y_t(\theta) - y_t(\theta')\| \;\le\; \frac{\tilde{C}_y}{1-\rho}\, \|\theta - \theta'\|$$

follow from the contraction property (Property 18): $\theta - \theta'$ changes the state by at most $\sum_\tau \gamma(\tau) \le \frac{1}{1-\rho}$, which directly yields the same factor for the state sensitivity. Collecting terms and folding $C_{\ell,h,(\theta,x)} C_y + L_{h,x} \tilde{C}_y$ (together with the through-policy channel) into the same symbol as the direct θ–channel for a convenient single coefficient, we obtain

$$\big\|\nabla F_t(\theta) - \nabla F_t(\theta')\big\| \;\le\; \Big(C_{\ell,h,(\theta,\theta)} + \frac{C_{\ell,h,(\theta,\theta)}}{1-\rho}\Big)\, \|\theta - \theta'\| \;\le\; \frac{2\, C_{\ell,h,(\theta,\theta)}}{1-\rho}\, \|\theta - \theta'\|.$$

Hence F_t is ℓ_F–smooth with $\ell_F \le \dfrac{2\, C_{\ell,h,(\theta,\theta)}}{1-\rho}$, as claimed. Then, for the projected update $\theta_{t+1} = \Pi_\Theta(\theta_t - \eta\, \nabla F_t(\theta_t))$, the standard descent lemma on an ℓ_F-smooth objective gives

$$F_t(\theta_{t+1}) \;\le\; F_t(\theta_t) - \eta\, \big\langle \nabla F_t(\theta_t),\, \nabla_{\Theta,\eta} F_t(\theta_t)\big\rangle + \frac{\ell_F}{2}\, \eta^2\, \big\|\nabla_{\Theta,\eta} F_t(\theta_t)\big\|^2.$$

To keep this stable (nonexpansive up to the inner product term), it suffices to require $\eta \le 1/\ell_F$. Using the bound on ℓ_F above, a convenient sufficient condition is

$$\eta \;\le\; \frac{1-\rho}{2\, C_{\ell,h,(\theta,\theta)}}\,.$$

Combining the two requirements yields the final stepsize condition

$$\boxed{\eta \;<\; \min\left\{ \frac{(1-\rho)\,\epsilon_\theta}{C_{L,h,\theta}} \;,\; \frac{1-\rho}{2\,C_{\ell,h,(\theta,\theta)}} \right\}.}$$

This choice simultaneously (i) guarantees the slowly time–varying update $\|\theta_{t+1} - \theta_t\| \le \epsilon_\theta$ and (ii) keeps the projected gradient step stable under the ℓ_F-smooth surrogate.

6. Comparison with the M-GAPS

Originally, M-GAPS considers online policy optimization in a discrete-time dynamical system that varies over time with dynamics $x_{t+1} = \tilde{g}_t(x_t, u_t) + w_t$, where $\tilde{g}_t(x_t, u_t) := g_t(x_t, u_t, f_t(x_t, a_t^*))$, $x_t \in \mathbb{R}^n$ denotes the system state, $u_t \in \mathbb{R}^m$ denotes the control input, and g_t is the dynamical function. Here, $f_t(x_t, a_t^*) \in \mathbb{R}^k$ is a nonlinear residual term of which the online agent can make (noisy) observations. It has a known function form f_t and an unknown parameter $a_t^* \in \mathcal{A} \subseteq \mathbb{R}^p$. The disturbance term $w_t \in \mathcal{W} \subseteq \mathbb{R}^n$ does not depend on the states or the control inputs.

Standalone correspondence. The standalone variant corresponds to the special case of the original formulation in which

- the exact a_t^* are always known;
- the nonlinear residual is absent (equivalently, $f_t(x_t, a_t^*) \equiv 0$), so the known model is $g_t(x_t, u_t)$ and no residual estimation is required;
- no exploration/algorithmic perturbation is used (i.e., $\zeta_t \equiv 0$);
- exogenous disturbances are ignored (i.e., w_t is set to zero for analysis);

Under these conditions, the working dynamics are reduced to $x_{t+1} = g_t(x_t, u_t)$ with policy $u_t = \pi_t(x_t, \theta_t)$.

7. Conclusion

Conclusion. We studied online policy optimization in a simplified "standalone" regime where the controller has direct access to the maps it needs and no estimator or exogenous disturbance is present. In this setting we formalized the M-GAPS update with constant memory and per-step computation, and analyzed the induced joint recursion of (x_t, y_t, θ_t). Under standard smoothness and a time-varying contraction condition, the update is stable, the parameter path remains slowly varying with an appropriate stepsize, and the algorithm achieves sublinear local regret with an explicit dependence on system variation. These guarantees mirror those known for GAPS Lin et al. (2023) while removing the estimation layer, thereby isolating which effects are intrinsic to the policy update mechanism.

Outlook. Manifold–aware M-GAPS: A natural next step is to place the policy parameter on a Riemannian manifold (Θ, g) and replace the Euclidean update by a manifold step.

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Outline of the appendices.

- In Appendix **??**, we provide a formal treatment of the projected gradient, including a proof of its equivalence to the KKT conditions under convex inequality and affine equality constraints.
- In Appendix B, we provide a notation table and a constant table to summarize the important notations in this paper.

Appendix A. Zero Projected Gradient is Equivalent to KKT

Let $\Theta \subset \mathbb{R}^d$ be closed and convex, $\eta > 0$. Define the projected gradient

$$\nabla_{\Theta,\eta} F(\theta) \;:=\; \frac{1}{\eta}(\theta - \Pi_\Theta(\theta - \eta \nabla F(\theta))).$$

Then the projected step can be written equivalently as

$$\theta^+ \;=\; \theta - \eta\, \nabla_{\Theta,\eta} F(\theta) \;=\; \Pi_\Theta(\theta - \eta \nabla F(\theta)).$$

By the metric projection optimality condition, $y - \Pi_\Theta(y) \in N_\Theta(\Pi_\Theta(y))$, where

$$N_\Theta(p) \;:=\; \{\, w : \; \langle w,\, v - p \rangle \le 0 \;\; \forall v \in \Theta \,\}$$

is the normal cone. Taking $y = \theta - \eta \nabla F(\theta)$ yields the KKT-style characterization

$$\nabla_{\Theta,\eta} F(\theta) = 0 \iff \theta = \Pi_\Theta(\theta - \eta \nabla F(\theta)) \iff 0 \in \nabla F(\theta) + N_\Theta(\theta).$$

Two immediate facts: (i) if $\Theta = \mathbb{R}^d$, then $\nabla_{\Theta,\eta} F(\theta) = \nabla F(\theta)$; (ii) zeros of the projected gradient coincide with first–order stationarity under the constraint Θ.

Projected gradient and KKT conditions. In what follows, use the notation $[N] := \{1, \dots, N\}$.

Proposition 23 (Zero projected gradient $\iff$ KKT conditions) *Let $\eta > 0$ and let $F : \mathbb{R}^d \to \mathbb{R}$ be differentiable. Suppose the constraint set has the convex-inequality/affine-equality form*

$$\Theta \;=\; \Big\{ x \in \mathbb{R}^d : \; f_i(x) \le 0 \;\; \forall i \in [m], \;\; h_j(x) = 0 \;\; \forall j \in [p] \Big\},$$

where each $f_i : \mathbb{R}^d \to \mathbb{R}$ is convex and differentiable and each $h_j : \mathbb{R}^d \to \mathbb{R}$ is affine. Assume the constraint qualification (Slater): there exists $\bar{x} \in \mathbb{R}^d$ [Ryan: Should this be $\bar{x} \in \mathrm{relint}\Theta$? (relint is relative interior)] [Shengnan: I also observed that Wikipedia and some convex analysis references state Slater's condition in a general form: given a convex constraint set D, Slater requires the existence of a point $x \in \mathrm{relint}(D)$ that strictly satisfies all inequality constraints. The use of relint *is essential in that level of generality because D may be lower-dimensional (e.g., due to affine equalities), so* $\mathrm{int}(D)$ *can be empty even when a meaningful notion of strict feasibility holds. In our setting,*

$$\Theta = \{x \in \mathbb{R}^d : \; f_i(x) \le 0 \; \forall i, \;\; h_j(x) = 0 \; \forall j\},$$

with convex differentiable f_i and affine h_j. Let

$$\mathcal{A} := \{x \in \mathbb{R}^d : \; h_j(x) = 0, \; \forall j\},$$

the affine set defined by the equalities. Then $\Theta \subseteq \mathcal{A}$, where $\mathcal{A}$ is an affine set defined by the equality constraints. Since $\mathcal{A}$ is affine and contains Θ, we have $\operatorname{aff}(\Theta) \subseteq \mathcal{A}$. Moreover, the strict inequalities $f_i(\bar{x}) < 0$ ensure that Θ is not contained in any proper affine subset of $\mathcal{A}$, so its affine hull coincides with $\mathcal{A}$, i.e.,

$$\operatorname{aff}(\Theta) = \mathcal{A}.$$

Under the stated Slater assumption, there exists $\bar{x} \in \mathcal{A}$ such that $f_i(\bar{x}) < 0$ for all i. By continuity of each f_i, there exists $\varepsilon > 0$ such that for all $x \in \mathcal{A}$ with $\|x - \bar{x}\| < \varepsilon$, we still have $f_i(x) < 0$ for all i, hence $x \in \Theta$. Therefore $\bar{x}$ is an interior point of Θ relative to $\mathcal{A} = \operatorname{aff}(\Theta)$, i.e.,

$$\bar{x} \in \operatorname{relint}(\Theta) \quad \textit{(relative to } \operatorname{aff}(\Theta) = \mathcal{A}\textit{)}.$$

So while it is not necessary to explicitly write $\bar{x} \in \operatorname{relint}(\Theta)$ in the statement, the strict-feasibility form we use is equivalent to the relative-interior formulation in this affine-equality setting] such that $f_i(\bar{x}) < 0$ for all $i \in [m]$ and $h_j(\bar{x}) = 0$ for all $j \in [p]$. Then for any $\theta \in \Theta$,

$$\nabla_{\Theta,\eta} F(\theta) = 0 \quad \Longleftrightarrow \quad \exists\, \lambda_1, \ldots, \lambda_m \geq 0,\ \nu_1, \ldots, \nu_p \in \mathbb{R} \textit{ s.t.}$$

$$f_i(\theta) \leq 0 \ \ \forall i \in [m], \qquad h_j(\theta) = 0 \ \ \forall j \in [p], \tag{83}$$

$$\lambda_i f_i(\theta) = 0 \ \ \forall i \in [m], \tag{84}$$

$$\nabla F(\theta) + \sum_{i=1}^{m} \lambda_i \nabla f_i(\theta) + \sum_{j=1}^{p} \nu_j \nabla h_j(\theta) = 0. \tag{85}$$

Proof Step 1 (projected-gradient zeros $\Longleftrightarrow$ normal-cone stationarity). Write the projected step as

$$\Pi_\Theta(y) \ \in \ \arg\min_{v \in \Theta} \ \frac{1}{2}\|v - y\|^2.$$

A standard first-order optimality condition for this strongly convex problem is: for $p = \Pi_\Theta(y)$,

$$\langle p - y,\ v - p \rangle \geq 0 \qquad \forall v \in \Theta. \tag{86}$$

Equivalently,

$$y - p \in N_\Theta(p), \qquad N_\Theta(p) := \{\, w : \ \langle w,\ v - p \rangle \leq 0 \ \ \forall v \in \Theta \,\}.$$

Now take $y = \theta - \eta \nabla F(\theta)$. By Definition 5,

$$\nabla_{\Theta,\eta} F(\theta) = 0 \quad \Longleftrightarrow \quad \theta = \Pi_\Theta(\theta - \eta \nabla F(\theta)).$$

With $p = \theta$ and $y = \theta - \eta \nabla F(\theta)$, condition (86) becomes

$$\langle \theta - (\theta - \eta \nabla F(\theta)),\ v - \theta \rangle \leq 0 \quad \forall v \in \Theta,$$

i.e.

$$-\nabla F(\theta) \in N_\Theta(\theta) \quad \Longleftrightarrow \quad 0 \in \nabla F(\theta) + N_\Theta(\theta). \tag{87}$$

Thus it remains to characterize $N_\Theta(\theta)$ for the given Θ.

Step 2 (explicit description of $N_\Theta(\theta)$). Define the active set $I(\theta) := \{i \in [m] : f_i(\theta) = 0\}$. We claim that, under Slater,

$$N_\Theta(\theta) = \Big\{\sum_{i=1}^{m} \lambda_i \nabla f_i(\theta) + \sum_{j=1}^{p} \nu_j \nabla h_j(\theta) : \ \lambda_i \geq 0, \ \lambda_i f_i(\theta) = 0 \ \forall i \in [m], \ \nu_j \in \mathbb{R}\Big\}. \tag{88}$$

(a) "$\supseteq$" in (88). Fix any $\lambda_i \geq 0$ and $\nu_j \in \mathbb{R}$ with $\lambda_i f_i(\theta) = 0$. Let

$$w \ := \ \sum_{i=1}^{m} \lambda_i \nabla f_i(\theta) + \sum_{j=1}^{p} \nu_j \nabla h_j(\theta).$$

Take any $v \in \Theta$. Since each f_i is convex and differentiable,

$$f_i(v) \ \geq \ f_i(\theta) + \langle \nabla f_i(\theta), \, v - \theta \rangle \quad \Longrightarrow \quad \langle \nabla f_i(\theta), \, v - \theta \rangle \ \leq \ f_i(v) - f_i(\theta).$$

Because $v \in \Theta$ implies $f_i(v) \leq 0$ and $\theta \in \Theta$ implies $f_i(\theta) \leq 0$, we have $f_i(v) - f_i(\theta) \leq -f_i(\theta)$. Hence

$$\langle \nabla f_i(\theta), \, v - \theta \rangle \ \leq \ -f_i(\theta).$$

Multiplying by $\lambda_i \geq 0$ and summing over $i \in [m]$ yields

$$\sum_{i=1}^{m} \lambda_i \langle \nabla f_i(\theta), \, v - \theta \rangle \ \leq \ -\sum_{i=1}^{m} \lambda_i f_i(\theta) \ = \ 0,$$

where the last equality uses $\lambda_i f_i(\theta) = 0$ for all i. Next, since each h_j is affine, for all v,

$$h_j(v) - h_j(\theta) = \langle \nabla h_j(\theta), \, v - \theta \rangle.$$

Because $v, \theta \in \Theta$ imply $h_j(v) = h_j(\theta) = 0$, we get $\langle \nabla h_j(\theta), \, v - \theta \rangle = 0$ for all $j \in [p]$. Therefore,

$$\langle w, \, v - \theta \rangle = \sum_{i=1}^{m} \lambda_i \langle \nabla f_i(\theta), \, v - \theta \rangle + \sum_{j=1}^{p} \nu_j \langle \nabla h_j(\theta), \, v - \theta \rangle \ \leq \ 0.$$

Since this holds for every $v \in \Theta$, we conclude $w \in N_\Theta(\theta)$, proving the "$\supseteq$" direction of (88).

(b) "$\subseteq$" in (88). Let $w \in N_\Theta(\theta)$. By definition of the normal cone with x denoting an arbitrary point in Θ,

$$\langle w, \, x - \theta \rangle \leq 0 \quad \forall x \in \Theta.$$

Rearranging the inequality yields

$$\langle w, x \rangle \leq \langle w, \theta \rangle \quad \forall x \in \Theta,$$

or equivalently,

$$\langle -w, x \rangle \geq \langle -w, \theta \rangle \quad \forall x \in \Theta.$$

Hence, among all $x \in \Theta$, the linear functional $x \mapsto \langle -w, x \rangle$ attains its minimum at $x = \theta$, that is,

$$\theta \in \arg\min_{x \in \Theta} \langle -w, x \rangle.$$

Consider the convex optimization problem

$$\min_{x\in\mathbb{R}^d} \langle -w, x\rangle \quad \text{s.t.} \quad f_i(x) \le 0\ \forall i \in [m],\ \ h_j(x) = 0\ \forall j \in [p]. \tag{89}$$

The objective is linear (hence convex), the inequalities are convex, and the equalities are affine. By Slater's condition, strong duality holds and there exist KKT multipliers for (89); since the problem is convex, these KKT conditions are sufficient for optimalityTherefore, since θ is optimal, there exist multipliers $\lambda_1, \dots, \lambda_m \ge 0$ and $\nu_1, \dots, \nu_p \in \mathbb{R}$ such that

$$\begin{aligned}
&\text{(stationarity)} \qquad -w + \sum_{i=1}^{m} \lambda_i \nabla f_i(\theta) + \sum_{j=1}^{p} \nu_j \nabla h_j(\theta) = 0,\\
&\text{(complementary slackness)} \qquad \lambda_i f_i(\theta) = 0\ \ \forall i \in [m],
\end{aligned}$$

together with feasibility of $\theta \in \Theta$. Rearranging stationarity gives exactly

$$w = \sum_{i=1}^{m} \lambda_i \nabla f_i(\theta) + \sum_{j=1}^{p} \nu_j \nabla h_j(\theta),$$

with $\lambda_i \ge 0$ and $\lambda_i f_i(\theta) = 0$, proving the "$\subseteq$" direction of (88). This completes the characterization.

Step 3 (conclude the KKT equivalence). Combining (87) with (88), we obtain

$$\begin{aligned}
\nabla_{\Theta,\eta} F(\theta) = 0 &\iff -\nabla F(\theta) \in N_\Theta(\theta)\\
&\iff \exists\, \lambda_1, \dots, \lambda_m \ge 0,\ \nu_1, \dots, \nu_p \in \mathbb{R} \text{ s.t.}\\
&\qquad -\nabla F(\theta) = \sum_{i=1}^{m} \lambda_i \nabla f_i(\theta) + \sum_{j=1}^{p} \nu_j \nabla h_j(\theta), \qquad \lambda_i f_i(\theta) = 0\ \ \forall i \in [m],
\end{aligned}$$

together with feasibility $\theta \in \Theta$. Equivalently, (83)–(85) hold. ■

This result establishes the equivalence between projected-gradient stationarity and the KKT conditions. The proof first reformulates a zero projected gradient as a normal-cone stationarity condition via the first-order optimality of the projection operator. Under Slater's condition, the normal cone admits an explicit representation in terms of constraint gradients, yielding the standard KKT system.

Appendix B. Table of Terms

As there are many parameters and constants in our theoretical analysis, we provide a notation table (Table 1) and a constant table (Table 2) here that summarize the important terms in this paper.

Table 1: Important notations in this paper.

Notation	Meaning
$t_1 : t_2$	The integer sequence $\{t_1, \ldots, t_2\}$;
$a_{t_1:t_2}$	A sequence of variables $\{a_t\}_{t=t_1,\ldots,t_2}$;
$\|\cdot\|$	ℓ_2 (Euclidean) norm;
$\|\cdot\|_F$	Frobenius norm;
$\|\cdot\|_P$	Norm induced by matrix P;
$\mathbb{Z}_{\geq 0}$	The set of non-negative integers;
$\mathbb{R}_{\geq 0}$	The set of non-negative reals;
$\sigma(z_{1:t}, z'_{1:t})$	Product sigma-algebra generated by sequences $z_{1:t}$ and $z'_{1:t}$;
x_t	$x_t \in \mathbb{R}^n$ is the system state;
u_t	$u_t \in \mathbb{R}^m$ is the control input;
R^L_η	Local regret with learning rate η;
$q_t(x_t, y_t, \theta_t)$	The joint dynamics of the system at time t;
$g_{t\|\tau}$	Multi-step dynamics between two time steps $\tau \leq t$;
$h_{t\|\tau}$	Multi-step cost between two time steps $\tau \leq t$;
$\Pi_\Theta(y)$	Euclidean projection of y to set Θ;
$\nabla_{\Theta,\eta} F(\theta)$	The (Θ, η)-projected gradient of F;
$S_{\epsilon_\theta}(t_1 : t_2)$	The set of policy parameter sequences with ϵ_θ-constrained step size;
$B_n(\cdot, \cdot)$	Closed Euclidean ball in $\mathbb{R}^n$ (center, radius);

Table 2: Important constants in this paper.

Notation	Meaning
Γ	The Contraction Stability Constant Γ associated with γ, defined in Property 18;
$L_{g,x}$	State-wise Lipschitz Constant of Dynamics, defined in Assumption 11;
$L_{g,u}$	Input-wise Lipschitz Constant of Dynamics, defined in Assumption 11;
$\ell_{g,x}$	State-Jacobian Smoothness of Dynamics, defined in Assumption 11;
$\ell_{g,u}$	Input-Jacobian Smoothness of dynamics, defined in Assumption 11;
$L_{\pi,x}$	State-wise Lipschitz Constant of Policy, defined in Assumption 11;
$L_{\pi,\theta}$	Parameter-wise Lipschitz Constant of Policy, defined in Assumption 11;
$\ell_{\pi,x}$	State-Jacobian Smoothness of Policy, defined in Assumption 11;
$\ell_{\pi,\theta}$	Parameter-Jacobian Smoothness of Policy, defined in Assumption 11;
$L_{h,x}$	Cost Lipschitz Constant in x, defined in Assumption 11;
$L_{h,u}$	Cost Lipschitz Constant in u, defined in Assumption 11;
$L_{h,\theta}$	Cost Lipschitz Constant in θ, defined in Assumption 11;
$\ell_{h,x}$	Gradient Smoothness Constant of cost in x, defined in Assumption 11;
$\ell_{h,u}$	Gradient Smoothness Constant of cost in u, defined in Assumption 11;
$\ell_{h,\theta}$	Gradient Smoothness Constant of cost in θ, defined in Assumption 11;
R_C	Contraction Region Radius, defined in Assumption 12;
R_S	Uniform Stability Bound, defined in Assumption 12;
S_0	Baseline Variation Constant, defined in Property 7;
S_1	Linear Perturbation Gain, defined in Property 7;
S_2	Quadratic Perturbation Gain, defined in Property 7;
$\bar{C} = C_{L,g,x}$	Multi-Step State-Sensitivity Lipschitz Constant, defined in Lemma 16;
$C_{L,g,\theta}$	Multi-Step Parameter-Sensitivity Lipschitz Constant, defined in Lemma 16;
$C_{\ell,g,(x,x)}$	Self-Smoothness of the State-Jacobian in x, defined in Lemma 16;
$C_{\ell,g,(x,\theta)}$	Cross-Smoothness of the State-Jacobian w.r.t. Parameter Change, defined in Lemma 16;
$C_{\ell,g,(\theta,x)}$	Cross-Smoothness of the Parameter-Jacobian w.r.t. State Change, defined in Lemma 16;
$C_{\ell,g,(\theta,\theta)}$	Self-Smoothness of the Parameter-Jacobian in θ, defined in Lemma 16;
$C_{L,h,x}$	Multi-Step Cost Gradient Lipschitz in State, defined in Corollary 17;
$C_{L,h,\theta}$	Multi-Step Cost Gradient Lipschitz in Parameter, defined in Corollary 17;
$C_{\ell,h,(x,x)}$	Self-smoothness of the Cost State-Gradient in x, defined in Corollary 17;
$C_{\ell,h,(x,\theta)}$	Cross-Smoothness of the Cost State-Gradient w.r.t. Parameter Change, defined in Corollary 17;
$C_{\ell,h,(\theta,x)}$	Cross-Smoothness of the Cost Parameter-Gradient w.r.t. State Change, defined in Corollary 17;
$C_{\ell,h,(\theta,\theta)}$	Self-smoothness of the Cost Parameter-Gradient in θ, defined in Corollary 17;
$\hat{C}_0$	Constant η-Bias Coefficient, defined in Theorem 21;
$\hat{C}_1$	Linear η-Bias Coefficient, defined in Theorem 21;
$\hat{C}_2$	Quadratic η-Bias Coefficient, defined in Theorem 21;